MM-HEALTHFUSION

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ABSTRACT

Multi-modal artificial intelligence (MMAI) can combine different sources of information collected during a patient's care, but routine clinical records are rarely complete. Investigations are ordered for clinical reasons, so the availability and timing of each data modality may carry clinical signals. Building on prior NHS England research, **mm-healthfusion** developed a shared modelling framework for evaluating multi-modal fusion strategies on heterogeneous and incomplete clinical records. We leveraged the MIMIC-IV open dataset to define two clinical prediction pathways: short-term deterioration following emergency department attendance and outcomes associated with inpatient hospitalisation. Both pathways contained sparse and incomplete records. We extended the range of clinical data by incorporating ECG waveforms and chest X-ray images alongside structured EHR data, vital signs, laboratory tests and clinical notes. Within this framework, a common interface allowed simple concatenation to be compared with more complex fusion architectures, including sparse mixture-of-experts routing and graph-based imputation of missing modalities. Beyond predictive performance, the framework supported modality permutation to separate the information carried by each modality from its availability, Shapley-based attribution at the modality and feature level, spatial explanations for chest X-rays and fairness evaluation. Our results suggest that additional architectural complexity does not guarantee better prediction. We found that predictive performance depended mainly on recorded clinical information, although modality availability made a smaller contribution to certain outcomes. The resulting framework provides a reusable foundation for evaluating multi-modal fusion strategies in future healthcare research. This work was conducted as part of an NHS England PhD Internship. The code is openly available on GitHub.

Keywords: multi-modal fusion, missing modalities, mixture-of-experts, graph convolutional network, fairness, explainability, electronic health records

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Chapter 1

Introduction

It was a busy Tuesday afternoon when a woman walked into the emergency department (ED). After the receptionist took her details and basic symptoms, she sat down and waited to be called. A few minutes later, a triage nurse brought her to the assessment room and asked what had brought her in. The lady complained about having chest pain since the morning. A cuff tightened around her arm, while a pulse oximeter clipped onto her finger. The nurse noticed that her pulse was irregular, so an ECG was arranged. She remained in the department for further assessment and observation. An hour later, she became increasingly short of breath. A repeated measurement indicated dropping blood oxygen, so oxygen was administered and a chest X-ray was requested. Vitals, waveforms, images and notes accumulated at different times, each capturing a different part of the same clinical picture. With each step, her hospital record grew. While she remained under observation, the cardiac monitor began to alarm as her heart rhythm suddenly deteriorated. Moments later, she went into cardiac arrest.

Clinicians rely on experience and different sources of information to make clinical decisions, often under the pressure of time, labour and resources. No single source tells the whole story. But taken together, could the information collected shortly after a patient's arrival have warned clinicians that he or she was at risk of deteriorating? Such scenarios have motivated research into multi-modal clinical prediction, where information from different data sources is combined to support risk assessment. Previous studies, including the HAIM framework [1], have demonstrated improvements in predictive performance through multi-modal integration. However, the relative contribution of each modality varies across clinical prediction tasks.

Despite the recent progress, important challenges remain. Even in the most comprehensive and tidy healthcare research datasets, hospital records are often "incomplete". In fact, this reflects the reality of each patient's journey. Measurements are taken at irregular intervals, tests are ordered for different clinical reasons, and serious deteriorations are rare. The availability and timing of each data modality may therefore carry important clinical meaning. A clinically useful multi-modal model must be able to handle these natural variations in care rather than assuming each patient follows the same patient journey. Any well-performing method should demonstrate robustness and fairness in different patient groups, and provide reasonable explanations for its predictions.

This study aimed to investigate how different multi-modal fusion paradigms perform across heterogeneous and incomplete clinical records. Building on previous NHS England research, **mm-healthfusion** developed a shared modelling framework to compare simple feature concatenation

with two more complex fusion architectures, FuseMoE and M3Care. The evaluation covered two retrospectively defined clinical prediction pathways: short-term deterioration following emergency department attendance and outcomes associated with inpatient hospitalisation. Beyond predictive performance, the study examined how architectural choices influence the handling of missing modalities, the contribution of different information sources, fairness across patient groups and model explainability. These pathways differ in their prediction objectives, observation windows and availability of clinical information, allowing fusion strategies to be evaluated under distinct clinical data conditions. Our study was guided by the following research questions:

- Could the proposed fusion methods outperform simple concatenation of patient embeddings across multiple modalities?
- Do training adaptations such as curriculum learning and contrastive learning improve model learning?
- Does modelling irregularly collected time-series data as temporal sequences improve model performance?
- How does each modality contribute to the model predictions?
- Do modality availability and fusion methods affect fairness across patient groups?
- Do findings from one pathway generalise to the other?

This work extended from previous **MM-HealthFair** [2, 3] research, which investigated fairness and explainability in multi-modal clinical prediction. We extended the range of clinical data by incorporating ECG waveforms and chest X-ray images alongside structured EHR data, time-series measurements and clinical notes. The study also introduced an emergency department deterioration pathway adapted from **MDS-ED** [4] and accommodated more heterogeneous patterns of modality availability. Using a shared modelling and evaluation framework, we compared different fusion architectures across clinical prediction tasks and examined their robustness to missing information, modality contributions, fairness and explainability. The resulting **mm-healthfusion** framework provides a reusable foundation for evaluating multi-modal fusion strategies in future healthcare research.

Chapter 2

Background

2.1 Multimodality in healthcare

Healthcare decision-making utilises various sources of patient information, including demographics, medical history, observations, laboratory tests, medications, ECGs, imaging, and clinical notes. Multimodal predictions via machine learning (ML) represent these clinical modalities as digital data modalities, such as tabular data, time-series, physiological signals, images and text, and attempt to integrate them dynamically to form patient representations that are useful for different downstream tasks [5].

Tabular data comprises structured information that can be organised into rows and columns, including patient- and episode-level variables such as demographics (e.g., age, sex and ethnicity), administrative information (e.g., route of admission, insurance and billing information), medical history, diagnoses, medications and clinical scores [6, 7]. Such structured EHR data provide an important fundamental source of information for clinical prediction and have been extensively analysed using conventional ML approaches, including logistic regression, decision trees, support vector machines and gradient-boosting methods such as XGBoost [8]. More recently, deep-learning (DL) approaches such as Deep Neural Networks (DNNs) and Graph Neural Networks (GNNs) have increasinal EHR data due to their angly been applied to structured EHR data to learn patient representations for downstream clinical prediction tasks [9, 10].

Time-series data consist of information collected sequentially at multiple time points over a given period. Common examples in hospital settings include vital signs and laboratory test results, which provide a dynamic representation of patients' physiological and clinical status. Temporal patterns in such measurements have been utilised for clinical prediction and monitoring, including the detection of patient deterioration [11, 12]. Despite the widespread availability of time-series data in healthcare, irregular sampling, sparsity and missingness remain common challenges for their analysis and may require appropriate preprocessing and modelling strategies [13].

Physiological signals are another form of temporal data that capture dynamic biological processes. Common modalities include electrocardiograms (ECGs), electroencephalograms (EEGs), electromyograms (EMGs) and electrooculograms (EOGs) [14]. A standard 12-lead ECG uses ten electrodes to derive twelve leads that record the electrical activity of the heart from different orientations. It is widely used in clinical practice to identify cardiovascular abnormalities, including cardiac arrhythmias and myocardial ischaemia or infarction [15].

Time-series data can be analysed using conventional statistical and ML methods, often by first summarising temporal measurements into engineered features such as means, extrema, trends or measurements within predefined time windows. More recently, DL methods have enabled temporal representations to be learned directly from sequential data. Recurrent neural networks (RNNs), particularly long short-term memory (LSTM) and gated recurrent unit (GRU) networks, have been widely applied to longitudinal EHR data due to their ability to model dependencies across successive observations [13, 12]. Convolutional neural networks (CNNs) have also been adapted to capture local temporal patterns, while attention mechanisms and Transformer-based architectures have been increasingly adopted to model longer-range dependencies within patient trajectories [13, 16]. These approaches have been applied to clinical tasks including disease and clinical-event prediction, patient risk prediction, and representation learning [13]. DL methods have also demonstrated considerable capacity to extract clinically relevant information from physiological signals, particularly ECGs, supporting applications in cardiovascular analysis and disease diagnosis [14, 17].

Medical imaging provides visual representations of anatomical structures and physiological processes and encompasses a range of modalities, including X-ray, computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), and ultrasound. These modalities play an important role in disease screening, diagnosis, treatment planning, and monitoring across clinical specialties, including oncology, cardiology, neurology, respiratory medicine, and musculoskeletal care [18, 19]. Interpretation of medical images has traditionally relied heavily on trained clinicians and radiologists. However, manual interpretation can be time-consuming and subject to intra- and inter-observer variability [20]. Consequently, computer-aided detection and diagnosis (CAD) systems have been developed to assist clinicians in identifying and characterising abnormalities in medical images [21].

While conventional ML methods made early contributions to medical image analysis, they often relied on manually engineered image features. In contrast, DL enables hierarchical representations to be learned directly from imaging data and has been widely applied to tasks including image classification, detection, segmentation, registration, reconstruction, and enhancement [19]. CNNs have been the predominant DL architecture for medical image analysis, while Vision Transformers (ViTs) and related Transformer architectures have increasingly been investigated for their ability to capture long-range spatial relationships [22]. Generative approaches, including generative adversarial networks (GANs) and diffusion models, have further expanded the applications of DL to areas such as image synthesis and reconstruction [23]. With the increasing availability of published imaging datasets, large-scale foundation models have emerged as a promising direction, using extensive pretraining to learn generalisable representations that can be adapted to multiple medical imaging modalities and downstream clinical tasks [24, 25].

Textual data constitute a major source of unstructured information within EHRs and could take the form of clinical notes, admission and progress notes, radiology reports, and discharge summaries. Unlike structured EHR fields, clinical narratives can capture detailed information about patients' symptoms, medical history, diagnoses, treatments, clinical reasoning, and care trajectories that may not be fully represented in coded or tabular data [26, 27]. Discharge summaries are particularly information-rich documents that consolidate information from a hospital episode, including the reason for admission, clinical findings, diagnoses, treatments, hospital course, and follow-up plans. However, the unstructured and heterogeneous nature of clinical text, together with variations in terminology, abbreviations, writing styles, and documentation practices, presents challenges for automated analysis [26]. Natural language processing (NLP) methods have therefore been widely

applied to clinical text for tasks including information extraction, named entity recognition, classification, phenotyping, summarisation, and clinical prediction [26, 27]. Transformer-based pretrained language models have become increasingly prominent, enabling contextual representations learned from large text corpora to be transferred to downstream biomedical and clinical tasks. Domain-specific models such as BioBERT, which adapts BERT through pretraining on biomedical literature, demonstrate how large-scale pretrained representations can be tailored to biomedical text analysis [28]. This development has subsequently expanded towards larger foundation language models and large language models (LLMs), although challenges relating to reliability, comprehensiveness, interpretability, bias, and patient privacy remain important considerations for their clinical application [29].

2.2 Multimodal AI

This section provides an overview of multimodal AI (MMAI), with specific focuses on advanced multimodal fusion strategies and data missingness as a key challenge to develop useful and clinically relevant MMAI. For a more detailed introduction to the fundamental concepts of MMAI, please refer to the reports of **MM-HealthFair**.

2.2.1 Overview of Multimodal AI

MMAI refers to the development of AI models that integrate information from multiple heterogeneous data modalities to support learning and decision-making [5]. Unlike unimodal approaches that rely on a single data source, MMAI aims to exploit complementary information and relationships across modalities to develop more comprehensive representations of the underlying data. In healthcare, different modalities capture distinct yet interconnected aspects of patients' physiological and clinical conditions. By combining these sources, MMAI has demonstrated potential to improve predictive performance and support applications such as disease diagnosis, prognosis and clinical decision support [5].

A typical MMAI framework involves extracting informative representations from individual modalities, integrating these representations through a fusion mechanism, and subsequently using the combined information for downstream tasks. Modality-specific encoders are commonly employed to accommodate differences in data structure and characteristics, while fusion mechanisms aim to extract complementary information and cross-modal relationships [30, 31]. However, the integration of heterogeneous modalities introduces additional challenges, including differences in dimensionality, temporal misalignment, varying data quality and incomplete modality availability. Addressing these challenges is therefore essential for developing robust and effective MMAI systems in healthcare [5].

2.2.2 Multimodal representation learning and fusion

Based on the stage at which information is integrated, multimodal fusion strategies are commonly categorised into early, intermediate and late fusion, as previously introduced in **MM-HealthFair** and illustrated in Fig. 2.1. Early fusion enables joint learning from a combined feature space but can be challenging when integrating highly heterogeneous or poorly aligned modalities, as commonly encountered in healthcare. Intermediate fusion typically employs modality-specific encoders to extract learned representations, which are subsequently integrated within a shared

learning framework to capture cross-modal interactions. Late fusion aggregates predictions from independently processed unimodal models, offering greater flexibility and modularity, although interactions between modalities may be less extensively modelled [31].

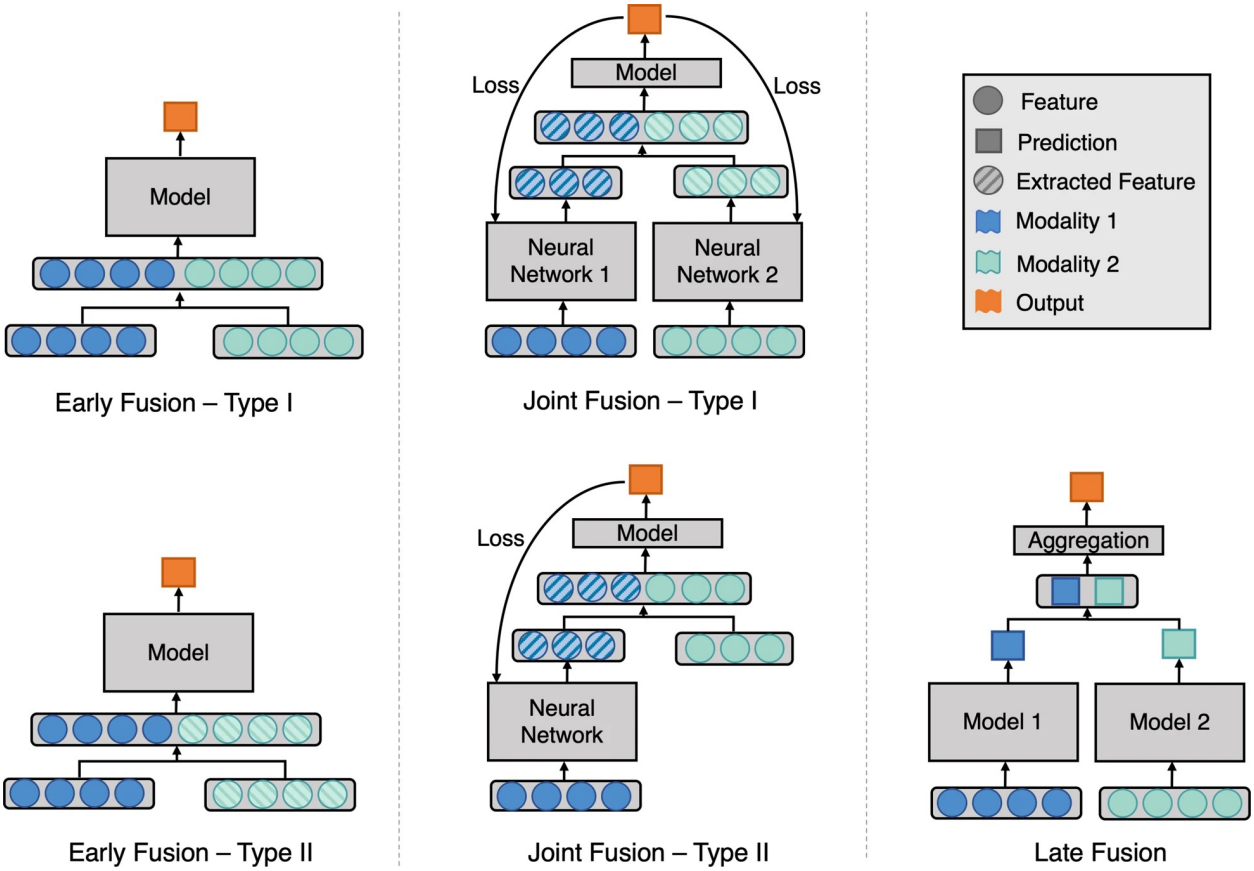


Figure 2.1: Categories of multimodal fusion strategies from Huang et al. [32]. Early fusion concatenates modalities at the input level. Intermediate fusion integrates modality-specific representations within a shared learning framework. Late fusion aggregates predictions from multiple unimodal models to generate a final prediction.

Beyond the stage of fusion, different mechanisms have been developed to integrate modality-specific information. Simple feature or representation concatenation, adopted as a baseline in **MM-HealthFair**, remains widely utilised due to its simplicity and relatively low computational overhead. More sophisticated DL-based approaches include attention-based mechanisms, which dynamically identify and emphasise relevant information within and across modalities. In particular, cross-attention allows representations from one modality to selectively incorporate information from another, facilitating the learning of cross-modal dependencies. For example, Amador et al. proposed a cross-attention-based framework integrating CT perfusion imaging and clinical metadata for stroke outcome prediction [33]. Alternatively, introduced by Yao et al., DrFuse integrates EHR data and chest X-rays through disentangled representation learning and disease-aware attention fusion, enabling patient- and disease-specific weighting of shared and modality-specific information while accommodating missing modalities [34].

Graph-based approaches provide an alternative for multimodal representation learning by explicitly modelling relationships between patients, modalities or clinical features through graph structures.

GNNs enable information exchange between connected nodes through message passing, allowing relational information to be incorporated into learned representations. For example, M3Care proposed by [35] constructs task-guided patient similarity graphs and employs graph convolutional networks (GCNs) to propagate information between similar patients, supporting the estimation of missing modality representations. More recently, Wu et al. introduced MUSE, which uses a patient–modality bipartite graph and mutual-consistent graph contrastive learning to learn patient representations in the presence of missing modalities and labels [36]. These studies demonstrate the flexibility of graph-based approaches in exploiting different forms of relational information to support multimodal learning, particularly when clinical data are incomplete.

Gating mechanisms and mixture-of-experts (MoE) architectures provide another approach to adaptive multimodal fusion. MoE models consist of multiple specialised neural networks, referred to as experts, and a learnable gating or routing mechanism that dynamically selects or weights expert contributions according to the input. Such architectures can accommodate heterogeneous information by allowing different experts to specialise in distinct data characteristics or modality combinations [37]. For example, Han et al. proposed FuseMoE, which incorporates sparsely gated MoE layers and a novel Laplace gating function to integrate varying numbers of modalities, including irregularly sampled and incomplete clinical data [38]. Similarly, Yun et al. introduced Flex-MoE, which combines a missing-modality bank with generalised and specialised routing mechanisms to support arbitrary combinations of observed modalities [39]. These approaches demonstrate the potential of adaptive expert routing for multimodal learning under heterogeneous and incomplete data conditions. Generative representation-learning approaches, such as multimodal variational autoencoders (VAEs), have also been explored to learn joint latent representations and support cross-modal generation or inference from partially observed modalities [40, 41].

The effectiveness of these fusion approaches depends on the characteristics of the data, the clinical task and the model architecture, and no single fusion strategy is universally optimal. For example, **MM-HealthFair** reported comparable predictive performance between simple concatenation and the gated-based Multimodal Adaptation Gate (MAG) mechanism [42], suggesting that increased architectural complexity does not necessarily translate into improved performance. Further details of this comparison are provided in the report by Dr Sophie Martin. Although advanced fusion mechanisms can capture complex cross-modal relationships, their effectiveness may be affected by differences in data quality, temporal alignment and modality availability.

2.2.3 Challenges of incomplete multimodal data

Incomplete data remain a major challenge in healthcare applications, where data availability and consistency vary across patients and care pathways. Missingness can occur at different levels, ranging from individual missing features or measurements within a modality to the complete absence of an entire modality. Importantly, missingness in healthcare is not necessarily random, and systematic patterns of missingness may contain clinically relevant information [13, 5]. Feature-level missingness is particularly common in longitudinal EHR data, where vital signs and laboratory measurements are often recorded irregularly or selectively according to clinical needs. In contrast, modality-level missingness occurs when a particular data source, such as an ECG, medical image or clinical note, is unavailable for a patient. Such missingness may arise from differences in clinical decisions, resource availability and patient care pathways.

Modality-level missingness poses additional challenges for multimodal fusion, as conventional mechanisms, such as concatenation, may assume that all input modalities are present for each patient. Excluding patients with missing modalities can reduce the available training population and potentially introduce selection bias, while simple imputation methods may fail to preserve complex relationships between modalities [43]. Various solutions have therefore been proposed, including modality masking, modality dropout, missingness-aware fusion and the estimation of missing representations from observed modalities [44, 45]. These approaches can be broadly distinguished by how they accommodate missing information. Some methods adapt the fusion process to utilise available modalities without explicitly reconstructing the missing ones, while imputation-based methods exploit cross-modal relationships to estimate missing information or representations [45, 46]. The effectiveness of these approaches depends on the missingness patterns, the informativeness of the observed modalities and the reliability of the estimated representations. Developing robust multimodal models capable of learning from heterogeneous and incomplete clinical information therefore remains an important research challenge.

2.3 XAI

Explainable artificial intelligence (XAI) encompasses techniques that aim to provide human-understandable insights into the predictions and behaviour of AI models. In healthcare, XAI can support model validation, error analysis and the identification of potential biases, contributing to the development of more transparent and trustworthy clinical prediction systems. A wide range of XAI approaches have been developed, including gradient-based attribution, perturbation-based methods, surrogate models, activation visualisation, relevance propagation and attention-based mechanisms [47, 48]. These approaches differ in their underlying mechanisms, the level of model access required and the types of explanations they produce. Their fundamental principles and general taxonomies have previously been discussed in the **MM-HealthFair** reports.

Explainability becomes particularly challenging in MMAI, where heterogeneous data sources are processed through modality-specific components and subsequently integrated for prediction. In this report, explanations are considered at two complementary levels: modality-level explanations, which investigate the influence of individual modalities on predictions, and within-modality explanations, which identify influential features, measurements, text elements or image regions. These explanations may be generated for individual predictions or aggregated to investigate model behaviour across patient populations.

2.3.1 Modality-level explanations

Modality-level explanations aim to assess how individual data modalities contribute to multimodal predictions. Modality ablation is one straightforward method in which selected modalities are removed or masked to examine changes in model predictions or predictive performance. This can help identify whether a model relies heavily on particular data sources and whether additional modalities provide complementary information. However, the observed changes may depend on the masking strategy, interactions between modalities and the model's ability to handle missing inputs.

Shapley-based approaches provide an alternative for quantifying modality contributions by considering their marginal effects across different combinations of available modalities. SHapley Additive exPlanations (SHAP) is a widely adopted attribution framework that assigns contributions to input features relative to a reference prediction, making it flexible to input types and dimensions

[49]. MM-SHAP, proposed by Parcalabescu and Frank [50], extends Shapley-based attribution to multimodal explanations by aggregating feature-level attribution magnitudes within each modality. The resulting normalised scores quantify the relative degree of model dependence on different modalities, enabling the investigation of modality dominance and utilisation. An adapted approach to aggregating SHAP values across modalities was also explored in **MM-HealthFair** for clinical risk prediction.

Attention-based mechanisms offer another perspective on how multimodal information is processed. Modality-level attention or gating weights can indicate how a model dynamically combines different representations, while cross-attention maps can reveal relationships between elements of different modalities, such as image regions and clinical text tokens. Such visualisations can provide insights into learned cross-modal associations and information flow [47, 48]. Nevertheless, attention weights should not automatically be interpreted as faithful measures of feature or modality contribution, as different attention distributions may produce similar predictions [51, 52]. Therefore, attention-based analyses are generally considered complementary to attribution- and perturbation-based explanations.

2.3.2 Within-modality explanations

Within-modality explanations provide more fine-grained insights into the information contributing to model predictions. For structured EHR data, feature-attribution methods such as SHAP can estimate the influence of individual variables, such as demographics, diagnoses and laboratory measurements. Other methods include Local Interpretable Model-agnostic Explanations (LIME) [53], which perturbs input features and fits an interpretable local surrogate model to approximate the original model's behaviour, and gradient-based approaches such as Integrated Gradients [54], which attributes predictions to input features by accumulating gradients along a path from a reference baseline. Alternatively, propagation-based methods, such as DeepLIFT [55] and Layer-wise Relevance Propagation (LRP) [56], assign feature contributions by propagating attribution or relevance scores backwards through neural network layers. These methods can be applied to individual predictions or summarised across patient populations to investigate influential features.

For temporal and textual modalities, explanations can be generated at different levels of granularity. In time-series data, attribution methods may identify influential clinical measurements or time points, although temporal dependencies and correlations complicate their interpretation. For clinical notes, SHAP and related methods can estimate the contributions of words, tokens or text segments, potentially identifying clinically relevant descriptions associated with a prediction [57]. However, token-level explanations may be sensitive to tokenisation and contextual dependencies, and individual token contributions do not necessarily reflect the meaning of complete clinical statements.

For medical imaging, visual explanation methods are commonly employed to identify spatial regions associated with model predictions. Gradient-based approaches, such as saliency maps [58] and Gradient-weighted Class Activation Mapping (Grad-CAM) [59], utilise gradients of model outputs to identify influential image pixels or regions. Grad-CAM generates class-specific localisation maps by weighting convolutional feature maps according to the gradients of a target prediction. Extensions such as Grad-CAM++ [60] refine the gradient-weighting procedure, while alternative approaches, including occlusion sensitivity [61] and RISE [62], employ input perturbations to

assess the influence of different image regions. In chest X-ray analysis, these approaches may help investigate whether a model's predictions are associated with clinically relevant anatomical regions.

Despite their usefulness, within-modality explanations have important limitations. Attribution estimates may depend on the selected baseline, perturbation strategy and correlations between input features, while image-based heatmaps may be spatially coarse or highlight regions unrelated to the underlying pathology. Also, explanations generated from an isolated unimodal model do not necessarily reflect the contribution of that modality within the complete multimodal prediction framework. Therefore, explanation methods should be evaluated for faithfulness, stability and clinical plausibility. Combining modality-level and within-modality explanations may provide a more comprehensive understanding of multimodal model behaviour, though neither level alone guarantees a faithful or clinically meaningful explanation [48].

2.4 Fairness in healthcare

Fairness is a growing concern in clinical-decison making [63]. Widely used clinical tools can exhibit systematic bias and lead to harmful practices. In a study of more than 113,000 hospitalised patients across 27 US hospitals, Ashana et al. found that two widely adopted clinical risk scores (SOFA and LAPS2) systematically overestimated in-hospital mortality for Black patients and underestimated it for White patients [64]. In crisis triage settings, where scarce resources may be prioritised towards patients judged more likely to survive, such bias could disadvantage Black patients. Bias grounded in clinical data can easily propagate into ML models. For example, Seyyed-Kalantari et al. reported consistent under-diagnosis of female and younger patients in state-of-the-art classifiers trained on four chest X-ray datasets [65]. These disparities may remain hidden under conventional model evaluation methods. As demonstrated by Röösli et al., aggregate performance can mask poor detection of high-risk patients in minority groups [66]. To this date, there is no consensus on how fairness should be addressed in healthcare ML [67].

Bias can enter clinical ML through several properties of healthcare data. First, demographic groups are often unevenly represented in training cohorts. This can lead to poorer performance for under-represented populations [68]. Second, bias can arise from the choice of outcome label. Obermeyer et al. showed that labels based on healthcare utilisation may reflect unequal access to care, rather than underlying clinical need [69]. Some recorded outcomes may capture prior decisions and existing inequalities as well as illness. Third, missingness adds another layer to the problem. In EHR data, whether and when a diagnostic test is performed can itself be informative. For example, Agniel et al. confirmed that the timing of 68% of laboratory tests was more predictive of 3-year survival than the test values themselves [70]. This may not itself demonstrate unfairness. However, when testing practices or access to investigations differ across patient groups, models may learn patterns of care that reproduce existing inequalities. This concern is particularly relevant to multi-modal prediction, where modality availability reflects both the patient's condition and their clinical pathway.

Given these sources of bias, fairness must be evaluated in ways that reflect how a model behaves across patient groups. Individual fairness [71] and causal fairness [72] provide useful frameworks, but they depend on similarity measures or causal assumptions that are often difficult to justify in multi-modal clinical data. Group-level fairness measures are more commonly used. Those measures generally assess three criteria:

- Whether the predicted risks are similarly calibrated across groups (example: calibration within groups [64])?
- Whether positive predictions are assigned at equitable rates across groups (example: demographic parity [71])?
- Whether true and false positive rates are equal across groups (example: equalised odds ratio, equal opportunity [73])?

However, no single fairness criterion is universally appropriate. In real healthcare settings, prevalence of outcomes are not uniform between patient groups. As a result, we cannot realistically expect both equal calibration and equal error rates simultaneously in a model. The choice of fairness measures must be justified for the clinical task. In addition, fairness estimates can be unstable when some subgroups are particularly small. That means uncertainty quantification should be provided alongside point estimates.

Fortunately, these group-level fairness evaluations are well supported by open-source toolkits, including `AIF360` [74], `Fairlearn` [75], `Aequitas` [76] and `OxonFair` [77]. Among them, `Fairlearn` is particularly flexible and supports user-defined metrics and subgroup analysis across multi-level patient attributes.

Chapter 3

Data

3.1 Data sources

This study utilised de-identified secondary care data from the Medical Information Mart for Intensive Care (MIMIC) database family, retrospectively collected from patients receiving care at Beth Israel Deaconess Medical Center (BIDMC) in Boston, Massachusetts, USA [78]. The databases contain routinely collected electronic health records (EHRs) and associated clinical data, enabling the investigation of patient trajectories across emergency department (ED) attendances, inpatient hospitalisations and intensive care unit (ICU) stays.

The primary data sources were **MIMIC-IV** [79] and **MIMIC-IV-ED** [80]. **MIMIC-IV** contains detailed hospital and ICU records, including patient demographics, admissions, diagnoses, laboratory investigations, medications and care provider orders, while **MIMIC-IV-ED** provides information on ED encounters, including triage assessments, vital signs and patient disposition. Together, these datasets provided structured patient information and longitudinal measurements, including ED vital signs and hospital laboratory results.

The **MIMIC-IV-Ext-BHC** dataset [81] provides extracted and standardised Brief Hospital Course (BHC) sections from discharge summaries, describing patients' clinical progression, investigations and treatments during hospitalisation. The source discharge summaries were obtained from **MIMIC-IV-Note** [82].

Imaging and physiological signals were obtained from the following datasets:

- **MIMIC-CXR-JPG** [83]: Chest X-ray (CXR) images in JPEG format, accompanied by associated study metadata.

- **MIMIC-IV-ECG** [84]: Standard 12-lead ECG waveforms from the diagnostic ECG matched subset, enabling linkage with patient records in MIMIC-IV.

Additionally, **MIMIC-IV-ECG-Ext-ICD** [85] provided ECG-to-diagnostic-code linkages and predefined dataset partitions. The pathway-specific splitting procedures are described in Section 3.5 below.

3.2 The two care pathways

This study investigated two clinical prediction pathways: emergency department (ED) deterioration (**deterioration**) and inpatient hospitalisation (**hospitalisation**). Both pathways utilised linked multimodal MIMIC datasets but differed in their clinical objectives, observation windows, prediction outcomes and included data modalities. The deterioration pathway focused on predicting short-term adverse events following ED arrival, while the hospitalisation pathway investigated outcomes associated with subsequent inpatient care using information from the current ED encounter and previous hospital admissions. The key characteristics of the two pathways are summarised and compared in Table 3.1.

Table 3.1: Comparison of the two clinical prediction pathways.

Characteristic	Deterioration	Hospitalisation
Clinical setting	ED admission	ED-to-inpatient admission
Unit of analysis	ED stay	Patient (one current admission)
Prediction point	90 minutes after ED arrival	Transfer from ED to inpatient ward
Observation window	First 90 minutes of ED attendance	Current ED phase and 90-day historical lookback
Prediction outcomes	Mechanical ventilation, cardiac arrest, one-day mortality, ICU admission within 24 hours, ICU admission during the hospital stay	Extended stay $>$ 7 days; in-hospital death; non-home discharge
Prediction horizon	Subsequent 24 hours for time-limited outcomes; ICU admission assessed over the associated stay	Current inpatient hospitalisation
Input modalities	Static EHR, time-series[a], ECG, CXR	Static EHR, previous/current time series, current ECG, previous/current CXR, previous discharge notes
Previous admissions	Not used as a separate observation window	Previous ED-linked admission discharged within 90 days, if available
Modality availability	Both static EHR and ECG required; CXR may be absent	Only static EHR required; other modalities may be absent

[a] Time-series treated as a separate fourth modality only in the temporal models.

The deterioration pathway focused on predicting acute clinical deterioration following ED admission. It was based on the **MIMIC-IV-Ext-MDS-ED (MDS-ED)** benchmark [4], which provides a framework for predicting adverse clinical events using routinely collected ED information. The unit of analysis was an individual ED stay, with clinical observations restricted to the first 90 minutes following arrival. Five binary prediction tasks were investigated: mechanical ventilation, in-hospital cardiac arrest, one-day mortality, ICU admission within 24 hours and ICU admission at any point during the hospital stay. The prediction window began at the end of the initial 90-minute observation period. These outcomes represent different aspects of acute deterioration, escalation of care and short-term mortality. Four data modalities were incorporated: static EHR features, time-series measurements[1], 12-lead ECG recordings and chest X-ray (CXR) images. CXR availability varied across ED stays, providing an opportunity to investigate multimodal prediction under incomplete data conditions.

The hospitalisation pathway focused on predicting outcomes associated with inpatient hospitalisation, extending the previous **MM-HealthFair** framework [3]. The unit of analysis was the patient, with one current hospital admission retained per patient. The prediction setting was defined at the transition from ED to inpatient care, using clinical information available before transfer to the hospi-

[1] Except in the temporal models, time-series measurements were summarised and combined with the tabular features.

tal ward. Two observation windows were considered to capture both current and previous-admission patient information and are described in Section 3.3 below.

Three binary prediction tasks were investigated: extended hospital stay > 7 days (EXT-ST), in-hospital death (IHD) and non-home discharge (NHD). These outcomes capture different aspects of inpatient care, including prolonged hospital resource utilisation, mortality and discharge destination. The pathway incorporated structured static EHR features, time-series measurements, ECG recordings, CXR images and BHC sections from discharge summaries. Time-series measurements and CXR images could be obtained from both observation windows, while ECG recordings were restricted to the current window. BHC notes were obtained from previous admissions within the previous window. This resulted in seven model input modalities: static EHR, previous and current time series, previous and current CXR, current ECG, and previous clinical notes.

Unlike the previous **MM-HealthFair** cohort, which required the availability of selected clinical measurements and historical notes, only the static EHR modality was mandatory for inclusion in the hospitalisation cohort. Consequently, the pathway accommodated patients with different combinations of observed modalities, providing a setting for evaluating multimodal models under heterogeneous and incomplete clinical information.

3.3 Cohort curation

As shown in Figure 3.1, separate cohorts were constructed for the two prediction pathways by adapting the cohort definitions of **MDS-ED** and **MM-HealthFair**. Data curation involved selecting eligible ED or hospital admissions, linking records to the relevant observation windows and deriving outcome labels.

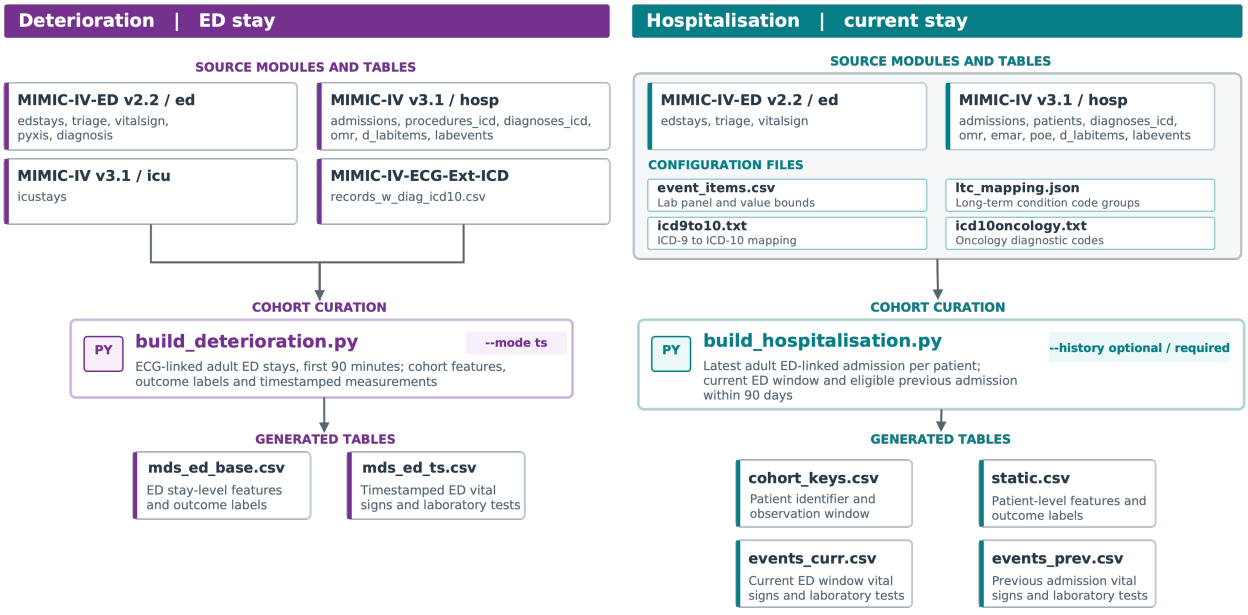


Figure 3.1: Overview of the data curation and extraction workflows for curating **deterioration** and **hospitalisation** pathways, levaraging **MIMIC-IV**, **MIMIC-IV-ED** and **MIMIC-IV-ECG-Ext-ICD**.

Deterioration cohort

The deterioration cohort was based on **MDS-ED** [4] and constructed using `build_deterioration.py`. We used hospital records from **MIMIC-IV** v3.1 [79], together with **MIMIC-IV-ED** v2.2 [80] and ECG records from **MIMIC-IV-ECG-Ext-ICD** [85]. The adult-only population was inherited from MIMIC-IV, with no additional age restrictions applied. The ED stay was the unit of analysis. Candidate stays were required to have at least one ECG recording linked within 90 minutes of ED arrival, and records without linked diagnostic codes were excluded. Static EHR information was also required, whereas CXR images could be absent. No minimum number of vital-sign or laboratory measurements was required.

The five binary outcome labels described in Section 3.2 were reconstructed by linking the eligible ED stays with subsequent hospital, ICU and outcome records. Deaths within the first 90 minutes and cases where all recorded ICU contacts occurred within this period were assigned missing labels for the respective outcomes, rather than being treated as negative cases.

We made several changes to the original **MDS-ED** extraction. First, as new data were released since the original work, we migrated from **MIMIC-IV** v2.2 to **MIMIC-IV** v3.1. Second, ICU outcomes were evaluated against all ICU stays associated with the linked hospital admission, rather than relying on a single ICU record. This allowed later qualifying ICU admissions to be captured consistently and the resultant labels to be reproducible.

Hospitalisation cohort

The hospitalisation cohort was adapted from **MM-HealthFair** [3] and constructed using `build_hospitalisation.py`. Here, the unit of analysis was the patient. Candidate hospital admissions were restricted to adults aged at least 18 years, with valid chronological ED registration, ED departure, admission and discharge timestamps and an associated ED stay. Recorded age, sex, marital status and insurance type were required.

One current admission was retained per patient, defined as the most recent hospital admission linked to an ED stay. Its **current window** started from the ED arrival to the recorded admission to the hospital ward. The next most recent eligible ED-linked hospital admission was considered for the **previous window**, provided that the patient had been discharged within a 90-day lookback period before the current ED arrival. This window covered information taken between ED arrival and hospital discharge. Patients without a qualifying previous admission remained eligible, with no previous-window records.

Outcome labels were derived from the current admission: extended stay was defined from admission-to-discharge duration, in-hospital death was identified from the recorded death timestamp, and non-home discharge was derived from discharge destination among patients not labelled as having died in hospital.

We made changes relative to **MM-HealthFair**. Most importantly, only static EHR was required for inclusion. Other modalities were allowed to be missing. The earlier **MM-HealthFair** cohort required at least two recorded measurements in vital signs and lab tests, as well as discharge notes. Second, a previous admission is no longer mandatory. However, when it was available, we now restricted the previous window to admissions discharged within 90 days before the current ED stay. Third, the extraction was extended to include recent admission history, comprising the previous length of stay, discharge location and days since discharge. The total number of prior admissions was included separately as a measure of previous hospital use. Fourth, triage acuity, arrival transport,

and ED admission location were also added to characterise the patient's condition. Finally, ECG and CXR modalities were incorporated as additional modalities.

3.4 Per modality preparation

To improve reusability, our extraction pipeline of CXR, ECG and notes modalities was designed to be cohort-independent. We ran the frozen pre-trained foundation models once for all records available and stored the master modality-level embeddings for future use. For each of the three modalities, one key file was stored alongside the embeddings to streamline subsequent processing. Once the deterioration and hospitalisation cohorts were defined, we retrieved the relevant records from the master files and assembled them into the pathway-specific data bundles.

3.4.1 Tabular data pipeline

Tabular data underwent standard feature selection, cleaning and engineering before they were added to the pathway-specific bundles. Several variables were shared across the two pathways, including sex, ethnicity, biometrics and triage acuity. Ethnicity was mapped to five categories: White, Black, Hispanic/Latino, Asian and Other. Triage acuity represented the urgency score assigned by the triage nurse, ranging from 1 (most urgent) and 5 (least urgent). Height, weight and BMI were derived from outpatient records following the original pipeline. Insurance type and marital status were included only in the hospitalisation pathway.

The hospitalisation pathway contained additional features that described medical use and care activities. Medication features were derived from electronic medication administration records, with a status of *Administered*, *Started* or *Confirmed*. The 50 most common drugs within the cohort were retained. Care provider orders were also counted across 7 specialities: cardiology, respiratory, radiology, neurology, nutrition, total parenteral nutrition and hemodialysis. Clinical diagnoses were used to derive features relevant to long-term conditions. ICD-9 diagnoses were converted to ICD-10, after which hospital diagnoses were matched to 25 long-term conditions (22 physical and 3 mental health). From these mappings, we derived the presence of individual conditions and a multimorbidity count.

3.4.2 Time-series data pipeline

The time-series pipeline included routinely collected vital signs and laboratory measurements. Six vital sign features were used across both pathways: temperature, heart rate, respiratory rate, oxygen saturation (SpO2), and systolic and diastolic blood pressure. The laboratory panel was inherited from the source pipelines of either cohort. This resulted in 45 tests for the deterioration cohort and 50 tests from the hospitalisation cohort. Implausible values were removed using pre-defined thresholds for each test. This replaced **MM-HealthFair**'s approach of excluding values beyond two standard deviations from the population mean.

The processed time-series data were stored in two formats. For models using fixed-length inputs, each feature was summarised and flattened using the mean, median, minimum, maximum, standard deviation, first and last recorded values, rate of change, and the slope of a linear regression fit against time since arrival. For temporal modelling, we retained the raw measurements as sequences and preserved the original measurement times.

3.4.3 Chest X-ray pipeline

Chest X-rays were processed using a pre-trained **DenseNet-121** model from **TorchXRayVision** [86]. We followed the same feature extraction approach in **FuseMoE** [38]. Each image was normalised and resized to 224x224, then passed through the frozen network. The resulting feature maps were globally average pooled to produce a 1024-dimensional embedding vector.

3.4.4 ECG pipeline

12-lead ECG recordings were processed using **ECG-FM** [87]. We used its pre-trained checkpoint, which was trained on a large ECG corpus that included **MIMIC-IV-ECG**. As the pre-training did not use the downstream outcome labels defined in our study, the risk of label contamination was low. The model was loaded through the `fairseq-signals` framework and used in inference mode only. We followed the same preprocessing approach as **ECG-FM**. The leads were reordered to match the model's 12-lead configuration, resampled to 500 Hz and z-score normalised. Each recording was then divided into non-overlapping 5-second segments. **ECG-FM** produced one embedding for each segment. These segment-level representations were averaged to give one 768-dimensional vector for the full ECG recording.

3.4.5 Notes pipeline

We used the BHC segments provided in **MIMIC-IV-Ext-BHC** as the clinical text input. Admission identifiers and note timestamps were obtained from the **MIMIC-IV-Note** discharge table. Each BHC segment was processed using the pre-trained `Bio_Discharge_Summary_BERT` model [88]. Before encoding, de-identification placeholders ("___") were removed and repeated whitespaces were collapsed. The cleaned text was then split into sentences using the scispaCy `en_core_sci_md` pipeline [89]. Each sentence was truncated to a maximum of 128 tokens before encoding. An attention-weighted mean over the final hidden layer was used to obtain a sentence-level representation. These representations were then averaged across the note to produce a single 768-dimensional embedding vector. As there was one discharge note per admission, this gave one text embedding for each hospital admission.

3.5 Data preprocessing

This section describes how we assembled the constructed cohort and per-modality embeddings into pathway-specific multi-modal bundles (Figure 3.2). For this, `prepare_deterioration.py` and `prepare_hospitalisation.py` linked each cohort row to the corresponding modality records and embeddings. For deterioration, the unit of analysis remained the ED stay. For hospitalisation, modalities were aligned at the patient level across the current and previous windows. Time-series data were represented either as summary statistics as described in Section 3.4.2, or as timestamped raw sequences for temporal modelling.

The resultant bundles were then divided into training, validation and test sets using `split_deterioration.py` and `split_hospitalisation.py`. Splitting was performed at the patient level, so no patient appeared in more than one partition. For deterioration, we inherited the stratified folds of **MDS-ED**[2] [4]. Patients could contribute multiple ED stays to the training set.

[2]The stratified folds were provided in **MIMIC-IV-ECG-Ext-ICD** [85]

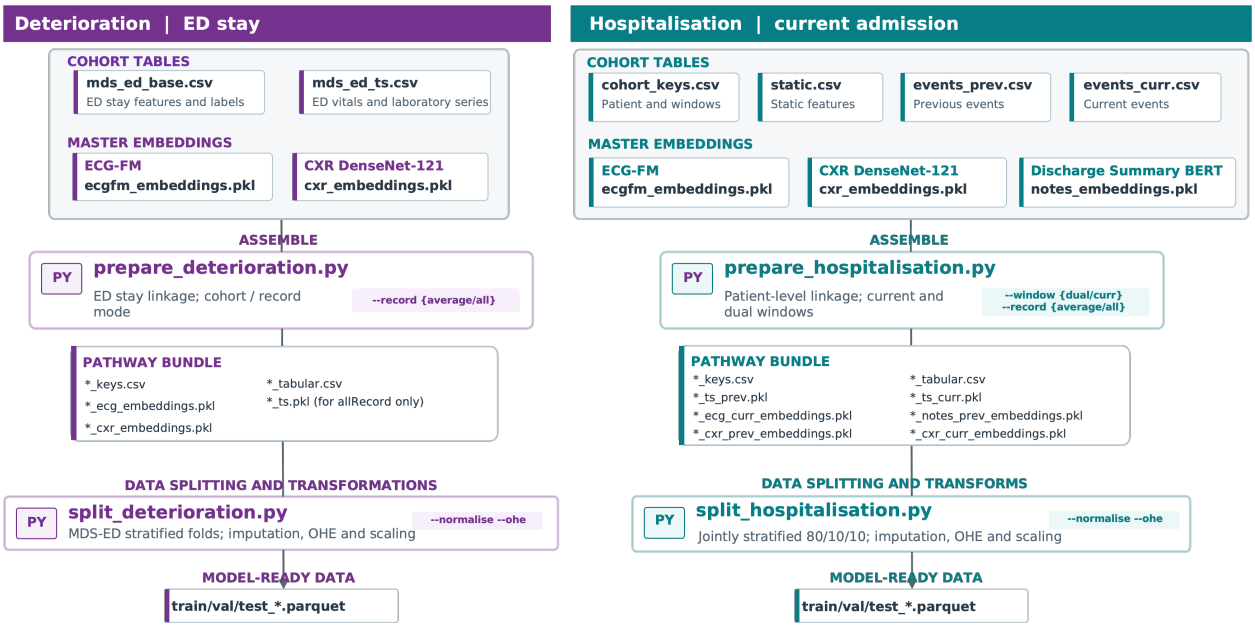


Figure 3.2: Overview of the data preprocessing workflows for **deterioration** and **hospitalisation** pathways. Cohort tables and precomputed modality embeddings were linked to assemble pathway-specific multi-modal datasets. The preparation scripts supported different record aggregation modes (`-record {average/all}`), while hospitalisation additionally supported current-only or dual observation windows (`-window {dual/curr}`). The assembled datasets underwent patient-level splitting and preprocessing, with optional normalisation (`-normalise`) and one-hot encoding (`-ohe`) to produce model-ready training, validation and test in `.parquet` format.

However, only the earliest eligible stay from each patient was kept in the validation and test sets. For hospitalisation, the original **MM-HealthFair** split could not be reused because the cohort definition had drifted. As such, a new 80/10/10 split was generated, jointly stratified on the outcome labels, sex and ethnicity group. A single patient-level partition was shared across the three hospitalisation outcomes, and reused across different window configurations within the same cohort.

The static EHR data underwent basic feature preparation before model training. Missing numeric values were replaced with the corresponding median calculated from the training set. Continuous features were then min-max normalised using the training-set minimum and maximum. Non-ordinal multi-level categorical features and two ordinal categorical features were one-hot encoded (triage acuity and count of care orders). Variables already represented as binary indicators were retained. All preprocessing parameters were fitted on the training split first and then applied to validation and test sets. Unlike **MM-HealthFair**, we did not remove highly correlated features using a Pearson correlation threshold. The full selected feature set was kept.

The time-series data required a different handling strategy. We did not apply the same filtering (≥ 2 measurements), resampling or forward filling as in **MM-HealthFair**. Instead, they followed a similar train-only preprocessing strategy like the static EHR data. For temporal modelling, timestamps were converted to relative time since the start of the corresponding observation window.

Missingness was retained explicitly throughout preprocessing. For tabular modality, a feature-level mask was created before median imputation, such that the original missingness remained identifiable. Time-series sequences used a mask of the same shape as the sequence, indicating which features were absent at each recorded timestamp. For ECG, CXR and notes, missingness was represented at the modality level.

As shown in Figure 3.2, we prepared two record representations for downstream modelling. In the `averageRecord` configuration, multiple ECG or CXR embeddings within the same observation window were mean pooled into a single vector, while time-series measurements were represented using the fixed-length summary statistics described in Section 3.4.2. The `allRecord` configuration retained the individual records instead. All time-series, ECG, CXR and note embeddings were stored as variable-length sequences with relative timestamps, which preserved their original temporal order. We prepared two window configurations `dualWindow` and `currWindow` for hospitalisation pathway.

Finally, all modality features, missingness masks and timestamps were aligned to a common identifier, which was an ED stay for deterioration and a patient for hospitalisation. The resulting training, validation and test sets were written as separate `.parquet` files and used directly by the multi-modal fusion and evaluation pipeline.

Chapter 4

Methods

4.1 Shared pipeline and experiment interface

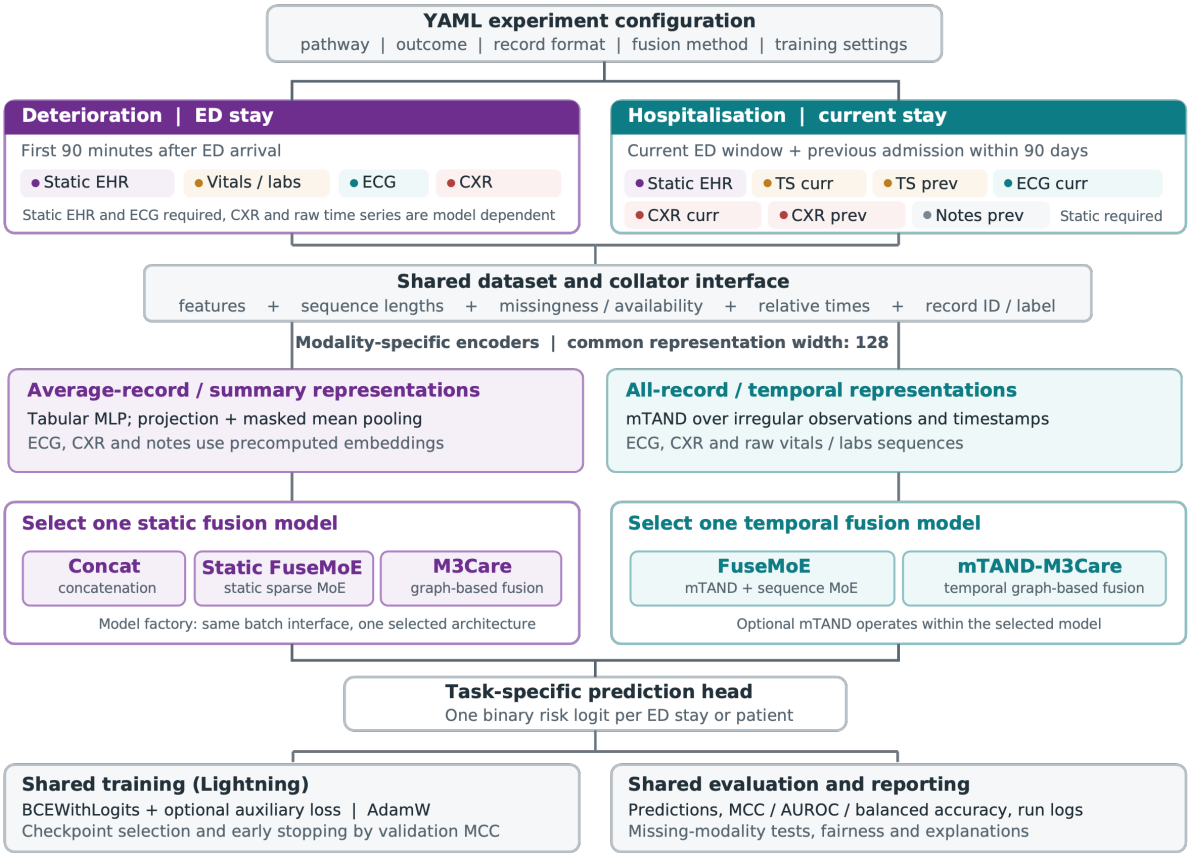


Figure 4.1: **Overview of the shared multi-modal pipeline and experiment interface.** Pathway-specific patient records are processed through a common data interface and modality-specific encoders to generate pooled or temporal representations. The selected fusion model produces predictions, while a shared training and evaluation framework enables consistent comparison across architectures, clinical tasks and modality availability patterns.

To compare fusion methods under consistent experiment conditions, we implemented a shared training and evaluation pipeline, as shown in Figure 4.1. Experiments were controlled throughYAML configuration files, in which a single fusion architecture was selected at a time. The surrounding framework remained unchanged whenever possible. This included the training loop, primary prediction loss, optimiser, evaluation procedure and experiment logging. This kept the comparison focused on the fusion architecture. This modular design built on earlier iterations of **MM-HealthFair**, where modality-specific components were developed separately but evaluated through a common multi-modal framework. The experimental setup and evaluation procedures are described in Section 4.3 and Section 4.5 respectively. Fairness evaluation and model explanations are described in Section 4.6 and Section 4.7. The software and other shared implementation details are described separately in Section 4.8.

Each modality entered the model through a common interface. Depending on the data source, the inputs consisted of the prepared embeddings, missingness information, and relative timestamps where temporal records were available.

The missingness information stored during preprocessing was converted into a modality availability signal for each sample. For architectures with explicit missing-modality handling, this signal was passed to the relevant fusion components such that unavailable modalities could be distinguished from the observed ones.

Before fusion, each modality was projected into a shared 128-dimensional representation. We deliberately kept these modality-specific encoders lightweight. The aim was not to build a separate high-capacity network for every data source, but rather to provide a consistent representation for comparing fusion mechanisms. Tabular data and summary representations of the vital signs and laboratory tests were passed through small feed-forward encoders with normalisation and dropout. The pre-computed embeddings of CXR images, ECG and note modalities were projected directly into the same 128-dimensional space.

For the time-aware variants, we retained the individual records rather than relying on pooled or summary representations. CXR and ECG were encoded using multi-time attention encoders (**mTAND** [90]), while the raw vitals or labs sequences were passed through the same temporal mechanism in place of summary statistics. Each record kept its relative timestamps. **mTAND** encoded these irregular sequences by attending from a fixed set of reference times to the observed timestamps. This produced a fixed-dimensional representation that could be passed to the fusion model.

4.2 Multi-modal fusion

4.2.1 FuseMoE

MoE models provide a natural way to handle patients with different combinations of available data. Rather than requiring every modality to be present, a routing component selects a small subset of expert networks for each patient and combines their outputs. We adapted this idea from FuseMoE [38] and implemented two related variants, as shown in Figure 4.1 above. The first was a **static sparse MoE** that fused the pooled patient representations. The second was a **temporal FuseMoE** that retained the individual temporal records and embedded the MoE operation within a sequence encoder. Unless otherwise stated, we will refer to both variants as FuseMoE in the remainder of this

report. Both variants used the same underlying routing mechanism, allowing us to investigate the effect of the fusion structure separately from the expert design.

Shared routing mechanism

Many components were shared across all variants of our MoE implementations. First, each router received an encoded patient representation and assigned it across a set of independently parameterised experts. Second, each expert consisted of a two-layer multi-layer perceptrons (MLP) with a non-linear activation and configurable dropout between the two linear layers. Each expert therefore learned its own transformation of the input rather than sharing parameters with the other experts.

We implemented routing based on the **cosine similarity** between the router input and a learnt vector associated with each expert. For patient n, the clean routing score for expert i was expressed as

$$g_i^{(n)} = \tau \frac{(x^{(n)})^\top v_i}{\|x^{(n)}\|_2 \|v_i\|_2} \tag{4.1}$$

where $x^{(n)}$ is the input representation for patient n, v_i is the learnt vector associated with expert i, and τ is a learnt positive scaling parameter. This design could reduce the dependence of the routing decision on the magnitude of the input embedding, while the learned scale controls how strongly the router separates the experts.

During training, Gaussian noise was added to these scores. The magnitude of the noise $\sigma_i^{(n)}$ was also learned from the router input:

$$\tilde{g}_i^{(n)} = g_i^{(n)} + \epsilon_i^{(n)}, \qquad \epsilon_i^{(n)} \sim \mathcal{N}(0, {\sigma_i^{(n)}}^2) \tag{4.2}$$

where $\tilde{g}_i^{(n)}$ is the noisy routing score and $\sigma_i^{(n)}$ is the learnt noise scale for expert i and patient n. This introduced some variation in expert selection during training. During evaluation, routing was deterministic and used the clean scores $g_i^{(n)}$.

The sparsity of the network came from **top-k expert selection**. For each patient n, let $\mathcal{K}^{(n)}$ denote the set of indices of the k experts with the highest routing scores. A softmax was applied only across these selected experts. The routing weight $G_i^{(n)}$ assigned to expert i was therefore:

$$G_i^{(n)} = \begin{cases} \dfrac{\exp \tilde{g}_i^{(n)}}{\sum_{j \in \mathcal{K}^{(n)}} \exp \tilde{g}_j^{(n)}}, & i \in \mathcal{K}^{(n)} \\ 0, & \text{otherwise} \end{cases} \tag{4.3}$$

The routing weights summed to one for each patient.

In addition to the routed experts, we introduced an **always-on shared expert**. This followed the shared-expert principle used in recent MoE architectures [91], where part of the model is available to every sample rather than being selected by the router. The shared expert had the same general MLP structure as the routed experts, but **it was evaluated for every patient.** Its contribution was controlled by a fixed mixture weight and was excluded from both top-k selection and load-balancing (will be described below).

Router layouts

We compared four router layouts: **joint**, **combination**, **permod**, and **disjoint**. These layouts used the same expert architecture and routing mechanism but differed in how the modality representations were presented to the router and how experts were shared.

With **joint routing**, all available modality representations were concatenated into one patient-level representation. A single router then selected experts from a shared expert pool. The experts were not explicitly assigned to individual modalities, allowing their roles to emerge during training.

The **combination router** instead introduced explicit supervision based on modality availability. The design was motivated by the combination-specific routing used in Flex-MoE [39]. With M modalities, there are $E = 2^M - 1$ possible non-empty modality combinations. For the three modalities used in the deterioration pathway, tabular data, ECG and CXR, this gave 7 combinations and 7 corresponding experts. For each patient, the observed modality mask identified the expert associated with that combination. The router was encouraged to assign a high score to this expert using an additional cross-entropy loss:

$$\mathcal{L}_{\text{comb}} = -\lambda_r \frac{1}{N} \sum_{n=1}^{N} \log \frac{\exp g_{t(n)}^{(n)}}{\sum_{j=1}^{E} \exp g_j^{(n)}} \tag{4.4}$$

where N is the number of patients in the training batch, $t(n)$ is the expert corresponding to the modality combination observed for patient n, and λ_r controls the strength of the combination-routing supervision. We included patients with all three modalities available in the same way.

Importantly, this supervision did not force the model to use only the designated combination expert. All experts remained available to the top-k router. The additional loss instead provided a soft training signal, encouraging the routing structure to reflect modality availability while preserving the ability to select another expert when useful.

The per-modality (**permod**) layout used a separate router for each modality. These routers made independent routing decisions, but all selected from the same underlying expert pool. An expert could therefore be used by several modalities, while the routing pattern could still differ between tabular data, ECG and CXR.

The **disjoint** layout separated the modality pathways further. Each modality had both its own router and its own private expert pool. Experts were therefore never shared between modalities, and top-k selection took place independently within each modality-specific pool.

Together, these layouts represented different assumptions about how expert specialisation should emerge. Joint routing placed the fewest restrictions on the experts. Permod routing separated the routing decisions while retaining expert sharing, whereas disjoint routing separated both. Combination routing provided a different form of structure by encouraging experts to specialise according to which modalities were available. The organisation of the shared experts followed the router structure. Joint and combination routing used a single shared expert, whereas permod and disjoint routing used a separate shared expert for each modality pathway.

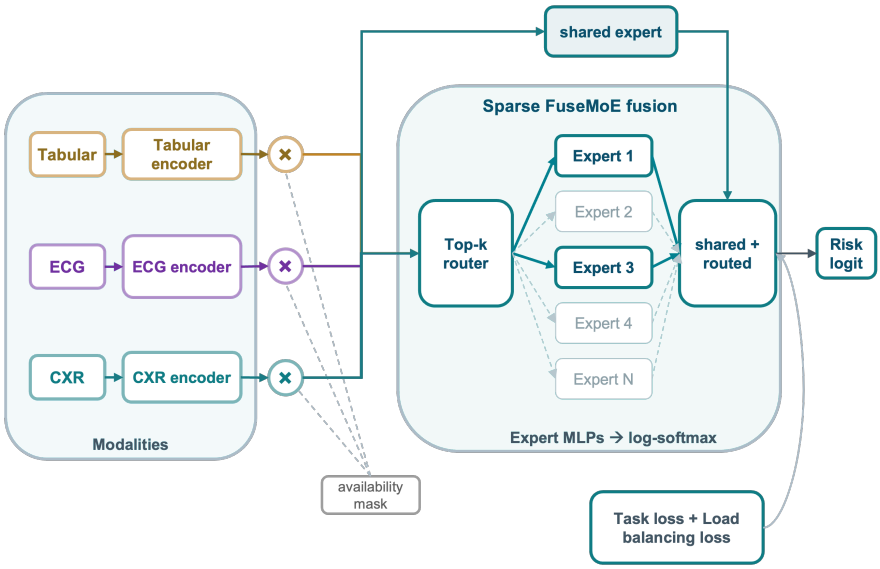


Figure 4.2: Illustration of static FuseMoE

Static FuseMoE

As shown in Figure 4.2, the static FuseMoE operated on the patient-level representations from Section 4.1. These included the encoded tabular (including vitals and labs) representation and the pooled ECG and CXR embeddings. In the standard implementation, unavailable modalities were masked before routing. In the models, each expert produced a two-class log-probability output. The outputs of the selected experts were converted to probabilities and combined according to the routing weights. The always-on shared expert was then mixed with this routed prediction. For a single routing pathway, the predicted probability for patient n was:

$$p^{(n)} = (1 - w) \sum_{i=1}^{E} G_i^{(n)} q_i^{(n)} + w q_s^{(n)} \tag{4.5}$$

where $q_i^{(n)}$ is the positive-class probability produced by routed expert i, $q_s^{(n)}$ is the positive-class probability from the shared expert, E is the number of routed experts, and w is the fixed mixture weight assigned to the shared expert.

Temporal FuseMoE

The static FuseMoE discarded the ordering of repeated measurements by operating on pooled patient-level representations. The temporal FuseMoE [38] retained the individual records instead. ECG, CXR and raw clinical time-series measurements were represented as sequences and first passed through the mTAND components described in Section 4.1. mTAND mapped the irregularly sampled records onto a fixed set of R reference positions.

The resulting modality sequences were then passed through a stack of temporal encoder layers. Each encoder layer contained three sequential sublayers. 1) Within-modality self-attention modelled relationships across the reference time points separately for ECG, CXR and raw time-series data. No information was exchanged between modalities at this stage. 2) The representations entered

the cross-modal sparse MoE sublayer, where fusion took place. Temporal FuseMoE used the same joint, permod and disjoint routing design as the static FuseMoE. Combination routing was not supported in the temporal variant. Each modality sequence was normalised and flattened into a patient-level vector. The router then selected the top-k experts once per patient for that encoder layer, rather than making a separate routing decision at every time point. The selected experts jointly processed the temporal modality representations. Their combined output was then split back into modality-specific sequences and added to the corresponding representations through residual connections. 3) Each updated modality sequence passed through its own feed-forward network before being passed to the next encoder layer. The first and third sublayers therefore operated within individual modalities, while the middle MoE sublayer provided the cross-modal interaction. This three-sublayer structure was repeated across the temporal encoder stack.

Unlike the static FuseMoE, the expert outputs in temporal FuseMoE did not directly form the class prediction. After the final encoder layer, the last reference-position representation from each temporal modality was concatenated with the static tabular representation and the temporal modality-availability indicators. A separate prediction head then produced the final binary output.

Auxiliary loss

Sparse routing can become concentrated on a small number of experts during training. If this happens, some experts receive very few samples and have little opportunity to specialise. We therefore used an auxiliary load-balancing loss to encourage the router to distribute patients more evenly across the expert pool. For expert i, let its "importance" I_i be defined as the sum of its routing weights $I_i = \sum_{n=1}^{N} G_i^{(n)}$, where N denotes the number of patients over which the loss was calculated. Let its "load" D_i represents how often each expert was selected. Imbalance was measured using the squared coefficient of variation CV^2:

$$\mathrm{CV}^2(a) = \frac{\mathrm{Var}(a)}{\bar{a}^{\,2} + \epsilon_{\mathrm{bal}}} \tag{4.6}$$

where a denotes the set of expert importance or load values, $\bar{a}$ is their mean, and ϵ_{bal} is a small constant for numerical stability. The balancing loss $\mathcal{L}_{\mathrm{load}}$ can be expressed as:

$$\mathcal{L}_{\mathrm{load}} = \lambda[\mathrm{CV}^2(I) + \mathrm{CV}^2(D)] \tag{4.7}$$

where λ denotes the contribution of the load-balancing loss. For the joint, per-modality and disjoint models, the overall loss function was therefore $\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{load}}$. $\mathcal{L}_{\mathrm{task}}$ denotes the primary classification loss. For the combination router, the modality-combination supervision was added as a further term $\mathcal{L} = \mathcal{L}_{\mathrm{task}} + \mathcal{L}_{\mathrm{load}} + \mathcal{L}_{\mathrm{combination-supervision}}$.

4.2.2 M3Care

While FuseMoE accommodates missing modalities through adaptive expert routing, M3Care employs graph-based representation imputation. Proposed by Zhang et al. [35], M3Care exploits task-guided patient similarity graphs to estimate unavailable modality representations by propagating information between clinically similar patients through graph convolutional networks (GCNs). Unlike conventional imputation methods that reconstruct missing raw measurements, M3Care operates within a learned embedding space, enabling the estimated representations to be incorporated into downstream multimodal prediction.

We adapted M3Care to the shared modelling framework described in Section 4.1, supporting different numbers and combinations of modalities across the two clinical pathways. The implementation comprised three main components: patient similarity graph construction, GCN-based representation refinement, and attention-based multimodal fusion.

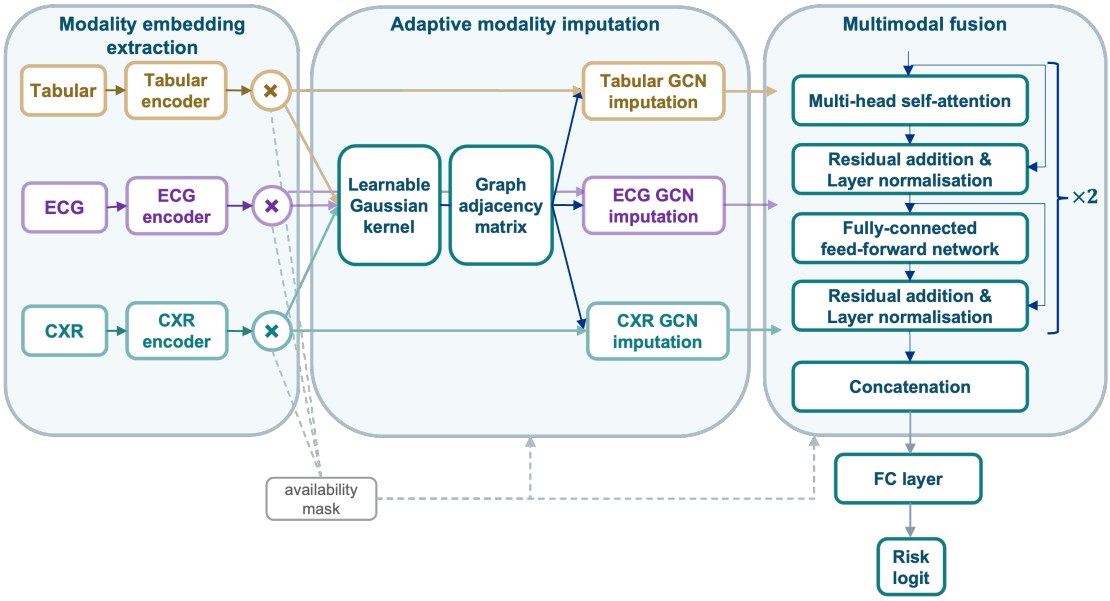


Figure 4.3: Illustration of M3Care

Patient similarity graph construction

After unimodal feature extraction, a batch-wise patient similarity graph was constructed to facilitate missing modality imputation. For a batch of B patients and M modalities, let $\mathbf{x}_i^{(m)} \in \mathbb{R}^d$ denote the encoded representation of modality m for patient i, and let $a_i^{(m)} \in \{0, 1\}$ indicate whether that modality was observed. Each modality was first processed through a separate multilayer perceptron (MLP) projection network to produce a transformed representation $\mathbf{p}_i^{(m)}$.

Patient similarity was estimated using Gaussian kernels applied to both the original and projected representations. For an embedding matrix $\mathbf{X}$, the Gaussian similarity between patients i and j was defined as

$$K(\mathbf{X})_{ij} = \frac{1}{L} \sum_{\ell=1}^{L} \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|_2^2}{\sigma_\ell}\right), \tag{4.8}$$

where L is the number of kernels and σ_ℓ denotes the bandwidth of kernel ℓ. Our implementation used three Gaussian kernels with geometrically scaled bandwidths, resembling the original M3Care [35].

The modality-specific similarity matrix combined similarities calculated from the original and projected representations:

$$\mathbf{S}^{(m)} = \left[(1 - \gamma_m)K(\widetilde{\mathbf{P}}^{(m)}) + \gamma_m\right] \odot K(\widetilde{\mathbf{X}}^{(m)}), \tag{4.9}$$

where $\widetilde{\mathbf{P}}^{(m)}$ and $\widetilde{\mathbf{X}}^{(m)}$ denote the batch-normalised projected and original embeddings, respectively, $\gamma_m = \sigma(\epsilon_m)$ is a learnable mixing coefficient, and $\odot$ denotes element-wise multiplication.

To account for modality-level missingness, similarity scores were retained only for patient pairs with the corresponding modality available. The combined similarity S_{ij} between patients was calculated by averaging across their jointly observed modalities.

A learnable threshold $\tau = \sigma(\theta)$ was subsequently applied to remove weak patient relationships. In the forward pass, the resulting graph adjacency matrix retained the original similarity values for edges exceeding the threshold:

$$A_{ij} = \begin{cases} S_{ij}, & S_{ij} > \tau, \\ 0, & \text{otherwise.} \end{cases} \tag{4.10}$$

During backpropagation, the threshold was restored using a detached term, allowing gradient-based optimisation of it through a surrogate gradient despite the non-differentiable edge-selection operation. The graph was constructed dynamically from patients within each processing batch instead of being fixed across the entire cohort.

Graph construction stability regularisation

To regularise the learned similarity projections, M3Care incorporated an auxiliary stability loss. We investigated three alternative formulations, controlled by the `stability_loss_type` configuration parameter.

The original **norm-based loss** (`norm`) penalised differences between the Frobenius norms of the original and projected embedding matrices:

$$\mathcal{L}^{(m)}_{\text{norm}} = \left| \|\mathbf{P}^{(m)}\|_F - \|\mathbf{X}^{(m)}\|_F \right|. \tag{4.11}$$

This encouraged the projection network to preserve the overall embedding magnitude, but did not explicitly constrain pairwise relationships between patients.

We additionally implemented a **cosine-similarity preservation loss** (`cosine`), which penalised changes in the pairwise cosine-similarity matrix:

$$\mathcal{L}^{(m)}_{\cos} = \frac{1}{B^2} \left\| C(\mathbf{X}^{(m)}) - C(\widetilde{\mathbf{P}}^{(m)}) \right\|_F^2, \tag{4.12}$$

where

$$C(\mathbf{X})_{ij} = \frac{\mathbf{x}_i^\top \mathbf{x}_j}{\|\mathbf{x}_i\|_2 \|\mathbf{x}_j\|_2}. \tag{4.13}$$

Lastly, we adopted a **Gaussian similarity-matrix preservation loss** (`similarity`), which encouraged the learned projections to retain the original pairwise Gaussian-kernel relationships:

$$\mathcal{L}^{(m)}_{\text{sim}} = \frac{1}{B^2} \left\| K(\mathbf{X}^{(m)}) - K(\widetilde{\mathbf{P}}^{(m)}) \right\|_F^2. \tag{4.14}$$

Unlike norm preservation, the latter two objectives explicitly regularised relationships between patient representations. Cosine-similarity preservation focused on maintaining their relative angular relationships, whereas Gaussian-kernel preservation aimed to maintain similarities derived from pairwise Euclidean distances.

The overall graph construction loss was defined as

$$\mathcal{L}_{\text{graph}} = \lambda_g \, \text{Agg}_{m=1}^{M} \left(\mathcal{L}_{\text{stab}}^{(m)} \right), \tag{4.15}$$

where λ_g controlled the regularisation strength and Agg represented either summation or averaging across modalities, according to the model configuration parameter `mean_graph_construction_loss`.

GCN-based representation refinement

The constructed patient similarity graph was used to propagate information between patients through two successive graph convolutional layers. For each modality selected for imputation, the graph-refined embedding matrix was computed as

$$\begin{aligned} \mathbf{H}_1^{(m)} &= \text{ReLU} \left(\mathbf{A}\mathbf{X}^{(m)}\mathbf{W}_1^{(m)} + \mathbf{b}_1^{(m)} \right), \\ \widehat{\mathbf{X}}^{(m)} &= \text{ReLU} \left(\mathbf{A}\mathbf{H}_1^{(m)}\mathbf{W}_2^{(m)} + \mathbf{b}_2^{(m)} \right), \end{aligned} \tag{4.16}$$

where $\mathbf{A}$ is the shared batch-level adjacency matrix and $\mathbf{W}_1^{(m)}$ and $\mathbf{W}_2^{(m)}$ are modality-specific trainable weight matrices.

The resulting GCN-generated embeddings could be combined with the original representations using three alternative refinement strategies, controlled by `gcn_refine_mode`.

In the **adaptive** strategy, the relative contributions of the graph-refined and original representations were determined by learnable gating functions:

$$\begin{aligned} u_i^{(m)} &= \sigma \left(f_{g,m}(\widehat{\mathbf{x}}_i^{(m)}) \right), \\ v_i^{(m)} &= \sigma \left(f_{o,m}(\mathbf{x}_i^{(m)}) \right), \\ \alpha_i^{(m)} &= \frac{u_i^{(m)}}{u_i^{(m)} + v_i^{(m)} + \varepsilon}. \end{aligned} \tag{4.17}$$

The final representation was then

$$\mathbf{z}_i^{(m)} = \begin{cases} \alpha_i^{(m)} \widehat{\mathbf{x}}_i^{(m)} + (1 - \alpha_i^{(m)}) \mathbf{x}_i^{(m)}, & a_i^{(m)} = 1, \\ \widehat{\mathbf{x}}_i^{(m)}, & a_i^{(m)} = 0. \end{cases} \tag{4.18}$$

The **imputed-only** strategy instead used the GCN output for every patient, regardless of modality availability:

$$\mathbf{z}_i^{(m)} = \widehat{\mathbf{x}}_i^{(m)}. \tag{4.19}$$

The **imputed-for-missingness** strategy preserved the original representation when the modality was available and used the GCN output only when it was missing:

$$\mathbf{z}_i^{(m)} = a_i^{(m)}\mathbf{x}_i^{(m)} + (1 - a_i^{(m)})\widehat{\mathbf{x}}_i^{(m)}. \tag{4.20}$$

These strategies enabled the investigation of whether GCN-generated representations were most useful as replacements for all modality embeddings, as substitutes for missing modalities only, or as adaptively weighted refinements of observed representations. The implementation also allowed GCN refinement to be restricted to selected modalities, reducing the number of modality-specific GCN and gating components when imputation was unnecessary.

Multimodal fusion and prediction

Following GCN refinement, the modality representations were integrated through a two-stage multi-head self-attention (MHSA) fusion module. Each modality was represented by a token formed from its refined embedding and a learnable modality-type embedding. An additional learnable missingness embedding could optionally be incorporated to distinguish observed from unavailable modalities.

The modality tokens were processed by two successive Transformer-style encoder blocks containing self-attention, feed-forward layers and residual connections. The first block used configurable multi-head attention, while the second employed a single attention head. The outputs corresponding to individual modalities were concatenated and passed through a fully connected prediction head to produce the final binary classification logit.

The fusion mechanism supported optional missingness-aware attention masking through the `mask_missingness_fusion` setting. When enabled, unavailable modalities were masked at the key positions in both self-attention stages, preventing their value representations from contributing directly to the attention-weighted outputs. However, these modalities remained active as queries, allowing their GCN-refined representations to attend to available modalities and contribute to the final fused representation. This enabled missingness-aware attention while retaining the potential contribution of graph-imputed modality representations.

For the standard M3Care implementation, the training objective combined the task-specific loss and the graph construction regularisation loss:

$$\mathcal{L}_{\text{M3Care}} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{graph}}. \tag{4.21}$$

The graph regularisation term was optional and could be disabled by setting its weight to zero.

Temporal M3Care: incorporating mTAND into M3Care

To investigate the influence of temporal representation learning on M3Care, we incorporated the same mTAND mechanism employed in temporal FuseMoE as described in Section 4.1. Rather than aggregating ECG and CXR observations into pooled patient-level representations before fusion, mTAND-M3Care used time-aware encoders to map these observations onto a fixed number of reference time points. Raw clinical time-series measurements such as vital signs and laboratory tests could also be included as a separate modality, with feature-level observation masks distinguishing recorded from missing measurements. To accommodate differences in embedding dimension, the

patient similarity projection networks and GCN layers were adapted to operate on modality-specific embedding dimensions. The resulting graph construction and representation-refinement procedures otherwise followed the M3Care formulation described above.

A further architectural modification concerned multimodal fusion. The static tabular representation was excluded from the MHSA fusion module, which instead operated on the temporal modalities, including ECG, CXR and, when enabled, clinical time series. Following attention-based fusion, the temporal representations were concatenated with the static tabular embedding before being passed to the prediction head. This design enabled the investigation of time-aware multimodal learning while retaining M3Care's graph-based approach to handling missing modalities.

Incorporating contrastive learning

To further improve the quality of graph-based modality representations, contrastive learning (CL) was incorporated into M3Care as an auxiliary training strategy. The motivation was to encourage consistency between GCN-generated representations and their corresponding target embeddings, particularly when the original modality was unavailable. In addition to the primary clinical prediction objective, CL provided supplementary supervision for learning patient-specific modality representations. Two complementary contrastive objectives were implemented: observed-modality alignment and missing-modality alignment. Furthermore, a learned cross-modal target mechanism was introduced to provide supervision for missing modalities using information from other available modalities.

Contrastive alignment objectives CL encourages representations of related samples to become more similar while distinguishing unrelated samples within the embedding space. In this implementation, a symmetric InfoNCE objective was adopted, with positive pairs formed from corresponding representations of the same patient and negative pairs drawn from other eligible patients within the processing batch.

For a target modality m, let $\widehat{\mathbf{x}}_i^{(m)}$ denote the GCN-generated embedding of patient i and $\mathbf{x}_i^{(m)}$ the original modality embedding when observed. Separate projection networks were applied to the GCN-generated and target embeddings, followed by L2 normalisation. For N eligible patient pairs, let $\mathbf{u}_i$ and $\mathbf{v}_i$ denote the resulting normalised representations. The symmetric InfoNCE loss was defined as

$$\mathcal{L}_{\mathrm{NCE}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{\exp\left(\mathbf{u}_i^\top \mathbf{v}_i/T\right)}{\sum_{j=1}^{N}\exp\left(\mathbf{u}_i^\top \mathbf{v}_j/T\right)} + \log\frac{\exp\left(\mathbf{v}_i^\top \mathbf{u}_i/T\right)}{\sum_{j=1}^{N}\exp\left(\mathbf{v}_i^\top \mathbf{u}_j/T\right)}\right], \tag{4.22}$$

where T denotes the temperature hyperparameter controlling the concentration of the similarity distribution. A minimum number of eligible samples was required to calculate the contrastive objective, ensuring that positive pairs could be compared against negative samples. For simplicity, the notation $\mathcal{L}_{\mathrm{NCE}}(\mathbf{A}, \mathbf{B})$ denotes the symmetric contrastive loss calculated after applying the corresponding projection networks and L2 normalisation in the following paragraphs.

For **observed-modality alignment**, the GCN-generated representation was contrasted against the original embedding of the same modality. The objective encouraged the GCN branch to preserve information from directly observed modalities:

$$\mathcal{L}_{\text{obs}}^{(m)} = \mathcal{L}_{\text{NCE}}\left(\widehat{\mathbf{X}}_{\text{obs}}^{(m)}, \mathbf{X}_{\text{obs}}^{(m)}\right), \tag{4.23}$$

where the subscript obs denotes patients for whom modality m was available.

For **missing-modality alignment**, the corresponding original embedding was unavailable. Instead, the GCN-generated representation was contrasted against a cross-modal target estimated from other observed modalities:

$$\mathcal{L}_{\text{miss}}^{(m)} = \mathcal{L}_{\text{NCE}}\left(\widehat{\mathbf{X}}_{\text{miss}}^{(m)}, \mathbf{T}_{\text{miss}}^{(m)}\right), \tag{4.24}$$

where $\mathbf{T}_{\text{miss}}^{(m)}$ represents the cross-modal target embeddings for patients with modality m missing. Only patients with at least one eligible context modality available were considered. Both contrastive losses were calculated separately for each selected modality and averaged across modalities with sufficient eligible samples.

Cross-modal target construction Two strategies were implemented to construct the target representations used for missing-modality alignment. The first employed a simple aggregation of available context modalities, while the second introduced learnable networks to estimate target-specific representations.

In the **aggregation-based strategy**, the target was calculated as the mean of the observed context-modality embeddings:

$$\mathbf{t}_i^{(m)} = \frac{\sum_{k \in \mathcal{C}_m} a_i^{(k)} \mathbf{x}_i^{(k)}}{\max\left(\sum_{k \in \mathcal{C}_m} a_i^{(k)}, 1\right)}, \tag{4.25}$$

where $\mathcal{C}_m$ denotes the selected context modalities excluding the target modality, and $a_i^{(k)} \in \{0, 1\}$ indicates the availability of modality k for patient i. This approach provided a straightforward estimate of the missing representation from observed information, but assumed that different context modalities could be meaningfully combined through averaging.

Alternatively, the **learned cross-modal target strategy** employed a modality-specific MLP to estimate each target modality representation from the available context modalities. The network received concatenated context embeddings together with their corresponding availability indicators, which allowed the network to distinguish between observed and missing context inputs.

To train the learned target networks, patients with both the target modality and at least one eligible context modality available were used for direct supervision. The predicted target embeddings were compared against the corresponding original modality representations using a reconstruction loss. Two loss functions were supported: mean squared error (MSE) and cosine distance.

The MSE reconstruction objective was

$$\mathcal{L}_{\text{rec,MSE}}^{(m)} = \frac{1}{N_m d_m} \sum_{i=1}^{N_m} \left\| \mathbf{t}_i^{(m)} - \mathbf{x}_i^{(m)} \right\|_2^2, \tag{4.26}$$

where N_m is the number of eligible patients and d_m is the target embedding dimension.

The cosine reconstruction objective was

$$\mathcal{L}_{\text{rec,cos}}^{(m)} = \frac{1}{N_m} \sum_{i=1}^{N_m} \left(1 - \frac{\mathbf{t}_i^{(m)\top} \mathbf{x}_i^{(m)}}{\|\mathbf{t}_i^{(m)}\|_2 \|\mathbf{x}_i^{(m)}\|_2} \right). \tag{4.27}$$

An additional contrastive supervision objective was introduced to encourage patient-discriminative target representations. This applied symmetric InfoNCE between the learned target embeddings and the corresponding observed modality embeddings:

$$\mathcal{L}_{\text{target,NCE}}^{(m)} = \mathcal{L}_{\text{NCE}} \left(\mathbf{T}_{\text{obs}}^{(m)}, \mathbf{X}_{\text{obs}}^{(m)} \right). \tag{4.28}$$

The reconstruction and contrastive supervision losses were calculated over eligible observed-modality samples, enabling the target networks to learn cross-modal relationships before being used to guide missing-modality alignment.

Overall training objective The contrastive-learning extensions were incorporated into the primary M3Care optimisation objective through weighted auxiliary losses. The complete objective was expressed as

$$\begin{aligned} \mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \mathcal{L}_{\text{graph}} \\ + \lambda_{\text{obs}} \mathcal{L}_{\text{obs}} + \lambda_{\text{miss}} \mathcal{L}_{\text{miss}} \\ + \lambda_{\text{target,rec}} \mathcal{L}_{\text{target,rec}} + \lambda_{\text{target,NCE}} \mathcal{L}_{\text{target,NCE}}. \end{aligned} \tag{4.29}$$

The new terms correspond to observed-modality contrastive alignment, missing-modality contrastive alignment, learned-target reconstruction and learned-target contrastive supervision, respectively. The auxiliary losses were calculated separately for each selected modality and aggregated across modalities with eligible samples in a given batch. The enabled objectives were then combined using configurable weighting coefficients, allowing their relative contributions to the overall training loss to be investigated independently.

4.3 Experiment setup

4.3.1 General experimental framework

The experiments were conducted across the two clinical prediction pathways described in Chapter 3. Each pathway comprised multiple binary prediction tasks, with separate models trained for each outcome. A shared modelling framework was adopted to support consistent data handling, model training and experimental comparisons across different multimodal fusion architectures.

For each pathway, a representative prediction outcome was selected for initial hyperparameter tuning and model configuration ablation studies. Mechanical ventilation was selected for the

deterioration pathway, while extended hospital stay > 7 days was selected for the hospitalisation pathway. Each fusion architecture was tuned independently using the corresponding representative outcome. Following the identification of the preferred configuration for each architecture, the same model settings were applied to the remaining prediction outcomes within the pathway, with separate models trained for each task.

Three multimodal fusion frameworks were investigated: simple concatenation, FuseMoE and M3Care. Concatenation, previously adopted as a baseline in **MM-HealthFair**, was retained as a reference for assessing the more advanced fusion architectures. FuseMoE and M3Care were implemented as described in Section 4.2, with additional experiments investigating their temporal extensions through the incorporation of mTAND. The mTAND experiments were restricted to mechanical ventilation prediction in the deterioration pathway. As incorporating mTAND substantially increased the dimensionality of the learned representations and associated computational requirements, this extension was not investigated for the hospitalisation pathway, which included a larger number of temporally structured modalities.

All models were implemented using PyTorch and trained through PyTorch Lightning, with a common interface for loading modality-specific inputs, applying availability masks and producing binary classification outputs. This enabled different fusion mechanisms to be investigated using consistent pathway-specific datasets and training procedures. For ECG recordings, CXR images and the time-series measurements, all experiments used the dataset with averaged embeddings (`averageRecord`), except for those incorporating mTAND, which employed the dataset with unpooled embeddings of all instances (`allRecord`) to retain the full sequence of observations for temporal modelling. The construction and differences between these two dataset configurations are described in Chapter 3.

For the hospitalisation pathway, two modality configurations were investigated to assess the incorporation of clinical information from current and previous hospital encounters. The **current-window** (`currWindow`) configuration included static EHR features, time-series measurements, ECG recordings and CXR images associated with the ED phase of the current admission. The **dual-window** (`dualWindow`) configuration additionally incorporated time-series measurements, CXR images and clinical notes from the previous hospital admission within the 90-day lookback window. Previous ECG recordings were excluded from the dual-window configuration to prioritise physiological information from the current encounter and limit the complexity of the previous-window modality set. The distinction between the two configurations primarily concerned the inclusion of historical time-dependent and textual modalities, as static EHR features could contain information derived from previous clinical encounters in both settings. These configurations enabled the investigation of different levels of temporal and multimodal information while retaining the same clinical prediction objectives.

4.3.2 Model training

Model training was managed through a shared PyTorch Lightning module, which handled model execution, loss calculation, optimisation and training-stage logging. All models were trained using mini-batch gradient-based optimisation with the AdamW optimiser. The primary prediction objective was binary cross-entropy (BCE) with logits. To account for class imbalance, positive-class weighting was calculated from the training labels and incorporated into the classification loss, unless explicitly disabled.

Where applicable, model-specific auxiliary losses were incorporated into the primary training objective. For FuseMoE, these included the expert load-balancing loss and the modality-combination supervision loss when the combination router was enabled. For M3Care, auxiliary objectives included graph construction stability regularisation and the optional contrastive-learning-related losses. The combined objectives were optimised during training, while individual auxiliary losses were recorded separately to facilitate the analysis of their behaviour.

Each experiment followed a configuration-driven workflow, with model architecture, modality selection, training settings and optional components specified in YAML configuration files. Within each pathway and data configuration, the same training, validation and test partitions were used across model architectures to support consistent experimental comparisons. Each outcome and model configuration was trained independently, with its own configuration, checkpoint and experimental outputs. Random seeds were fixed to support reproducibility, and experiment-specific configurations and outputs were stored in separate run directories. Models were trained for a maximum of 50 epochs, with early stopping applied after 10 consecutive epochs without improvement in validation Matthews correlation coefficient (MCC). The checkpoint achieving the highest validation MCC was retained for subsequent analysis.

4.3.3 Contrastive-learning training strategies

Two training strategies were implemented to incorporate contrastive learning (CL) into M3Care, depending on the availability of complete multimodal samples: a two-stage curriculum-learning procedure and a single-stage joint training procedure. As an initial investigation of the CL extension, experiments were restricted to mechanical ventilation prediction in the deterioration pathway and extended hospital stay prediction in the hospitalisation pathway. The choice of training strategy was informed by the availability of complete-case samples and the feasibility of learning reliable cross-modal representations from the restricted cohort.

Two-stage curriculum learning The two-stage strategy was designed to separate the initial learning of modality representations and cross-modal targets from their subsequent use in missing-modality alignment. During the first stage, referred to as **warm-up**, the model was trained on a complete-case subset containing all required modalities. The training objective combined the primary classification loss, graph construction stability regularisation, observed-modality contrastive alignment and learned-target supervision, including reconstruction and contrastive objectives. Missing-modality CL was disabled during this stage.

In the second stage, referred to as **fine-tuning**, the model was initialised using the best checkpoint obtained during warm-up and trained on the broader cohort containing naturally missing modalities. The pretrained modality encoders were frozen, while both observed- and missing-modality contrastive alignment objectives were enabled. Direct supervision of the learned cross-modal targets was disabled during fine-tuning, allowing the previously learned target representations to guide missing-modality alignment.

The two stages were executed as separate training runs, with model parameters transferred through checkpoint initialisation. The optimiser and learning-rate scheduler were reinitialised for fine-tuning rather than resumed from the warm-up stage.

Single-stage joint training When the complete-case subset was too small to support reliable warm-up training, a **single-stage joint training** strategy was adopted. In this setting, the model was trained directly on the broader cohort containing incomplete modality combinations, without a separate pretraining stage or freezing of modality encoders.

All four auxiliary objectives were enabled simultaneously: observed-modality contrastive alignment, missing-modality contrastive alignment, learned-target reconstruction and learned-target contrastive supervision. Each objective was calculated using its eligible patient subset for the corresponding modality, allowing the model to learn from both observed and incomplete clinical information within a single training procedure.

Both strategies employed the same underlying M3Care architecture and contrastive objectives described in Section 4.2.2. Their principal differences concerned the scheduling of auxiliary supervision, the use of pretrained representations and the freezing of selected model components.

4.4 Hyperparameter tuning

Hyperparameter tuning was conducted in two sequential phases. First, training hyperparameters were optimised to establish suitable learning conditions for each fusion architecture. Subsequently, model configuration ablation studies were performed to investigate alternative architectural components and identify preferred model designs. As described in Section 4.3, mechanical ventilation and extended hospital stay were selected as the representative outcomes for the deterioration and hospitalisation pathways, respectively. Each fusion architecture was tuned independently, and the resulting preferred configurations were subsequently applied to the remaining prediction outcomes.

For FuseMoE, the reference architecture employed a joint router with seven experts, top-k expert selection ($k = 2$) and an auxiliary load-balancing loss weighted by 0.01. For M3Care, the reference configuration enabled GCN-based modality imputation, missingness-aware attention masking and adaptive embedding refinement, with norm-based graph stability regularisation weighted by 0.01. Additional missingness embeddings were disabled. These configurations served as the starting points for training hyperparameter tuning and subsequent model configuration ablation studies.

4.4.1 Training hyperparameter tuning

The training hyperparameter search focused on two parameters of the AdamW optimiser: learning rate (LR) and weight decay (WD). A two-stage tuning procedure was adopted to balance computational efficiency with the assessment of training variability across random seeds. The Matthews correlation coefficient (MCC) calculated on the validation set was used as the selection criterion throughout both stages.

In the first stage, referred to as **screening**, a grid search was performed over predefined combinations of learning rates and weight decay values using a fixed random seed of 42. Each combination was trained independently using the corresponding reference model configuration. The experiments were ranked according to their validation MCC, and the three highest-performing LR-WD combinations were retained for further investigation.

In the second stage, referred to as **stability assessment**, each of the three shortlisted combinations was assessed using three random seeds (42, 43 and 44). The validation MCC values were aggregated across seeds to calculate the mean and standard deviation for each combination. The combination

achieving the highest mean validation MCC was selected as the preferred training configuration for the corresponding fusion architecture.

The tuning procedure was automated through a collection of Bash scripts. The `tune_one.sh` script coordinated the two stages for a selected fusion architecture by invoking `tune_models.sh` to execute the training runs and `summarise_tuning.sh` to aggregate their results. The underlying tuning runner read the model configurations, parameter grids, random seeds and fixed experimental settings from a YAML specification. For each hyperparameter combination, the script generated a distinct run identifier, applied the corresponding configuration overrides and launched the shared training pipeline. Experiment metadata and results were stored in structured directories, allowing multiple configurations to be executed sequentially and summarised automatically.

To reduce redundant computation, the stability assessment could reuse the screening run for seed 42 when the model configuration and data settings were unchanged. This allowed the three-seed assessment to be completed without unnecessarily repeating an identical training run.

The tuning experiments used the `averageRecord` dataset configurations with all ED stays of a patient included for deterioration, and the `dualWindow` and `averageRecord` configurations for hospitalisation. The LR search spaces differed between fusion architectures and pathways, while the same two WD values were investigated throughout. The complete search spaces are summarised in Table 4.1.

Table 4.1: Learning-rate and weight-decay search spaces for each fusion architecture and clinical prediction pathway.

Pathway	Model	Learning rate	Weight decay
Deterioration	Concatenation	10^{-5}, 5×10^{-5}, 10^{-4}, 5×10^{-4}, 10^{-3}	10^{-4}, 10^{-3}
	FuseMoE	5×10^{-5}, 10^{-4}, 5×10^{-4}	10^{-4}, 10^{-3}
	M3Care	10^{-5}, 5×10^{-5}, 10^{-4}, 5×10^{-4}	10^{-4}, 10^{-3}
Hospitalisation	Concatenation	10^{-5}, 5×10^{-5}, 10^{-4}, 5×10^{-4}	10^{-4}, 10^{-3}
	FuseMoE	10^{-5}, 5×10^{-5}, 10^{-4}, 5×10^{-4}	10^{-4}, 10^{-3}
	M3Care	10^{-5}, 5×10^{-5}, 10^{-4}, 5×10^{-4}, 10^{-3}	10^{-4}, 10^{-3}

4.4.2 Model configuration ablation

Following the identification of preferred training hyperparameters, model configuration ablation studies were conducted to investigate the influence of individual architectural components and modelling strategies. Unlike the initial training hyperparameter search, these experiments focused on model-specific design choices, with the selected LR and WD retained as the reference training settings.

The ablation experiments were organised through dedicated Bash scripts that launched multiple model configurations within a single execution. Each experiment was based on a common reference configuration, with selected parameters overridden to isolate the effect of a particular architectural choice. For example, the `run_ablation_router_types.sh` script investigated alternative FuseMoE routing mechanisms, including joint, per-modality and disjoint routing. This configuration-driven approach enabled systematic comparisons while reducing the need to maintain separate YAML files for every model variant.

The ablation studies were conducted using the representative outcome for each pathway. Model variants were compared using their test-set MCC, and the highest-performing configuration was selected for subsequent experiments on the remaining prediction outcomes. For hospitalisation, the best configuration was also adopted for experiments using the `currWindow` dataset. The architectural variations investigated in the ablation studies are summarised in Table 4.2.

Table 4.2: Model configurations investigated in the ablation studies. Reference settings are listed first and shown in bold. "Same" indicates identical configurations to deterioration, while "Not investigated" indicates no corresponding experiments. An asterisk (*) denotes a fixed setting. Italicised rows specify conditions held during the corresponding ablations.

Model	Ablation category	Deterioration configurations	Hospitalisation configurations
FuseMoE	Router type	**Joint**, per-modality, disjoint, combination	**Joint**, per-modality, disjoint
	Shared-expert weight	**0.333**, 0, 0.25, 0.5, 0.75	Same
	Load-balancing loss weight	**0.01**, 0, 0.001, 0.1, 0.3	Same
	Standard dropout	**0**, 0.1, 0.3	Same
	Modality dropout	**0**, 0.1	Same
	Shared-expert weight for dropout ablations	***0.333**, 0.5*	***0.333****
	Modality embedding width (d_{model})	**128**, 64	Same
	Expert hidden width	**256**, 128	Same
	Top-k expert selection	**2**, 1, 3	Same
	Number of experts	**7**, 5, 9	**7**, 9, 14
Temporal FuseMoE	Reference time points (tt_max)	**8**, 16	Not investigated
	Time embedding width (embed_time)	**16**, 8, 32	Not investigated
M3Care	Modality embedding width (d_{model})	**128***	Same
	Standard dropout	**0.1***	Same
	GCN-based imputation	**Yes**, no	Same
	Mask missingness during fusion	**Yes**, no	Same
	Impute only missing modalities	**No**, yes	Same
	GCN refinement mode	**Adaptive**, imputed-only, imputed-for-missingness	Same
	Missingness embedding	**No**, yes	Same
	Graph stability loss type	**Norm**, cosine similarity, Gaussian similarity	Same
	Graph stability loss weight	**0.01**, 0, 0.001, 0.1	Same
	GCN refinement for graph stability ablations	***Imputed-for-missingness****	*Same*
	Mean graph construction loss	**False***	**True***
Temporal M3Care	Reference time points (tt_max)	**8**, 16	Not investigated
	Time embedding width (embed_time)	**16***	Not investigated

4.4.3 Contrastive-learning loss-weight tuning

Additional ablation experiments were conducted to investigate the influence of auxiliary loss weights in the CL extension of M3Care. Four weighting coefficients were considered, corresponding

to observed-modality contrastive alignment, missing-modality contrastive alignment, learned-target reconstruction and learned-target contrastive supervision, as defined in Section 4.2.2. A one-factor-at-a-time approach was adopted, in which each loss weight was varied independently while the remaining three were held at their reference values. These experiments were conducted using mechanical ventilation for the deterioration pathway and extended hospital stay for the hospitalisation pathway, with the latter investigated under both `currWindow` and `dualWindow` configurations.

For deterioration and current-window hospitalisation, each loss-weight configuration was investigated using the two-stage curriculum-learning procedure, comprising complete-case warm-up followed by fine-tuning on the broader cohort. Dual-window hospitalisation employed single-stage joint training, with all four auxiliary objectives enabled throughout. The reference values and alternative loss weights investigated were identical across the three settings, as summarised in Table 4.3.

Table 4.3: Loss-weight configurations investigated for contrastive learning in M3Care. The reference weight is shown in bold and listed first, followed by the alternative values. The same configurations were investigated for deterioration, current-window hospitalisation and dual-window hospitalisation.

Auxiliary objective	**Weight parameter**	**Configurations**
Observed-modality contrastive alignment	λ_{obs}	**1.0**, 0.5
Missing-modality contrastive alignment	λ_{miss}	**1.0**, 0, 0.5
Learned-target reconstruction	$\lambda_{\text{target,rec}}$	**1.0**, 0.5
Learned-target contrastive supervision	$\lambda_{\text{target,NCE}}$	**0.5**, 0, 0.25

4.5 Evaluation

The trained models were evaluated across several complementary dimensions to assess their predictive performance, statistical reliability and robustness to incomplete multimodal information. In addition to conventional classification metrics, further analyses were conducted to compare model predictions, investigate the effects of modality availability and assess architecture-specific behaviours, including expert utilisation in FuseMoE and graph-based representation imputation in M3Care.

4.5.1 Predictive performance evaluation

Predictive performance was evaluated independently for each binary prediction task using five metrics: area under the receiver operating characteristic curve (AUROC), recall (sensitivity), specificity, Matthews correlation coefficient (MCC) and balanced accuracy (BA). AUROC measures the ability of a model to discriminate between positive and negative cases across different classification thresholds, while recall and specificity quantify the proportions of correctly identified positive and negative cases, respectively. Balanced accuracy is calculated as the arithmetic mean of recall and specificity, providing a measure of classification performance that accounts for differences in class prevalence.

MCC provides a correlation-based measure of binary classification performance by considering all four components of the confusion matrix: true positives (TP), true negatives (TN), false positives (FP) and false negatives (FN). It is defined as

$$\mathrm{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}. \tag{4.30}$$

MCC ranges from -1 to 1, where 1 indicates perfect classification, 0 represents no correlation between predictions and true labels, and -1 indicates complete disagreement. Given the potential class imbalance in clinical prediction tasks, MCC was used as the principal metric for model selection and comparison, alongside AUROC and the other classification metrics.

For the standard evaluation, predicted probabilities were converted into binary predictions using a fixed threshold of 0.5. The best checkpoint from each training run, selected according to validation MCC, was used to calculate the final predictive performance on the corresponding dataset partitions. Predictions and ground-truth labels were retained at the individual encounter level to support subsequent statistical comparisons and robustness analyses.

4.5.2 Statistical analysis and model comparison

To quantify uncertainty in predictive performance estimates, 95% confidence intervals (CIs) were calculated using non-parametric bootstrap resampling. For each model and dataset partition, encounters were randomly drawn with replacement to construct bootstrap samples containing the same number of encounters as the original dataset. The five classification metrics were recalculated for each bootstrap sample, and bias-corrected and accelerated (BCa) CIs were estimated from the resulting bootstrap distributions. The procedure used 1,000 bootstrap resamples with a fixed random seed of 42.

Statistical comparisons between models were performed using paired predictions from the same dataset partition. Prediction records were aligned using their corresponding encounter identifiers, ensuring that comparisons involved the same clinical cases and ground-truth labels. The principal comparison employed a paired permutation test to assess differences in MCC between two models. For each permutation, each stay's two predictions were randomly exchanged with a probability of 0.5, and the difference in MCC was recalculated. The resulting null distribution was used to estimate a two-sided p-value for the observed performance difference. By default, 10,000 permutations were performed.

Additional statistical tests included McNemar's test for comparing paired classification errors and the Wilcoxon signed-rank test for comparing paired predicted probabilities. When multiple candidate models were compared against a common reference model, the Benjamini-Hochberg (BH) procedure was applied to the MCC permutation-test p-values to control the false discovery rate at 0.05. These analyses provided complementary assessments of whether observed performance differences were consistent with systematic differences between model predictions.

4.5.3 Robustness to missing modalities

To assess the robustness of multimodal models to incomplete clinical information, modality ablation and modality permutation were performed using trained model checkpoints without retraining the models.

Modality ablation Modality ablation evaluated model performance under artificially restricted modality availability. For each evaluation scenario, selected modalities were retained while the

remaining modalities were masked, allowing the model to generate predictions using only the permitted information. This enabled the investigation of performance changes when particular data sources were unavailable.

For the deterioration pathway, all non-empty combinations of the configured modalities were considered, resulting in seven scenarios for the three-modality setting (tabular/ECG/CXR) and 15 scenarios when a separate time-series modality was included. For hospitalisation, where the full dual-window configuration contained seven modality slots, a more compact set of 15 scenarios was adopted. These comprised seven single-modality scenarios, seven scenarios in which one modality was removed while the others were retained, and one reference scenario without forced ablation.

To account for natural missingness, an encounter was considered eligible for a scenario only when all modalities required by that scenario were originally available. Encounters not meeting the availability requirements were excluded from the corresponding scenario. Consequently, the evaluated encounter subset could differ between scenarios, which was considered when interpreting performance differences.

In addition to the standard metrics calculated using a fixed threshold of 0.5, the ablation framework supported scenario-specific threshold calibration. For each scenario, the threshold maximising validation MCC was identified and subsequently applied to the corresponding test predictions. This provided an additional assessment of performance under modality restrictions without using test labels to determine the decision threshold.

Modality permutation Modality permutation provided a complementary approach to investigating whether model predictions depended on the information contained within a modality rather than solely on its availability. Unlike modality ablation, which removes both modality information and its availability signal, permutation preserved the original modality-availability patterns while disrupting the association between modality content and individual patients.

For each target modality, its input records were randomly reassigned between patients sharing the same modality-availability pattern. Where applicable, measurement values and their associated timestamps were permuted together to preserve their internal correspondence. This ensured that modality-availability indicators remained unchanged while the clinical information associated with individual patients was altered.

The evaluation considered a reference scenario without permutation, individual modality-permutation scenarios and a scenario in which all configured modalities were independently permuted. Each permuted scenario was repeated over 100 random permutations, and the resulting classification metrics were summarised using their means, standard deviations and CIs. All encounters were retained across scenarios because modality availability remained unchanged, allowing direct comparisons against the reference performance.

To assess the extent to which models utilised modality-specific information, the mean performance under permutation was compared against both the original performance and the corresponding modality-ablation result. Performance remaining close to the original level suggested limited dependence on the permuted modality's content, potentially indicating greater reliance on its availability or other modalities. Conversely, performance approaching the corresponding modality-ablation level suggested that the removed information contributed substantially to prediction. Intermediate results indicated partial sensitivity to modality content. In some cases, permutation

could reduce performance below the modality-ablation level, as incorrect but apparently available information may be more disruptive than an explicitly missing modality.

Statistical significance was assessed using one-sided empirical permutation tests with scenario-specific null hypotheses. For individual modality-permutation scenarios, the reference performance was used as the null value, and the empirical p-value was calculated from the proportion of permutation draws achieving a metric value at least as high as the original performance. A small p-value therefore indicated that disrupting the modality's content consistently reduced predictive performance.

For the all-modality permutation scenario, the null value was instead defined as the corresponding metric's no-skill reference level, such as MCC = 0 or AUROC = 0.5. The empirical p-value was calculated from the proportion of permutation draws producing performance at or below this level. A small p-value in this setting indicated that predictive performance remained above the no-skill reference despite disruption of all modality content, suggesting that the model might retain predictive information from modality-availability patterns or other unchanged inputs.

A finite-sample correction was applied when calculating empirical p-values, ensuring that the minimum attainable value was $1/(R+1)$, where R denotes the number of permutation draws. With 100 permutations, the minimum attainable p-value was approximately 0.0099. All reference and permutation metrics were calculated using the same binary classification procedures and a fixed decision threshold of 0.5.

4.5.4 Expert-routing diagnostics

For FuseMoE, additional analyses were conducted to investigate how the routing mechanism distributed patient representations across the available experts. The trained models were evaluated on the test set, and the routing weights assigned to individual experts were recorded for each patient. These were subsequently aggregated to examine expert utilisation, including the mean routing weight, the frequency with which each expert received a non-zero weight, and the proportion of patients for whom each expert received the highest routing weight.

Expert utilisation was also examined across different combinations of naturally available modalities. This enabled the investigation of whether routing behaviour varied according to the clinical information available for each patient and whether particular experts were preferentially selected for specific modality combinations. These diagnostics provided additional insights into expert specialisation and routing balance beyond conventional predictive performance metrics.

4.5.5 Imputed embedding quality

To evaluate the quality of graph-based missing-modality imputation in M3Care, additional analyses compared GCN-generated modality embeddings against their corresponding observed representations. For each target modality, patients were selected if the modality was originally available and at least one other context modality was observed. The target modality was then artificially masked, and the model generated its estimated representation using the remaining available modalities. The original embedding, obtained before masking, was retained as the reference representation. This enabled direct evaluation of imputation quality using observed modality information as a reference in the learned embedding space.

The primary comparison assessed the agreement between the raw GCN-generated embedding and the corresponding original target-modality embedding. Four metrics were calculated for each eligible patient: cosine similarity, mean squared error (MSE), mean absolute error (MAE) and relative embedding-norm error. Let $\widehat{\mathbf{x}}_i \in \mathbb{R}^d$ denote the GCN-generated embedding and $\mathbf{x}_i \in \mathbb{R}^d$ the corresponding original embedding for patient i. The metrics were defined as

$$\mathrm{Cosine}_i = \frac{\widehat{\mathbf{x}}_i^\top \mathbf{x}_i}{\|\widehat{\mathbf{x}}_i\|_2 \|\mathbf{x}_i\|_2}, \tag{4.31}$$

$$\mathrm{MSE}_i = \frac{1}{d}\sum_{j=1}^{d}(\widehat{x}_{ij} - x_{ij})^2, \tag{4.32}$$

$$\mathrm{MAE}_i = \frac{1}{d}\sum_{j=1}^{d}|\widehat{x}_{ij} - x_{ij}|, \tag{4.33}$$

$$\mathrm{RelativeNormError}_i = \frac{|\|\widehat{\mathbf{x}}_i\|_2 - \|\mathbf{x}_i\|_2|}{\max(\|\mathbf{x}_i\|_2, \varepsilon)}. \tag{4.34}$$

Here, ε is a small constant for numerical stability. Higher cosine similarity and lower values of the remaining three metrics indicate greater agreement between the estimated and original embeddings. Cosine similarity measures directional agreement, while MSE and MAE quantify differences in individual embedding components. Relative norm error assesses the discrepancy in embedding magnitude. The same metrics were also calculated for the final refined embedding to distinguish the quality of the direct GCN output from the representation subsequently used for multimodal fusion.

To investigate the influence of contrastive learning (CL), imputation quality was compared between M3Care models trained with and without the CL extension. For each target modality, results were matched at the patient level, and the difference in GCN-to-original cosine similarity was calculated between the two models:

$$\Delta_i^{(m)} = \mathrm{Cosine}_{i,\mathrm{CL}}^{(m)} - \mathrm{Cosine}_{i,\mathrm{baseline}}^{(m)}. \tag{4.35}$$

Positive values indicated improved directional agreement following CL. The patient-level differences were summarised using means, medians and standard deviations. Normal-approximation 95% CIs for the mean difference were calculated as

$$\mathrm{CI}_{95\%}^{(m)} = \overline{\Delta}^{(m)} \pm 1.96\frac{s_\Delta^{(m)}}{\sqrt{N_m}}, \tag{4.36}$$

where $\overline{\Delta}^{(m)}$ is the mean paired difference, $s_\Delta^{(m)}$ is its sample standard deviation, and N_m denotes the number of eligible paired patients. This uncertainty-estimation procedure differed from the BCa bootstrap CIs used for predictive performance evaluation.

For models employing learned cross-modal targets, further analyses examined three relationships: the learned target against the original observed embedding, the GCN-generated embedding against the learned target, and the GCN-generated embedding against the original observed embedding. The first comparison assessed the quality of the learned cross-modal target, the second investigated

whether the GCN output aligned with its contrastive supervision target, and the third measured the actual imputation quality. These comparisons were conducted in the original embedding space using reconstruction metrics and, where applicable, in the contrastive projection space using cosine similarity, patient retrieval accuracy and symmetric InfoNCE loss. Patient retrieval accuracy measured whether the corresponding target representation achieved the highest similarity among the eligible candidates within the evaluation batch.

The analyses were conducted separately for selected target modalities and observation-window configurations in the deterioration and hospitalisation pathways. Metrics were calculated for individual eligible patients before aggregation to avoid disproportionately weighting larger batches. Nevertheless, M3Care constructs patient similarity graphs dynamically within processing batches, meaning that the estimated representations may depend on the characteristics of other patients in the batch. Similarly, contrastive retrieval accuracy depends on the number and composition of eligible candidates. These factors were considered when interpreting differences in imputation quality between models and modalities.

4.6 Fairness evaluation

Building on the fairness considerations discussed in 2.4, we evaluated whether model predictions differed across clinically and demographically relevant sensitive attributes or patient groups. We identified three demographic attributes in both pathways: sex, age, and ethnicity. Sex was divided into male and female groups, age into five groups (under 30, 30-49, 50-64, 65-79 and 80+), and ethnicity into White, Black, Hispanic, Asian and Other. For the hospitalisation pathway, we additionally evaluated insurance type across Medicare, Private, Medicaid and Other groups[1]. For the deterioration pathway, we also stratified by the pathway-related grouping of chest X-ray availability.

All fairness metrics were calculated from predictions on the test set. A probability threshold of 0.5 was applied consistently across groups. We focused on the group-level criteria described in Section 2.4: whether positive predictions were made at similar rates, and whether prediction errors were distributed similarly between groups. These criteria were quantified using three ratio-based measures.

Let A denote the sensitive attribute, $\mathcal{A}$ the set of its possible values, and $a \in \mathcal{A}$ a particular subgroup.

Demographic Parity Ratio (DPR)

$$\mathrm{DPR} = \frac{\min_{a\in\mathcal{A}} P(\hat{Y} = 1 \mid A = a)}{\max_{a\in\mathcal{A}} P(\hat{Y} = 1 \mid A = a)} \tag{4.37}$$

Equal Opportunity Ratio (EOP)

$$\mathrm{EOP} = \frac{\min_{a\in\mathcal{A}} P(\hat{Y} = 1 \mid Y = 1, A = a)}{\max_{a\in\mathcal{A}} P(\hat{Y} = 1 \mid Y = 1, A = a)} = \frac{\min_{a\in\mathcal{A}} \mathrm{TPR}_a}{\max_{a\in\mathcal{A}} \mathrm{TPR}_a} \tag{4.38}$$

Equalised Odds Ratio (EOR)

$$\mathrm{EOR} = \min\left(\frac{\min_{a\in\mathcal{A}} \mathrm{TPR}_a}{\max_{a\in\mathcal{A}} \mathrm{TPR}_a}, \frac{\min_{a\in\mathcal{A}} \mathrm{FPR}_a}{\max_{a\in\mathcal{A}} \mathrm{FPR}_a}\right) \tag{4.39}$$

[1]The separate No Charge category contained only a few patients and was therefore excluded from the group comparison.

where

$$\mathrm{TPR}_a = P(\hat{Y} = 1 \mid Y = 1, A = a) \tag{4.40}$$

and

$$\mathrm{FPR}_a = P(\hat{Y} = 1 \mid Y = 0, A = a) \tag{4.41}$$

For all three ratios, a value of 1 indicates perfect parity across groups, while a value of 0 indicates complete disparity. These measures were calculated using the `Fairlearn` library, which was inherited from the previous **MM-HealthFair** study for subgroup fairness analysis.

Fairness estimates can also become unstable when some subgroups contain only a few positive cases. We therefore reported 95% **bias-corrected and accelerated (BCa)** confidence intervals for every fairness metric using 1,000 bootstrap resamples. BCa adjusts for bias and skewness in the bootstrap distribution, which is useful for bounded ratio metrics when estimates cluster near 0 or 1. Samples were drawn with replacement within each subgroup, while preserving the original subgroup size. This maintained the subgroup structure of the test set. The same approach was used in the previous **MM-HealthFair** study for uncertainty estimation of fairness metrics.

4.7 Multi-modal explanations

We applied the explainability module after training to examine how each model used its inputs to make a prediction. We organised the analysis at three levels of granularity: modality contribution, individual features, and spatial attribution within chest X-rays. The main explanation pipeline was implemented in `explain.py`, while spatial explanations were implemented separately in `explain_cxr.py`.

Two of these analyses were based on Shapley values. Shapley-value-based attribution, particularly `SHAP` library [49], is widely adopted for post-hoc explanation of ML models. It builds on cooperative game theory. Model inputs are treated as "players", and the prediction is attributed according to each input's marginal contribution across different combinations of the remaining inputs. Broadly speaking, the "players" may represent modalities, individual features, or other components of the input. In this study, we applied this principle at both modality and individual feature levels. The estimation strategy varied between them because the number and granularity of the inputs were substantially different.

Modality-level contribution

We adapted the idea behind MM-SHAP [50] to estimate relative modality contributions. Earlier **MM-HealthFair** study estimated modality importance by applying DeepSHAP to separate modality encoders before aggregating the resulting attributions. Here, we treated each modality as a Shapley player. Our implementation reused the existing missing-modality infrastructure: modalities excluded from a given combination were masked using the same mechanism used during model training. This allowed us to evaluate the trained fusion model directly across different modality combinations.

Formally, let $\mathcal{M}$ denote the set of modalities used by the model and $S \subseteq \mathcal{M}$ a subset of those modalities. For a sample n, we define $v^{(n)}(S)$ as the predicted probability when only the modalities in S are available. The Shapley value of modality $m \in \mathcal{M}$ for encounter n is then calculated as:

$$\phi_m^{(n)} = \sum_{S \subseteq \mathcal{M} \setminus \{m\}} \frac{|S|!(|\mathcal{M}| - |S| - 1)!}{|\mathcal{M}|!} [v^{(n)}(S \cup \{m\}) - v^{(n)}(S)] \tag{4.42}$$

The term $v^{(n)}(S \cup \{m\}) - v^{(n)}$ measures how much the predicted probability changes when modality m is added to a particular combination S. The factorial term weights changes according to the number of possible modality orderings associated with that subset. We evaluated all possible modality combinations. The resulting Shapley values can be written as:

$$\sum_{m \in \mathcal{M}} \phi_m^{(n)} = v^{(n)}(\mathcal{M}) - v^{(n)}(\varnothing) \tag{4.43}$$

where $v^{(n)}(\varnothing)$ is the predicted probability for sample n when no modalities are available. To compare modality contributions on a common scale, we then normalised the absolute Shapley values within each sample:

$$C_m^{(n)} = \frac{|\phi_m^{(n)}|}{\sum_{k \in \mathcal{M}} |\phi_k^{(n)}|}, \qquad C_m = \frac{1}{N} \sum_{n=1}^{N} C_m^{(n)} \tag{4.44}$$

Here, $C_m^{(n)}$ is the relative contribution of modality m for sample n, while C_m is its mean contribution across the N explained samples. The relative contributions sum to one within each sample. This allowed modality contributions to be compared directly across models and prediction tasks.

Feature-level importance

We applied the same Shapley attribution principle at the level of individual structured features. Unlike the modality-level analysis, however, evaluating every possible feature combination was not computationally practical. We therefore used the `SHAP` library [49] to estimate the contribution of each feature to the predicted outcome. For the deterioration pathway, the explained inputs included the tabular features together with the summary time-series features, where applicable. For hospitalisation, the analysis focused on the static features. Only these structured inputs were perturbed during feature explanation, while other modalities were held fixed at their observed values for the same sample.

We used `GradientExplainer` as the default SHAP estimator. It approximated feature contributions using expected gradients, which were calculated along paths between a set of background samples and the inputs being explained [92]. Across our fusion methods, this was substantially more computationally efficient than repeatedly evaluating all feature perturbations with `KernelExplainer`. However, the latter remained available as a model-agnostic alternative and was generally more accurate.

For a sample with structured input x, the resulting explanation can be expressed as

$$\hat{p}^{(n)} \approx \phi_0^{(n)} + \sum_{j=1}^{F} \phi_j^{(n)}, \qquad \phi_0^{(n)} = \frac{1}{|B|} \sum_{x' \in B} \hat{p}(x';\, z^{(n)}) \tag{4.45}$$

where $\hat{p}^{(n)}$ is the predicted outcome probability for sample n and F is the number of explained features. The value $\phi_j^{(n)}$ represents the estimated contribution of feature j to the prediction. The

reference value $\phi_0^{(n)}$ is the mean predicted probability over the background set B, which contained $|B|$ training samples. During this calculation, all other modalities (denoted by $z^{(n)}$) remained fixed at the observed values of the sample being explained.

The sign of attribution $\phi_j^{(n)}$ represents feature j's direction of influence. A positive value means that the feature increased the predicted probability relative to the background reference, whereas a negative value indicates that it reduced the prediction. The magnitude reflects the strength of feature j's contribution.

Because `GradientExplainer` estimates expected gradients from a finite number of sampled paths, the feature contributions do not necessarily reconstruct the original prediction exactly. This is reflected by the approximation in Equation 4.45. The routing of sparse MoE can introduce an additional source of discrepancy. A small perturbation to an input feature may change the selected top-k experts, producing a discrete change in the model output that is not fully captured by local gradients.

Our pipeline examined these feature attributions at both global and local levels. For global interpretation, features were ranked by their mean absolute contribution across the explained test samples:

$$I_j = \frac{1}{N} \sum_{n=1}^{N} |\phi_j^{(n)}| \tag{4.46}$$

where I_j denotes the global importance of feature j and N is the total number of explained samples. These values were visualised using SHAP summary plots. Local explanations retained the signed values $\phi_j^{(n)}$ for an individual sample and were displayed using decision plots. Each plot started from the background reference $\phi_0^{(n)}$ and added the estimated feature contributions sequentially. Its endpoint therefore corresponded to the reconstructed value $\phi_0^{(n)} + \sum_{j=1}^{F} \phi_j^{(n)}$ on the right-hand side of Equation 4.45, which may differ slightly from the model's original prediction.

Spatial explanations

The fusion models operated on pre-computed embeddings rather than raw chest X-ray images. As a result, the gradients of the model outputs could not be propagated back to the image pixels. For mitigation, we used a widely adopted gradient-based explanation method called **Grad-CAM** [59]. **Grad-CAM** identifies image regions associated with a prediction by weighting the final convolutional feature maps according to their influence on the target output. This post-hoc approach allowed us to propagate a prediction back to raw images without retraining the model.

Each preprocessed chest X-ray was passed through the same pre-trained DenseNet121 to generate the original embeddings. The model weights remained frozen, while gradients with respect to intermediate activations were retained for explanation. For image r from sample n, let $A_c^{(n,r)} \in \mathbb{R}^{H \times W}$ denote channel c of the final convolutional feature map, where $c = 1, \ldots, C$. In our implementation, $C = 1024$ and $H = W = 7$.

The feature maps were passed through global average pooling to recover the image embedding used by the original pipeline. Where multiple images contributed to a single sample, their embeddings were averaged before applying the same normalisation and encoding used during model training. The resulting representation was passed through the trained fusion model, together with the other

observed modalities. For each image, **Grad-CAM** first calcualted the importance of feature-map channel c from the gradient of the model's outcome logit $\ell^{(n)}$ with respect to that channel:

$$\alpha_c^{(n,r)} = \frac{1}{HW} \sum_{h=1}^{H} \sum_{w=1}^{W} \frac{\partial \ell^{(n)}}{\partial A_c^{(n,r)}[h,w]} \tag{4.47}$$

Here, $\alpha_c^{(n,r)}$ represents the importance of channel c for image r from sample n. The corresponding **Grad-CAM** heatmap was then calculated as:

$$L_{\text{GradCAM}}^{(n,r)} = \text{ReLU}(\sum_{c=1}^{C} \alpha_c^{(n,r)} A_c^{(n,r)}) \tag{4.48}$$

In addition to conventional **Grad-CAM**, we implemented a SHAP-weighted variant to relate the spatial explanation to the contribution of the CXR embedding. Let $e_{\text{CXR}}^{(n)} \in \mathbb{R}^C$ denote the sample-level CXR embedding before it entered the CXR encoder of the fusion model. We applied `GradientExplainer` to this representation and estimated a SHAP value $\phi_c^{(n)}$ for each embedding dimension c. These values indicate how strongly each CXR embedding dimension contributed to the model output relative to the background distribution. The SHAP-weighted spatial map for image r was then calculated as:

$$L_{\text{SHAP}}^{(n,r)} = \text{ReLU}(\sum_{c=1}^{C} \phi_c^{(n)} A_c^{(n,r)}) \tag{4.49}$$

The resulting 7×7 heatmaps were normalised to $[0, 1]$ and resized to 224×224 using bilinear interpolation, before being overlaid on the corresponding X-ray images. These maps show which image regions were associated with the model output. However, as their spatial resolution is limited by the final DenseNet121 feature map and the model was trained for outcome prediction rather than lesion localisation, they should never be interpreted as accurate anatomical localisation.

M3Care-specific explanation handling

M3Care required additional handling for model explanation. This was due to its missing-modality imputation that relied on similarities between samples within the same batch. As a result, the prediction from one sample can depend partially on the representation of other samples in that batch. Standard SHAP evaluations perturb one patient at a time, which would undermine the imputation process.

For M3Care models with imputation enabled, we therefore implemented a frozen-batch procedure. The original test batch was first passed through the model to construct the patient-similarity graph and obtain the representations of the remaining samples. During explanation of sample n, only the target input was perturbed. The graph and the representations of the other samples were held constant.

Let $\mathcal{G}_B$ denote the similarity graph constructed for batch B, and let $H_{B \setminus n}$ denote the stored representations of all samples in the batch other than sample n. Each perturbed prediction can then be represented as:

$$\hat{p}^{(n)}(\tilde{x}^{(n)}; \mathcal{G}_B, H_{B\setminus n}) \tag{4.50}$$

where $\tilde{x}^{(n)}$ denotes the perturbed input for the sample being explained.

Our mitigation preserved the context used by M3Care during the original prediction, while still allowing the target sample to be perturbed for SHAP estimation. The procedure was only required when M3Care's similarity-based imputation was enabled. Other fusion methods were explained through their standard forward pass.

4.8 Implementation tools

The **mm-healthfusion** framework was designed to support flexible multi-modal clinical prediction, providing a shared pipeline for data preparation, feature extraction, model training, evaluation and explanation. The framework was implemented in Python (3.10.20) with *PyTorch* (2.5.1) (CUDA 12.1 build) used for embedding extraction, model training, evaluation and explanation. All experiments were conducted on a Microsoft Azure virtual machine running Ubuntu 22.04 LTS with the following specifications: 1 x NVIDIA Tesla T4 GPU and 4 x vCPUs (28 GiB memory). Each experiment was controlled through a YAML configuration file. A resolved `config.yaml` was stored alongside the corresponding model checkpoint upon training. Key packages used to create this pipeline are described below.

For data extraction and preparation, we used *Polars* (0.20.31) library[2] for efficient scanning of the large MIMIC-IV event tables. Pre-trained encoders were applied to the source records with their weights frozen, and the resulting embeddings were stored for downstream modelling. Specifically, clinical notes were processed using *spaCy* (3.8.15) library[3] with the biomedical language model `en_core_sci_md` (0.5.4) [89]. Text embeddings were subsequently extracted using the pre-trained `Bio_Discharge_Summary_BERT` [88], loaded through Hugging Face Transformers (4.57.6). For CXR images, we used the DenseNet-121 encoder from *TorchXRayVision*(1.4.0) [4] [86], with the pre-trained weights `densenet121-res224-chex`. Image loading and resizing were handled by scikit-image and OpenCV. ECG embeddings were extracted using ECG-FM [87], with the `mimic_iv_ecg_physionet_pretrained.pt` checkpoint obtained from Hugging Face[5] and loaded through *fairseq-signals* (1.0.0a0) [6] framework.

Model training was implemented using *PyTorch Lightning* (2.6.1)[7], with a shared LightningModule supporting all fusion architectures. *Weights & Biases* library (0.27.0) [8] was available for experiment tracking. Predictive performance metrics (AUROC, MCC, recall and specificity) were implemented using *TorchMetrics* (1.4.0) [9], alongside a custom balanced accuracy metric. Fairness analysis used *Fairlearn* (0.12.0) [10] [75].

[2]Polars: https://pola.rs
[3]spaCy: https://allenai.github.io/scispacy
[4]TorchXRayVision: https://github.com/mlmed/torchxrayvisionnote
[5]ECG-FM: https://huggingface.co/wanglab/ecg-fm
[6]fairseq-signals: https://github.com/Jwoo5/fairseq-signals
[7]PyTorch Lightning: https://lightning.ai/docs
[8]Weights & Biases: https://wandb.ai
[9]TorchMetrics: https://lightning.ai/docs/torchmetrics
[10]Fairlearn: https://fairlearn.org

For model explanation, we used *SHAP* (0.47.2)[11] [49], with `GradientExplainer` as the default estimator. Grad-CAM and SHAP-weighted Grad-CAM were implemented separately using `torch.autograd` to calculate gradients through the frozen DenseNet encoder. Random seeds were controlled using Lightning's `seed_everything`. Seed 42 was specified in any standard model configurations and evaluations, with the exception of hyperparameter stability assessment using seeds 43 and 44 additionally. The MCC permutation tests used a fixed seed of 0.

[11] SHAP: https://shap.readthedocs.io

Chapter 5

Results

5.1 Cohort summary

The cohort selection pipelines for deterioration and hospitalisation pathways are summarised in Figure 5.1.

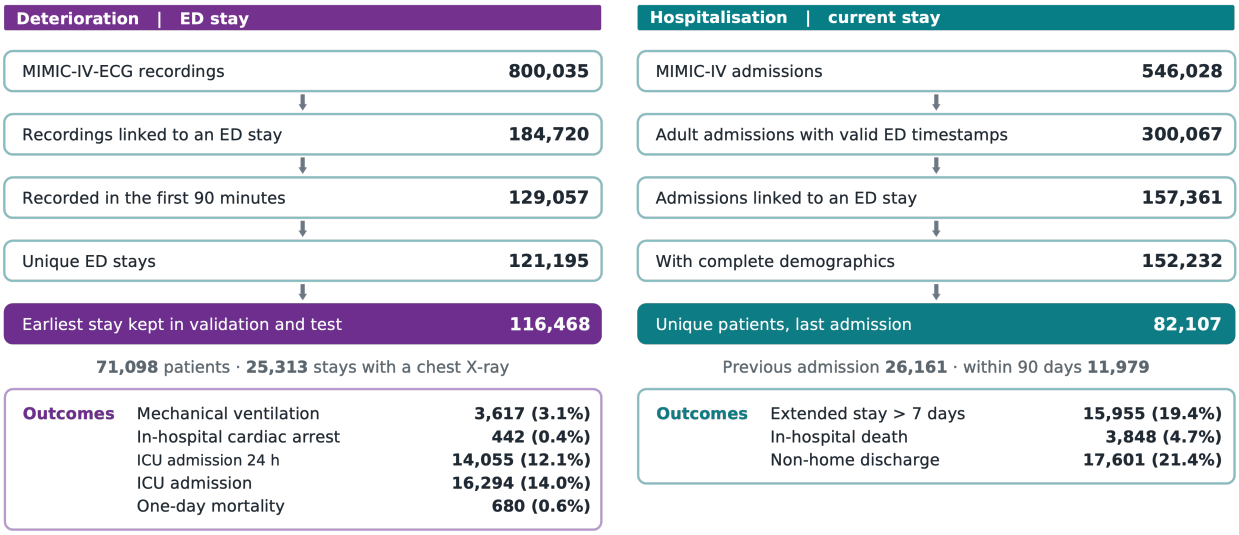


Figure 5.1: Overview of the hospitalisation and deterioration cohort selection

In the deterioration pathway, 129,057 of the initial 800,035 ECG recordings were obtained within the first 90-minute window of an ED stay. This corresponded to 121,195 unique ED stays. Among those, we followed the **MDS-ED** data partitioning strategy based on **MIMIC-IV-ECG-Ext-ICD**, and retained only the earliest eligible stay per patient in the validation and test sets. The final cohort comprised 116,468 ED stays from 71,098 patients.

For hospitalisation pathway, the initial **MIMIC-IV** admissions were reduced to 82,107 patients, each represented by their most recent eligible admission. Previous admissions were identified for 26,161 patients (31.9%), but only 11,979 (14.6% of the full cohort) had an admission within the 90-day lookback window. Compared with the 20,596 patients in **MM-HealthFair**, the hospitalisation cohort was substantially larger. This increase largely reflected the removal of earlier inclusion

requirements for previous admissions, previous clinical notes and a minimum number of time-series measurements.

Table 5.1: Baseline characteristics and outcomes of the deterioration cohort

Variable	All (n=116,468)	MV (n=3,617)	CA (n=442)	ICU-24h (n=14,055)	ICU (n=16,294)	MORT-1D (n=680)
Age, median [Q1,Q3]	63 [49,76]	66 [53,78]	67 [55,79]	68 [56,80]	68 [56,80]	76 [64,86]
Women, n (%)	62,188 (53.4%)	1,564 (43.2%)	168 (38.0%)	6,437 (45.8%)	7,425 (45.6%)	342 (50.3%)
Ethnicity, n (%)						
Asian	4,302 (3.7%)	104 (2.9%)	12 (2.7%)	496 (3.5%)	569 (3.5%)	25 (3.7%)
Black	24,767 (21.3%)	462 (12.8%)	51 (11.5%)	1,982 (14.1%)	2,316 (14.2%)	66 (9.7%)
Hispanic/Latino	8,361 (7.2%)	105 (2.9%)	8 (1.8%)	517 (3.7%)	614 (3.8%)	16 (2.4%)
White	70,742 (60.7%)	1,861 (51.5%)	208 (47.1%)	9,035 (64.3%)	10,647 (65.3%)	399 (58.7%)
Other	8,296 (7.1%)	1,085 (30.0%)	163 (36.9%)	2,025 (14.4%)	2,148 (13.2%)	174 (25.6%)
Triage acuity, n (%)						
1	14,346 (12.3%)	1,445 (40.0%)	143 (32.4%)	5,048 (35.9%)	5,331 (32.7%)	292 (42.9%)
2	58,868 (50.5%)	856 (23.7%)	45 (10.2%)	6,374 (45.4%)	7,853 (48.2%)	179 (26.3%)
3	40,341 (34.6%)	104 (2.9%)	4 (0.9%)	990 (7.0%)	1,445 (8.9%)	22 (3.2%)
4 or 5	310 (0.3%)	1 (0.0%)	0 (0.0%)	1 (0.0%)	1 (0.0%)	0 (0.0%)
Missing	2,603 (2.2%)	1,211 (33.5%)	250 (56.6%)	1,642 (11.7%)	1,664 (10.2%)	187 (27.5%)
BMI, median [Q1,Q3]	27.6 [23.8,32.6]	26.5 [22.7,31.5]	25.5 [22.9,30.7]	26.9 [23.2,31.6]	27.1 [23.4,31.8]	26.1 [21.9,30.2]
Modality availability, n (%)						
ECG	116,468 (100.0%)	3,617 (100.0%)	442 (100.0%)	14,055 (100.0%)	16,294 (100.0%)	680 (100.0%)
Chest X-ray	25,313 (21.7%)	1,749 (48.4%)	233 (52.7%)	5,060 (36.0%)	5,580 (34.2%)	308 (45.3%)
Vital signs or lab tests	107,289 (92.1%)	3,582 (99.0%)	434 (98.2%)	13,882 (98.8%)	16,014 (98.3%)	675 (99.3%)

Outcomes: MV: mechanical ventilation, CA: in-hospital cardiac arrest, ICU-24h: ICU admission within 24 hours, ICU: ICU admission during the hospital stay, MORT-1D: one-day mortality.
Triage acuity is the Emergency Severity Index, from 1 (most urgent) to 5 (least urgent).
99,785 stays (85.7%) had none of the five outcomes and a stay with several outcomes is counted in each of its columns.

The outcome prevalence differed considerably between the two pathways (deterioration: Table 5.1, hospitalisation: Table 5.2). In deterioration, individual prevalence ranged from 0.4% for cardiac arrest to 14.0% for ICU admission. Cardiac arrest and one-day mortality were particularly rare, with 442 and 680 positive cases respectively. The hospitalisation pathway had a more balanced outcome distribution. Extended stay, in-hospital death and non-home discharge occurred in 19.4%, 4.7% and 21.4% of patients respectively. Outcome groups could overlap, although in-hospital death and non-home discharge were mutually exclusive by definition. Overall, class imbalance was evident across both prediction pathways, supporting the use of imbalance-sensitive evaluation metrics (Section 4.5).

Modality availability also varied across pathways and outcomes. In deterioration, ECG was available for all stays by cohort definition, and most stays had vital signs or laboratory measurements. Chest X-rays coverage was much lower at 21.7%, although it was more frequent among outcome-positive stays, reaching 52.7% for cardiac arrest. A similar pattern was observed in hospitalisation, where current-window vital signs and laboratory measurements were widely available, but ECG and CXR were less common and more frequently present among patients with adverse outcomes. Previous-window modalities were particularly sparse, consistent with the small proportion of patients who had a previous admission within the lookback window.

The two cohorts therefore provided distinct prediction settings, differing in outcome prevalence, observation windows and modality coverage. Chest X-ray was frequently missing in both pathways, while previous-window information was limited in hospitalisation. Moreover, the higher availability of certain modalities among patients with adverse outcomes raised the question of whether their predictive contribution reflected the clinical information they contained or simply their availability. This distinction was examined in Section 5.6 below. The subsequent fairness analysis in Section 5.7

Table 5.2: Baseline characteristics and outcomes of the hospitalisation cohort

Variable	All (n=82,107)	EXT-ST (n=15,955)	IHD (n=3,848)	NHD (n=17,601)
Age, median [Q1,Q3]	61 [46,74]	66 [53,77]	73 [61,83]	73 [61,83]
Women, n (%)	42,148 (51.3%)	7,614 (47.7%)	1,817 (47.2%)	9,499 (54.0%)
Ethnicity, n (%)				
Asian	3,260 (4.0%)	613 (3.8%)	179 (4.7%)	461 (2.6%)
Black	11,014 (13.4%)	2,122 (13.3%)	486 (12.6%)	1,937 (11.0%)
Hispanic/Latino	4,162 (5.1%)	685 (4.3%)	126 (3.3%)	511 (2.9%)
White	58,602 (71.4%)	11,427 (71.6%)	2,767 (71.9%)	13,657 (77.6%)
Other	5,069 (6.2%)	1,108 (6.9%)	290 (7.5%)	1,035 (5.9%)
Marital status, n (%)				
Divorced	6,321 (7.7%)	1,373 (8.6%)	281 (7.3%)	1,532 (8.7%)
Married	36,512 (44.5%)	6,741 (42.3%)	1,735 (45.1%)	6,530 (37.1%)
Single	29,043 (35.4%)	5,534 (34.7%)	1,008 (26.2%)	5,395 (30.7%)
Widowed	10,231 (12.5%)	2,307 (14.5%)	824 (21.4%)	4,144 (23.5%)
Insurance, n (%)				
Medicaid	13,729 (16.7%)	2,646 (16.6%)	431 (11.2%)	2,007 (11.4%)
Medicare	38,701 (47.1%)	9,089 (57.0%)	2,760 (71.7%)	12,722 (72.3%)
Private	27,050 (32.9%)	3,705 (23.2%)	598 (15.5%)	2,480 (14.1%)
Other	2,627 (3.2%)	515 (3.2%)	59 (1.5%)	392 (2.2%)
Triage acuity, n (%)				
1	8,988 (10.9%)	2,427 (15.2%)	1,145 (29.8%)	2,685 (15.3%)
2	38,714 (47.2%)	8,194 (51.4%)	1,763 (45.8%)	9,153 (52.0%)
3	31,602 (38.5%)	4,536 (28.4%)	509 (13.2%)	4,911 (27.9%)
4 or 5	320 (0.4%)	27 (0.2%)	0 (0.0%)	46 (0.3%)
Missing	2,483 (3.0%)	771 (4.8%)	431 (11.2%)	806 (4.6%)
BMI, median [Q1,Q3]	27.1 [23.4,31.7]	26.6 [22.8,31.4]	25.6 [22.1,29.9]	26.0 [22.5,30.7]
Comorbidity history, n (%)				
$\geq$2 conditions	26,171 (31.9%)	6,610 (41.4%)	2,136 (55.5%)	7,473 (42.5%)
$\geq$4 conditions	15,762 (19.2%)	4,505 (28.2%)	1,572 (40.9%)	5,121 (29.1%)
$\geq$1 physical and $\geq$1 mental	10,859 (13.2%)	2,763 (17.3%)	806 (20.9%)	3,171 (18.0%)
Previous admission, n (%)				
Any	26,161 (31.9%)	6,557 (41.1%)	1,995 (51.8%)	6,976 (39.6%)
Within the 90-day lookback band	11,979 (14.6%)	3,560 (22.3%)	1,221 (31.7%)	3,684 (20.9%)
Current window				
Length in hours, median [Q1,Q3]	5.0 [3.2,7.6]	4.8 [2.9,7.3]	3.6 [2.1,5.9]	4.8 [3.0,7.4]
Vital signs or lab tests, n (%)	80,981 (98.6%)	15,792 (99.0%)	3,804 (98.9%)	17,395 (98.8%)
ECG, n (%)	36,433 (44.4%)	7,659 (48.0%)	2,519 (65.5%)	9,066 (51.5%)
Chest X-ray, n (%)	17,091 (20.8%)	3,693 (23.1%)	1,501 (39.0%)	4,168 (23.7%)
Previous window, n (%)				
Vital signs or lab tests	11,978 (14.6%)	3,560 (22.3%)	1,221 (31.7%)	3,684 (20.9%)
Discharge note	9,530 (11.6%)	2,809 (17.6%)	924 (24.0%)	2,948 (16.7%)
Chest X-ray	4,316 (5.3%)	1,299 (8.1%)	625 (16.2%)	1,503 (8.5%)

Outcomes: EXT-ST: extended hospital stay ($>$7 days); IHD: in-hospital death; NHD: non-home discharge.
Triage acuity is the Emergency Severity Index, from 1 (most urgent) to 5 (least urgent), with levels 4 and 5 merged.
Comorbidity history is derived from diagnoses recorded at earlier admissions.
Current window: the ED phase of the index admission. Previous window: the previous admission, used when it ended within 90 days of the index ED arrival.
54,004 patients (65.8%) had none of the three outcomes and a patient with several outcomes is counted in each of their columns.

also considered differences in model performance across patient groups, including those defined by CXR availability.

5.2 Main comparison

This section compares the predictive performance of the three multimodal fusion frameworks for the deterioration and hospitalisation pathways, with the latter further divided into current-window

Table 5.3: Best model configurations for M3Care and FuseMoE across the deterioration and hospitalisation pathways.

Pathway	M3Care	FuseMoE
Deterioration	GCN-based imputation	Joint router
	Imputed-for-missingness refinement	Top-k expert selection ($k = 2$)
	Gaussian similarity-based graph stability loss	Number of experts: 7
	Weight: 0.01	Load-balancing loss weight: 0.1
	Mask missingness during fusion	Shared-expert weight: 0.333
		Modality dropout rate: 0.1
Hospitalisation	GCN-based imputation	Joint router
	Imputed-for-missingness refinement	Top-k expert selection ($k = 2$)
	Norm-based graph stability loss	Number of experts: 7
	Weight: 0.01	Load-balancing loss weight: 0.1
	Mask missingness during fusion	Shared-expert weight: 0.333
		Modality dropout rate: 0 (disabled)

and dual-window configurations. The comparisons focus primarily on the Matthews correlation coefficient (MCC), supported by AUROC, balanced accuracy (BA), recall and specificity.

The selected configurations of M3Care and FuseMoE, identified through the model configuration ablation studies, are summarised in Table 5.3. For M3Care, both pathways adopted GCN-based modality imputation with imputed-for-missingness refinement and missingness-masked fusion. Gaussian similarity-based and norm-based graph stability losses were selected for deterioration and hospitalisation respectively. For FuseMoE, the preferred configurations were largely consistent across pathways, employing a joint router, seven experts with top-2 selection, and a shared-expert weight of 0.333. Modality dropout was set at 0.1 for ECG and CXR in the deterioration pathway and disabled in the hospitalisation pathway in the best models.

5.2.1 Deterioration

Table 5.4 summarises the test-set performance across the five deterioration outcomes. Overall, no single fusion framework consistently achieved the highest performance across all prediction tasks. FuseMoE obtained the highest MCC for three outcomes: ICU admission during the hospital stay (0.552), mechanical ventilation (0.574) and one-day mortality (0.223). M3Care achieved the highest MCC for in-hospital cardiac arrest (0.440), while concatenation performed best for ICU admission within 24 hours (0.597).

The differences between fusion architectures varied considerably across outcomes. For mechanical ventilation, FuseMoE achieved an MCC of 0.574, compared with 0.498 for concatenation and 0.542 for M3Care. Both advanced fusion frameworks showed statistically significant differences from concatenation according to the BH-adjusted paired MCC permutation tests. In contrast, for ICU admission within 24 hours, concatenation achieved the highest MCC (0.597), outperforming FuseMoE (0.573) and M3Care (0.548). The differences for both advanced architectures were also statistically significant, indicating that increased fusion complexity did not necessarily improve predictive performance.

For in-hospital cardiac arrest, M3Care achieved the highest MCC (0.440), compared with 0.349 for concatenation and 0.318 for FuseMoE. It also achieved the highest recall (0.957) and BA (0.967), suggesting a greater ability to identify positive cases at the selected classification threshold. However, these improvements were not statistically significant in the paired MCC comparisons. Similarly, for ICU admission during the hospital stay, the MCC values were relatively close across all three architectures, ranging from 0.543 to 0.552, without statistically significant differences from concatenation.

For one-day mortality, FuseMoE achieved the highest MCC (0.223), although all three models showed relatively low MCC values compared with the other deterioration outcomes. Despite achieving high AUROC values of approximately 0.95, the models demonstrated more limited performance according to MCC, indicating that strong overall discrimination did not necessarily translate into equally strong binary classification performance at the fixed decision threshold.

Across the five outcomes, the models exhibited different trade-offs between recall and specificity. For example, FuseMoE achieved the highest specificity for mechanical ventilation (0.962), while concatenation achieved the highest recall (0.856). These differences indicate that the fusion architectures did not consistently favour the same balance between identifying positive cases and avoiding false-positive predictions.

5.2.2 Hospitalisation

Table 5.5 presents the test-set performance for the three hospitalisation outcomes under the current-window and dual-window configurations. Compared with deterioration, the results showed smaller differences between the three fusion frameworks, with concatenation generally remaining competitive and frequently achieving the highest MCC.

Under the current-window configuration, concatenation achieved the highest MCC for extended stay (0.301), while M3Care performed the best in IHD prediction, with an MCC of 0.354. For NHD, concatenation and FuseMoE achieved identical MCC values of 0.424, while M3Care obtained 0.416. None of these comparisons showed statistically significant differences from concatenation.

Under the dual-window configuration, concatenation again achieved competitive performance across all three outcomes. For EXT-ST, the three models achieved roughly similar MCC of around 0.3. For IHD, concatenation achieved the highest MCC (0.355), followed by M3Care (0.336) and FuseMoE (0.332). The difference between FuseMoE and concatenation was statistically significant. Similarly, concatenation achieved the highest MCC for NHD (0.434), compared with 0.421 for M3Care and 0.418 for FuseMoE, with the difference between FuseMoE and concatenation again reaching statistical significance.

Comparing the two observation-window configurations, the inclusion of previous-window modalities did not consistently improve predictive performance. For EXT-ST, MCC remained nearly unchanged across the two configurations for all three models. For IHD, concatenation showed a small increase in MCC from 0.346 to 0.355, whereas M3Care and FuseMoE showed decreases. For NHD, MCC increased slightly for concatenation and M3Care but decreased for FuseMoE.

Differences between MCC and other classification metrics were also observed. For example, under the current-window configuration, M3Care achieved the highest specificity for extended stay (0.823) but substantially lower recall (0.501) than concatenation (0.652). Similarly, under the dual-window configuration, FuseMoE achieved the highest recall for IHD (0.802) but lower specificity

Table 5.4: Best model performance across five deterioration outcomes for concatenation, M3Care and FuseMoE, reported with 95% bootstrap confidence intervals. The highest performance for each metric within each outcome is highlighted in bold.

Outcome	Model	MCC	AUROC	BA	Recall	Specificity
CA	Concat	0.349 [0.240–0.463]	**0.981** [0.961–0.992]	0.838 [0.738–0.936]	0.696 [0.480–0.887]	**0.980** [0.975–0.984]
	M3Care	**0.440** [0.335–0.519]	0.978 [0.940–0.994]	**0.967** [0.903–0.990]	**0.957** [0.826–1.000]	0.977 [0.971–0.981]
	FuseMoE	0.318 [0.233–0.390]	0.979 [0.960–0.991]	0.915 [0.822–0.979]	0.870 [0.681–1.000]	0.961 [0.954–0.966]
ICU-24h	Concat	**0.597** [0.560–0.635]	**0.909** [0.894–0.923]	0.814 [0.793–0.834]	0.697 [0.658–0.739]	**0.931** [0.922–0.939]
	M3Care*	0.548 [0.511–0.587]	0.895 [0.879–0.911]	0.802 [0.781–0.822]	0.695 [0.653–0.734]	0.909 [0.899–0.920]
	FuseMoE*	0.573 [0.539–0.610]	0.904 [0.889–0.918]	**0.822** [0.803–0.841]	**0.739** [0.702–0.777]	0.905 [0.895–0.915]
ICU	Concat	0.543 [0.508–0.580]	**0.894** [0.880–0.909]	0.798 [0.778–0.816]	0.698 [0.658–0.731]	0.898 [0.887–0.908]
	M3Care	0.547 [0.514–0.584]	0.883 [0.867–0.899]	0.795 [0.777–0.816]	0.685 [0.648–0.722]	**0.905** [0.894–0.915]
	FuseMoE	**0.552** [0.518–0.587]	0.890 [0.875–0.905]	**0.807** [0.788–0.826]	**0.720** [0.681–0.756]	0.894 [0.883–0.905]
MV	Concat	0.498 [0.443–0.538]	0.945 [0.916–0.965]	0.889 [0.856–0.914]	**0.856** [0.790–0.909]	0.922 [0.913–0.931]
	M3Care*	0.542 [0.488–0.589]	**0.945** [0.919–0.964]	**0.892** [0.861–0.918]	0.843 [0.783–0.895]	0.940 [0.932–0.948]
	FuseMoE*	**0.574** [0.509–0.624]	0.941 [0.911–0.962]	0.860 [0.822–0.891]	0.758 [0.685–0.821]	**0.962** [0.955–0.968]
MORT-1D	Concat	0.206 [0.144–0.257]	0.956 [0.924–0.976]	**0.871** [0.768–0.934]	**0.826** [0.621–0.950]	0.916 [0.907–0.925]
	M3Care	0.197 [0.131–0.256]	0.947 [0.906–0.973]	0.833 [0.732–0.910]	0.739 [0.542–0.889]	0.927 [0.918–0.935]
	FuseMoE	**0.223** [0.151–0.291]	**0.956** [0.925–0.976]	0.841 [0.740–0.918]	0.739 [0.542–0.889]	**0.942** [0.934–0.950]

Notes: CA: in-hospital cardiac arrest; ICU-24h: ICU admission within 24 hours; ICU: ICU admission during the hospital stay; MV: mechanical ventilation; MORT-1D: one-day mortality. MCC: Matthews correlation coefficient; AUROC: area under the receiver operating characteristic curve; BA: balanced accuracy. An asterisk (*) denotes a statistically significant paired MCC difference from concatenation after Benjamini-Hochberg correction ($p_{\mathrm{adj}} < 0.05$).

(0.829) than concatenation (0.864) and M3Care (0.882). These findings highlight differences in classification behaviour that were not fully captured by MCC or AUROC alone.

Table 5.5: Best model performance for the hospitalisation pathway under current-window and dual-window configurations, reported with 95% bootstrap confidence intervals. The highest performance for each metric within each outcome and window configuration is highlighted in bold.

Outcome	Model	MCC	AUROC	BA	Recall	Specificity
currWindow						
EXT-ST	Concat	**0.301** [0.281–0.322]	**0.740** [0.728–0.753]	**0.682** [0.671–0.695]	**0.652** [0.628–0.672]	0.713 [0.702–0.723]
	M3Care	0.300 [0.278–0.323]	0.728 [0.714–0.740]	0.662 [0.650–0.674]	0.501 [0.476–0.526]	**0.823** [0.814–0.832]
	FuseMoE	0.288 [0.267–0.310]	0.737 [0.724–0.749]	0.673 [0.661–0.686]	0.630 [0.606–0.652]	0.717 [0.705–0.727]

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Outcome	Model	MCC	AUROC	BA	Recall	Specificity
IHD	Concat	0.346 [0.317–0.373]	0.899 [0.884–0.912]	0.811 [0.789–0.832]	0.768 [0.725–0.808]	0.854 [0.847–0.862]
	M3Care	**0.354** [0.324–0.382]	0.897 [0.881–0.911]	**0.814** [0.790–0.835]	0.768 [0.722–0.808]	**0.860** [0.853–0.868]
	FuseMoE	0.339 [0.313–0.367]	**0.900** [0.883–0.912]	**0.814** [0.791–0.835]	**0.789** [0.744–0.829]	0.840 [0.832–0.848]
NHD	Concat	**0.424** [0.404–0.445]	**0.829** [0.819–0.839]	**0.750** [0.739–0.762]	**0.767** [0.748–0.786]	0.733 [0.723–0.744]
	M3Care	0.416 [0.394–0.437]	0.825 [0.815–0.835]	0.738 [0.726–0.750]	0.704 [0.681–0.726]	0.772 [0.762–0.782]
	FuseMoE	0.424 [0.404–0.448]	0.825 [0.815–0.835]	0.733 [0.722–0.747]	0.653 [0.629–0.677]	**0.814** [0.805–0.824]
dualWindow						
EXT-ST	Concat	**0.300** [0.277–0.322]	**0.737** [0.723–0.749]	0.671 [0.658–0.683]	0.557 [0.532–0.581]	**0.784** [0.775–0.794]
	M3Care	0.299 [0.275–0.320]	0.733 [0.719–0.745]	0.671 [0.656–0.683]	0.560 [0.534–0.581]	0.782 [0.771–0.791]
	FuseMoE	0.300 [0.277–0.323]	0.732 [0.719–0.745]	**0.672** [0.658–0.684]	**0.568** [0.543–0.590]	0.776 [0.766–0.786]
IHD	Concat	**0.355** [0.324–0.384]	**0.900** [0.884–0.912]	0.812 [0.788–0.832]	0.760 [0.714–0.801]	0.864 [0.856–0.872]
	M3Care	0.336 [0.305–0.367]	0.889 [0.871–0.902]	0.780 [0.753–0.802]	0.677 [0.623–0.722]	**0.882** [0.876–0.890]
	FuseMoE*	0.332 [0.307–0.358]	0.900 [0.884–0.913]	**0.815** [0.793–0.834]	**0.802** [0.761–0.840]	0.829 [0.820–0.837]
NHD	Concat	**0.434** [0.413–0.455]	**0.829** [0.820–0.839]	**0.751** [0.739–0.762]	0.736 [0.715–0.757]	**0.765** [0.756–0.776]
	M3Care	0.421 [0.400–0.442]	0.824 [0.814–0.835]	0.748 [0.736–0.760]	**0.762** [0.739–0.782]	0.735 [0.724–0.745]
	FuseMoE*	0.418 [0.397–0.441]	0.825 [0.815–0.835]	0.742 [0.730–0.755]	0.724 [0.703–0.747]	0.760 [0.750–0.771]

Notes: EXT-ST: extended hospital stay > 7 days; IHD: in-hospital death; NHD: non-home discharge. MCC: Matthews correlation coefficient; AUROC: area under the receiver operating characteristic curve; BA: balanced accuracy. An asterisk (*) denotes a statistically significant paired MCC difference from concatenation after Benjamini-Hochberg correction ($p_{\mathrm{adj}} < 0.05$).

5.3 Does the MoE routing work?

In sparse MoE models, routing determines which experts process each patient's data and the weights assigned to their outputs. In this section, we examined routing behaviour on the test set to see whether the models assigned experts in a meaningful and structured way, rather than selecting them randomly or collapsing onto a few experts. We compared the four router layouts first on mechanical ventilation task in Section 5.3.1. The joint router, which achieved the highest test MCC, was then examined across all prediction tasks in Section 5.3.2.

5.3.1 Router layouts on mechanical ventilation

Table 5.6 compares four router layouts for mechanical ventilation prediction in the test set. The joint router, which selected the top two of seven routed experts from the concatenated multimodal input alongside an always-on shared expert, achieved the highest MCC of 0.574. This was followed by the combination router, disjoint routers and per-modality routers. Specifically, the joint router showed an advantage over the other router layouts without CXR.

Table 5.6: Comparison of router layouts for mechanical ventilation prediction on the test set ($n = 3{,}644$).

Router layout	Test MCC (95% CI)	Without CXR ($n = 2{,}855$)	With CXR ($n = 789$)	Mean top routing weight	Router output share (tabular / ECG / CXR)
Joint (best run)	0.574 [0.509, 0.624]	0.534 [0.444, 0.613]	0.585 [0.490, 0.661]	0.524	n/a
Combination	0.557 [0.497, 0.610]	0.500 [0.416, 0.578]	0.587 [0.496, 0.666]	0.978	n/a
Disjoint	0.522 [0.464, 0.567]	0.460 [0.380, 0.529]	0.560 [0.474, 0.623]	n/a[†]	Without CXR: 0.67 / 0.33 With CXR: 0.39 / 0.22 / 0.39
Permod	0.509 [0.453, 0.553]	0.445 [0.367, 0.512]	0.555 [0.463, 0.619]	n/a[‡]	Without CXR: 0.64 / 0.36 With CXR: 0.32 / 0.20 / 0.49

Notes: One run per layout. All layouts differ from the joint router only in the routing mechanism, except disjoint, which uses two private experts per modality and selects one ($k = 1$). MCCs are reported with 95% BCa bootstrap confidence intervals from 1,000 resamples. Mean top routing weight is the average weight of the higher-weighted selected expert. Router output share is each available modality router's proportion of the combined output magnitude, averaged within CXR availability groups; the shares sum to one per group. This measure applies only to layouts with separate modality routers. [†] Each disjoint router selects one expert with weight 1. [‡] Permod's saved routing weights are averages across its modality routers, so a comparable mean top routing weight is not reported.

The combination router explicitly linked expert selection to modality availability, as described in Equation 4.4. Its highest-weighted expert matched the observed modality combination for every test stay. As tabular and ECG data were required in the deterioration cohort (Table 3.1), only these two combinations occurred at test time. During training, the tabular and CXR, and tabular-only combinations were encountered only through the 10% ECG/CXR modality dropout, while combinations without tabular data (ECG-only, CXR-only, ECG and CXR) were never observed. The highest-weighted expert matched the observed modality combination for every test stay. The second selected expert received almost no weight, with the top expert receiving an average weight of 0.978. Thus, the modality-combination supervision behaved as intended, with routing determined by modality availability and only four of the seven experts ever selected.

The per-modality and disjoint layouts instead used separate routers for each available modality. Their relative contributions were measured by each router's share of the combined output magnitude (Table 5.6). Without a chest X-ray, the tabular router accounted for approximately 2/3 of the output in both layouts (0.64 for per-modality and 0.67 for disjoint). When a chest X-ray was available, it received the largest share under per-modality routing (0.49), while the disjoint layout assigned equal shares to CXR and tabular data (0.39 each). ECG consistently received the lowest share, ranging from 0.20 to 0.36 across the two layouts and availability groups.

Overall, the joint router achieved the highest test MCC without explicitly assigning experts to modalities or availability patterns. It showed an advantage over other router layouts when chest X-ray was absent. The best-performing joint configuration selected the top 2 of 7 routed experts for each patient, alongside an always-on shared expert. This configuration was used for the results reported in Section 5.2. and is retained for the remainder of this report. The relative contribution of modality information and availability patterns to model performance will be discussed further in Section 5.6.

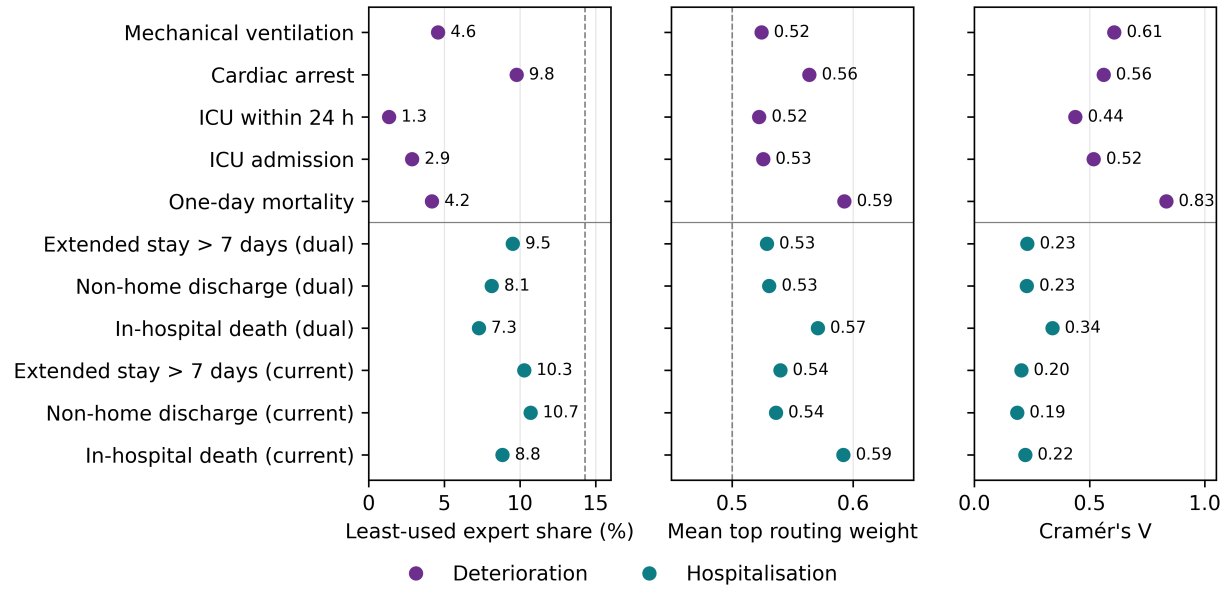


Figure 5.2: Routing behaviour of the joint router in the best run of each task in the test set. Left: share of top-2 selections assigned to the least-used of the seven routed experts, dashed line shows even use at 14.3%. Middle: mean routing weight of the higher-weighted selected expert, dashed line shows equal split at 0.5. Right: Cramér's V between the top expert and the modality-availability pattern. The deterioration cohort has two availability patterns, while the hospitalisation cohort has eight (current window) or 30 (dual window).

5.3.2 Joint routing behaviour across tasks

Figure 5.2 summarises expert usage and routing weights for the best-performing joint router in each prediction task. All seven routed experts were selected across every task, although their usage was not uniform. The least-used expert accounted for 1.3-10.7% of top-two selections, compared with 14.3% under an even distribution. The two selected experts generally received similar weights, with the mean weight of the higher-weighted expert ranging from 0.52 to 0.59 across outcomes. Routing weights varied with the load-balancing coefficient, λ. For mechanical ventilation, the mean top routing weight decreased from 0.95 at $\lambda = 0$ to 0.52 at $\lambda = 0.1$. At the latter setting, the selected experts contributed almost equally, while assignments remained distributed across all seven experts. Differences in routing between patients were therefore expressed primarily through which pair of experts was selected, rather than large differences in their assigned weights.

Expert selection was also associated with modality availability, despite the absence of explicit modality-combination supervision. Table 5.7 shows this relationship for the mechanical ventilation task. Expert 6 was the highest-weighted expert for 45.9% of stays with a CXR, compared with only 4.0% without one. Conversely, expert 0 was the most frequent top expert among stays without a CXR (31.1%), but accounted for only 11.0% of stays with a CXR. The two availability groups therefore exhibited distinct patterns of expert selection. Across tasks, this association was quantified using Cramér's V, calculated from the contingency table between the highest-weighted expert and the observed modality-availability pattern:

Table 5.7: Expert use of the joint router in the best mechanical-ventilation run, by chest X-ray availability in the test set.

	Without CXR ($n = 2{,}855$)			With CXR ($n = 789$)		
Expert	Mean routing weight	Selected (top-2)	Top expert	Mean routing weight	Selected (top-2)	Top expert
0	0.283	1,613 (56.5%)	**888 (31.1%)**	0.223	371 (47.0%)	87 (11.0%)
1	0.190	1,075 (37.7%)	643 (22.5%)	0.027	45 (5.7%)	9 (1.1%)
2	0.197	1,115 (39.1%)	580 (20.3%)	0.096	155 (19.6%)	63 (8.0%)
3	0.168	957 (33.5%)	504 (17.7%)	0.124	190 (24.1%)	109 (13.8%)
4	0.031	182 (6.4%)	30 (1.1%)	0.094	152 (19.3%)	60 (7.6%)
5	0.053	313 (11.0%)	95 (3.3%)	0.113	177 (22.4%)	99 (12.5%)
6	0.078	455 (15.9%)	115 (4.0%)	0.324	488 (61.9%)	**362 (45.9%)**
Total	1.000	5,710 (2 per stay)	2,855	1.000	1,578 (2 per stay)	789

Mean routing weight is averaged across all stays in each group, including zero for stays where the expert was not selected. Weights exclude the always-on shared expert and sum to one within each group. Selected counts the stays in which each expert was among the top two; percentages therefore sum to 200% within each group. Top expert counts stays in which the expert received the higher routing weight; percentages are relative to group size. Bold indicates the most frequent top expert in each group. Expert indices are arbitrary labels.

$$V = \sqrt{\frac{\chi^2}{n\,(\min(r, c) - 1)}} \tag{5.1}$$

where n is the number of stays, r is the number of availability patterns and c is the number of experts. Values range between 0 and 1, where 0 indicates no association and 1 indicates perfect association.

As shown in Figure 5.2, Cramér's V ranged 0.44-0.83 for deterioration outcomes and 0.19-0.34 for hospitalisation outcomes. However, the deterioration cohort contained only two availability patterns, compared with eight for the current-window and 30 for the dual-window hospitalisation cohorts. Differences in the contingency table dimensions should therefore be considered when comparing these values across cohorts.

These results show that the joint router learnt distinct expert-selection patterns associated with modality availability, without explicitly enforcing this relationship. Combined with the relatively balanced weights of the two selected experts, its routing behaviour was characterised mainly by the selection of different expert pairs. However, an association between expert selection and modality availability does not establish how much each modality contributed to the final prediction. This will be investigated separately through the modality contribution analysis in Section 5.8.1.

5.4 Does M3Care work?

5.4.1 Model architecture

To investigate the effectiveness of individual architectural components in M3Care, model configuration ablations were conducted using mechanical ventilation for the deterioration pathway and extended stay for the hospitalisation pathway. The results are summarised in Table 5.8.

Table 5.8: Ablation performance of M3Care model configuration variants for mechanical ventilation prediction in deterioration and extended hospital stay prediction in hospitalisation, reported with 95% bootstrap confidence intervals. The highest performance for each metric within each pathway is highlighted in bold.

	Deterioration: MV			**Hospitalisation: EXT-ST**		
Model configuration	**MCC**	**AUROC**	**BA**	**MCC**	**AUROC**	**BA**
Reference	0.513 [0.458–0.558]	0.940 [0.911–0.960]	0.879 [0.847–0.907]	0.291 [0.268–0.313]	0.730 [0.717–0.743]	0.670 [0.657–0.682]
w/o GCN-based imputation	0.501 [0.446–0.546]	0.944 [0.915–0.962]	0.887 [0.857–0.914]	0.273 [0.251–0.295]	0.724 [0.710–0.737]	0.665 [0.652–0.678]
w/o missingness-masked fusion	0.501 [0.452–0.548]	0.943 [0.918–0.961]	0.880 [0.851–0.908]	0.289 [0.267–0.312]	0.730 [0.717–0.743]	0.669 [0.657–0.682]
w/ missingness embedding	0.508 [0.454–0.555]	0.939 [0.905–0.960]	0.868 [0.835–0.898]	0.282 [0.263–0.305]	**0.734** [0.721–0.747]	0.671 [0.659–0.683]
Imputed-for-missingness refinement	0.535 [0.477–0.584]	0.940 [0.908–0.960]	0.861 [0.823–0.891]	**0.299** [0.275–0.320]	0.733 [0.719–0.745]	**0.671** [0.656–0.683]
+ w/ Cosine similarity-based graph stability loss	0.465 [0.417–0.508]	0.942 [0.912–0.960]	0.875 [0.844–0.904]	0.278 [0.254–0.299]	0.713 [0.699–0.727]	0.664 [0.650–0.676]
+ w/ Gaussian similarity-based graph stability loss	**0.542** [0.488–0.589]	**0.945** [0.919–0.964]	**0.892** [0.861–0.918]	0.293 [0.270–0.315]	0.733 [0.719–0.745]	0.670 [0.656–0.682]

Notes: MV: mechanical ventilation; EXT-ST: extended hospital stay > 7 days. Reference denotes the reference M3Care configuration used in these ablation experiments. The graph stability loss variants are separate comparisons with imputed-for-missingness refinement.

GCN-based imputation and missing-modality handling GCN-based modality imputation improved predictive performance in both pathways. For deterioration, removing GCN-based imputation reduced MCC from 0.513 to 0.501, while for hospitalisation, MCC decreased from 0.291 to 0.273. Disabling missingness masking during multimodal fusion also reduced MCC for both pathways. Although these differences were relatively small, the results suggest that incorporating GCN-based representations and explicitly accounting for modality availability during fusion can support predictive performance, particularly when clinical modalities are incomplete.

The choice of GCN refinement strategy also influenced model performance. Compared with adaptive refinement, imputed-for-missingness refinement raised MCC from 0.513 to 0.535 for MV and from 0.291 to 0.299 for EXT-ST, suggesting that preserving observed representations while selectively replacing missing ones may be preferable to adaptively mixing original and imputed representations in these prediction settings. Explicitly encoding modality absence through an additional learned embedding provided limited benefit beyond the existing missingness-aware mechanisms, as shown by our results with missingness embedding.

Graph stability regularisation The choice of graph stability loss affected predictive performance differently across the two pathways. For MV, Gaussian similarity-based regularisation achieved the highest MCC (0.542), compared with 0.513 for the norm-based reference configuration and 0.465 for cosine similarity-based regularisation. The model variant also achieved the highest AUROC (0.945) and BA (0.892) among the investigated stability loss variants. For EXT-ST, norm-based graph stability regularisation performed the best, with MCC of 0.299. Therefore, we selected Gaussian similarity-based regularisation for deterioration and norm-based regularisation for hospitalisation. Altering the graph stability loss weight led to small variations in model performance. For both pathways, setting the weight to 0.01 resulted in the best predictive performance. Our results showed

that the effectiveness of graph stability regularisation may depend on the characteristics of the clinical pathway and its modality composition.

Learnable graph-construction parameters In addition to predictive performance, the learned parameters governing patient similarity and graph construction were examined in the selected M3Care models. As described in Equation 4.9, a learnable mixing coefficient adaptively combines patient similarities calculated from the original and projected modality representations. The coefficient was initialised to 1 in the unconstrained parameter space, corresponding to approximately 0.731 after applying the sigmoid function.

Inspection of the trained models revealed that these coefficients remained close to their initial values. For the MV model, the sigmoid-transformed coefficients ranged from approximately 0.732 to 0.738 across the three modalities. Similarly, for the extended-stay model, the coefficients ranged from approximately 0.729 to 0.732 across the seven modalities. These values indicate that the relative weighting of the two similarity components changed only marginally during training.

A similar observation was made for the learnable graph threshold in Equation 4.10, which determines whether weak patient relationships are retained in the constructed similarity graph. The threshold parameter was initialised to 0, corresponding to a sigmoid-transformed value of 0.5. After training, the threshold remained close to this value, reaching approximately 0.501 for deterioration MV and 0.502 for hospitalisation EXT-ST.

The limited changes in both sets of parameters suggest that the adaptive components of graph construction underwent relatively little adjustment from their initial settings. Consequently, the learned graph-construction behaviour may have remained largely influenced by the initial parameterisation. Nevertheless, small parameter updates do not necessarily imply that these components were ineffective, as their influence also depends on the distributions of patient similarities and the resulting graph connectivity. Further investigation could be conducted in future to determine whether alternative initialisation strategies or optimisation settings could improve their adaptability.

5.4.2 Imputation quality and contrastive learning

Effects of contrastive learning on predictive performance Tables 5.9 and 5.10 summarise the predictive performance of M3Care under different CL configurations. In general, incorporating CL did not consistently improve predictive performance over the corresponding M3Care model trained without CL.

For deterioration, introducing the full CL framework reduced MCC to 0.507, AUROC to 0.937 and BA to 0.881. Removing missing-modality contrastive alignment resulted in similar performance (MCC = 0.503), while disabling learned-target supervision further reduced MCC to 0.494. These results indicate that the additional CL objectives did not provide a predictive advantage for MV, although learned-target supervision appeared to contribute positively within the CL-enabled configurations.

For hospitalisation, the effects of CL differed between the two observation-window configurations. Under the current-window setting, full CL reduced MCC but improved AUROC from 0.728 to 0.734 and BA from 0.662 to 0.675. Removing missing-modality alignment or learned-target supervision resulted in slightly lower MCC and BA than full CL. Under the dual-window configuration, full CL reduced MCC to 0.291, but full CL and the variant without missing-modality alignment achieved the highest AUROC (0.736). Removing learned-target supervision achieved the highest BA (0.674).

Table 5.9: Performance of M3Care under different contrastive-learning configurations for mechanical ventilation prediction in the deterioration pathway, with 95% bootstrap confidence intervals. The highest performance for each metric is highlighted in bold.

CL setting	MCC	AUROC	BA
w/o CL	**0.542** [0.488–0.589]	**0.945** [0.919–0.964]	**0.892** [0.861–0.918]
Full CL	0.507 [0.449–0.549]	0.937 [0.905–0.958]	0.881 [0.848–0.907]
CL w/o missing-modality alignment	0.503 [0.443–0.544]	0.938 [0.905–0.958]	0.880 [0.848–0.907]
CL w/o learned target	0.494 [0.441–0.537]	0.929 [0.894–0.953]	0.881 [0.848–0.908]

Notes: MV: mechanical ventilation. Full CL includes observed- and missing-modality contrastive alignment and learned-target supervision. The ablation variants disable the indicated CL components. Confidence intervals are shown in square brackets.

Table 5.10: Performance of M3Care under different contrastive-learning configurations for extended hospital stay prediction in the hospitalisation pathway, with 95% bootstrap confidence intervals. The highest performance for each metric within each window is highlighted in bold.

	Current-window			Dual-window		
CL setting	MCC	AUROC	BA	MCC	AUROC	BA
w/o CL	**0.300** [0.278–0.323]	0.728 [0.714–0.740]	0.662 [0.650–0.674]	**0.299** [0.275–0.320]	0.733 [0.719–0.745]	0.671 [0.656–0.683]
Full CL	0.288 [0.267–0.310]	**0.734** [0.721–0.746]	**0.675** [0.663–0.688]	0.291 [0.270–0.312]	0.736 [0.723–0.748]	0.669 [0.656–0.680]
CL w/o missing-modality alignment	0.285 [0.264–0.308]	0.733 [0.720–0.745]	0.670 [0.657–0.682]	0.292 [0.269–0.313]	**0.736** [0.723–0.748]	0.666 [0.654–0.679]
CL w/o learned target	0.275 [0.253–0.296]	0.730 [0.717–0.743]	0.668 [0.655–0.680]	0.288 [0.267–0.311]	0.727 [0.714–0.739]	**0.674** [0.661–0.687]

Notes: EXT-ST: extended hospital stay > 7 days. Full CL includes observed- and missing-modality contrastive alignment and learned-target supervision. The ablation variants disable the indicated CL components. Confidence intervals are shown in square brackets.

These findings suggest that CL influenced different aspects of classification performance, but did not consistently improve the principal evaluation metric, MCC.

Quality of GCN-generated modality embeddings To investigate whether CL improved the underlying imputation process, GCN-generated embeddings were compared against their corresponding original modality representations under artificial modality masking. Figures 5.3(A) and 5.4(A) present the resulting cosine similarity comparisons between M3Care models trained with and without full CL.

For deterioration, CL did not improve the directional agreement between the imputed and original CXR representations. As illustrated in Figure 5.3(A), the mean GCN-to-original cosine similarity decreased from 0.408 to 0.277 with CL. The comparison was conducted using 789 matched ED stays, with identical input observations and eligibility criteria verified across the two models. The paired analysis also revealed considerable variation in the change in cosine similarity across individual ED stays. These findings suggest that CL did not improve the reconstruction of the original CXR representations for the MV prediction task.

For hospitalisation, the effects of CL varied across modalities and observation-window configurations. Under the current-window configuration, which employed two-stage curriculum learning, CL reduced embedding similarity for both current CXR (`cxr_curr`) and current ECG (`ecg_curr`) despite the initial complete-case warm-up stage. In contrast, under the dual-window configuration using single-stage joint training, CL improved cosine similarity for `cxr_curr` from 0.225 to 0.293 and `ecg_curr` from 0.264 to 0.322. though similarity decreased slightly for previous clinical notes (`notes_prev`). These results indicate that the effects of CL on imputation quality were modality-dependent and differed between the two experimental configurations.

Improvements in embedding similarity did not consistently correspond to improvements in downstream predictive performance. For example, although CL improved GCN-to-original cosine similarity for current CXR and ECG in dual-window hospitalisation, the overall MCC decreased from 0.299 to 0.291. This suggests that improved directional agreement between imputed and original embeddings does not necessarily translate into better discrimination of clinical outcomes.

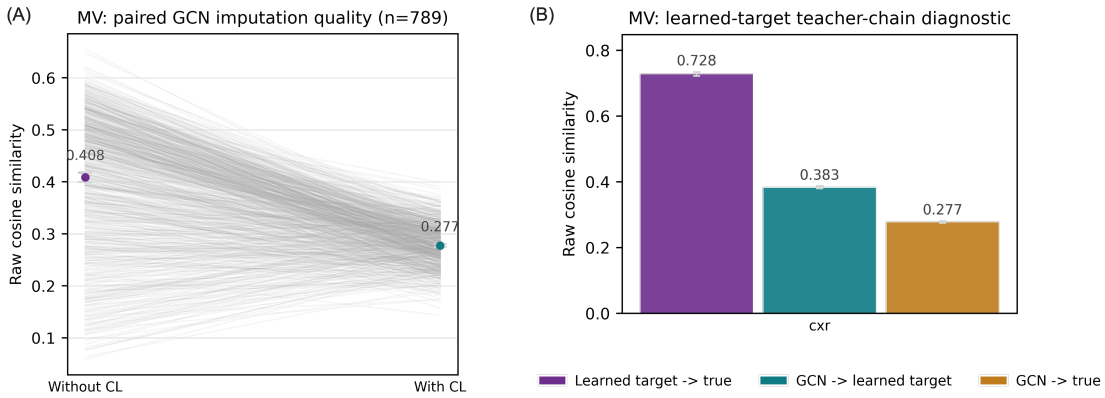


Figure 5.3: Evaluation of GCN-based modality imputation and learned-target alignment for mechanical ventilation (MV) prediction. (A) Stay-level comparison of GCN-to-original embedding cosine similarity between M3Care models trained with and without contrastive learning (CL), with individual lines representing paired stays and markers indicating the means with 95% confidence intervals. (B) Cosine similarity between learned cross-modal targets, GCN-generated embeddings and original observed embeddings for the CL-enabled model. Error bars represent 95% confidence intervals.

Learned cross-modal target quality The learned-target mechanism was further examined by comparing three relationships: the learned cross-modal target against the original observed embedding, the GCN-generated embedding against the learned target, and the GCN-generated embedding against the original embedding. Figures 5.3(B) and 5.4(B) also present these comparisons using cosine similarity in the original embedding space.

For deterioration, the learned target achieved a cosine similarity of approximately 0.728 with the original observed embedding, substantially higher than the GCN-to-target similarity (0.383) and GCN-to-original similarity (0.277). A similar pattern was observed for hospitalisation. Under the current-window configuration, learned-target-to-original cosine similarity was approximately 0.835 for current CXR and 0.904 for current ECG, while the corresponding GCN-to-original similarities were substantially lower, at 0.239 and 0.366, respectively. Under the dual-window configuration, learned-target-to-original similarity ranged from approximately 0.695 to 0.912 across `cxr_curr`,

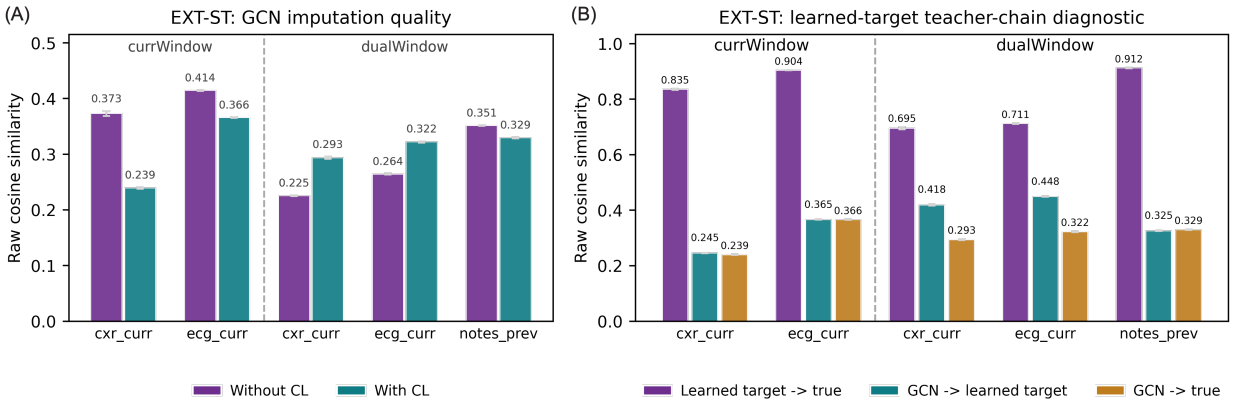


Figure 5.4: Evaluation of GCN-based modality imputation and learned-target alignment for extended hospital stay (EXT-ST) prediction. (A) Comparison of GCN-to-original embedding cosine similarity between M3Care models trained with and without contrastive learning (CL). Results are presented for current CXR (`cxr_curr`) and ECG (`ecg_curr`) under the current-window configuration, and additionally for previous clinical notes (`notes_prev`) under the dual-window configuration. (B) Cosine similarity between learned cross-modal targets, GCN-generated embeddings and original observed embeddings for the CL-enabled model. The current-window configuration employed two-stage curriculum learning, while the dual-window configuration used single-stage joint training. Error bars represent 95% confidence intervals.

`ecg_curr` and `notes_prev`. In comparison, GCN-to-target similarity ranged from approximately 0.326 to 0.448, while GCN-to-original similarity ranged from 0.293 to 0.329.

Across both pathways, the learned cross-modal targets generally showed substantially higher agreement with the original modality embeddings than the GCN-generated representations. Moreover, GCN-to-target and GCN-to-original similarities were often relatively close, despite the stronger agreement between the learned targets and original embeddings. These observations suggest that the learned-target mechanism was capable of constructing representations aligned with the original modality information, but that this alignment was not fully reflected in the GCN-generated embeddings.

In summary, the results demonstrate that CL did not consistently improve downstream predictive performance, while its effects on embedding quality varied across modalities and experimental configurations. Although the learned cross-modal targets exhibited relatively high similarity to the original representations, this did not consistently translate into improved GCN imputation quality or clinical outcome prediction. The findings therefore highlight the distinction between learning informative cross-modal supervision targets, aligning GCN-generated embeddings with those targets, and improving downstream predictive performance.

5.5 Does time-aware embedding help?

To investigate whether explicitly modelling irregular observation times could improve multimodal prediction, the mTAND temporal encoder was incorporated into both M3Care and FuseMoE. The experiments were conducted for MV prediction in the deterioration pathway. The preferred mTAND

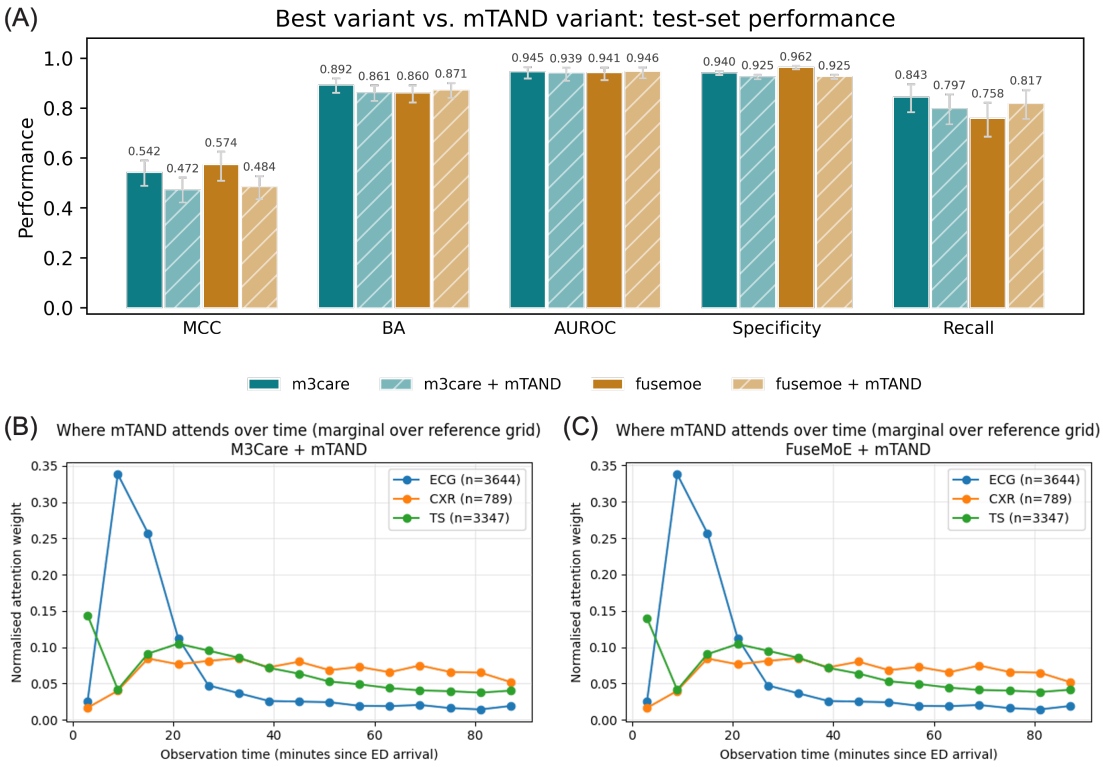


Figure 5.5: Effects of mTAND-based temporal encoding on mechanical ventilation prediction. (A) Test-set predictive performance of M3Care and FuseMoE with and without mTAND. Error bars represent 95% bootstrap confidence intervals. (B & C) Normalised mTAND attention weight across observation times for ECG, CXR and clinical TS modalities, obtained from temporal M3Care (B) and temporal FuseMoE (C). Attention weights were averaged across the mTAND reference-time grid and normalised within each modality. The curves represent aggregate attention distributions over eligible ED stays with available observations.

configurations for both architectures employed eight reference time points ($\mathrm{tt_max} = 8$) and a time embedding dimension of 16 ($\mathrm{embed_time} = 16$).

5.5.1 Effects of mTAND on predictive performance

Figure 5.5(A) compares the test-set performance of M3Care and FuseMoE with their mTAND extensions, showing that incorporating time-aware embeddings did not improve MCC for either fusion architecture. For M3Care, MCC decreased from 0.542 to 0.472 following the incorporation of mTAND, with similar reductions in the other metrics observed. These results indicate that the temporal extension did not provide a predictive advantage over the original M3Care configuration for MV. For FuseMoE, MCC similarly decreased from 0.574 to 0.484, while specificity declined from 0.962 to 0.925. However, mTAND improved AUROC from 0.941 to 0.946, BA from 0.860 to 0.871 and recall from 0.758 to 0.817. These results suggest that the temporal encoder altered the balance between sensitivity and specificity, improving the identification of positive cases at the expense of more false-positive predictions, which did not translate into a higher MCC at the selected classification threshold.

5.5.2 Temporal attention patterns

To investigate how mTAND utilised temporal information, attention weights were examined across observation times within the initial 90-minute ED observation window. Figures 5.5(B) and 5.5(C) present the normalised attention weight aggregated across the mTAND reference-time grid for ECG, CXR and clinical time-series (TS) measurements, using temporal M3Care and FuseMoE, respectively.

Both fusion architectures exhibited similar temporal attention patterns across the three modalities. For ECG, attention was strongly concentrated in the early part of the observation window, peaking at approximately 9 minutes after ED arrival, followed by a second relatively high value at approximately 15 minutes. Attention subsequently declined sharply after approximately 20 minutes and remained comparatively low throughout the remainder of the observation window. This pattern was observed in both temporal M3Care and FuseMoE, suggesting that their temporal encoders assigned greater attention to earlier ECG observations.

In contrast, CXR attention was distributed more evenly across the observation window. Although relatively little attention was assigned to the earliest observations, attention increased after approximately 10 minutes and remained comparatively stable across much of the subsequent period. Unlike ECG, CXR attention did not exhibit a pronounced concentration within a narrow temporal interval.

For TS measurements, the highest attention mass occurred in the earliest observation-time bin, at approximately 3 minutes after ED arrival. Attention then decreased before increasing again around 15-25 minutes, followed by a gradual decline towards the end of the observation window. Compared with ECG, TS attention was distributed across a broader range of observation times, though earlier measurements generally received greater weight.

The similarity of the attention distributions between temporal M3Care and FuseMoE suggests that both architectures learned comparable aggregate temporal weighting patterns despite their different multimodal fusion mechanisms. However, these distributions were aggregated across patients and mTAND reference time points, and may also reflect differences in observation frequency and modality availability. Our results indicate that mTAND was capable of learning modality-specific temporal attention patterns, but these patterns did not translate into improved MCC for either model.

5.6 How much of the observed performance is due to modality availability vs modality information?

As discussed in Section 5.1, modality availability in both cohorts reflects clinical care processes. The presence or absence of a recorded modality may therefore carry predictive information independently of its content. To distinguish these contributions, we conducted a permutation analysis in which modality content was shuffled between encounters sharing the same availability pattern, preserving the original missingness structure. Permuting individual modalities assessed the model's reliance on their content, while permuting all modalities examined the residual predictive performance associated with availability patterns. Each experiment was repeated over 100 shuffles.

For the deterioration cohort (Table 5.11), shuffling tabular content (ie. static EHR and time-series summaries) led to substantial reductions in MCC across all five prediction tasks and three models. CXR permutation produced smaller but significant reductions for mechanical ventilation, ICU admission and cardiac arrest prediction, while no significant reduction was observed for one-day

Table 5.11: Test MCC following permutation of modality content in the deterioration cohort.

Task	Model	No permutation	Tabular	ECG	CXR	All
CA	Concat	0.349	0.094^{*}	0.358	0.268^{*}	0.004
	M3Care	0.440	0.015^{*}	0.403^{*}	0.399^{*}	0.009
	FuseMoE	0.318	0.029^{*}	0.318	0.254^{*}	-0.002
ICU-24h	Concat	0.597	0.154^{*}	0.589	0.513^{*}	0.028
	M3Care	0.548	0.126^{*}	0.549	0.489^{*}	$0.033^{\dagger}$
	FuseMoE	0.573	0.145^{*}	0.566	0.524^{*}	$0.031^{\dagger}$
ICU	Concat	0.543	0.177^{*}	0.543	0.479^{*}	$0.029^{\dagger}$
	M3Care	0.547	0.111^{*}	0.546	0.484^{*}	0.030
	FuseMoE	0.552	0.123^{*}	0.545	0.498^{*}	0.025
MV	Concat	0.498	0.227^{*}	0.494	0.370^{*}	0.029
	M3Care	0.542	0.233^{*}	0.527^{*}	0.409^{*}	0.030
	FuseMoE	0.574	0.215^{*}	0.564	0.431^{*}	0.024
MORT-1D	Concat	0.206	0.007^{*}	0.206	0.216	0.001
	M3Care	0.197	0.011^{*}	0.190	0.208	0.002
	FuseMoE	0.223	0.009^{*}	0.223	0.231	-0.000

Notes: CA: in-hospital cardiac arrest; ICU-24h: ICU admission within 24 hours; ICU: ICU admission during the hospital stay; MV: mechanical ventilation; MORT-1D: one-day mortality. Tabular combines static EHR and time-series summaries. Permuted MCC values are averaged over 100 shuffles. * indicates a significant reduction relative to unpermuted MCC ($p < 0.05$). † indicates MCC significantly above zero after all modality contents were shuffled ($p < 0.05$).

Table 5.12: Test MCC following permutation of modality content in the hospitalisation cohort.

Task	Model	No permutation	Static	TS curr	ECG curr	CXR curr	TS prev	CXR prev	Notes prev	All
dualWindow										
EXT-ST	Concat	0.300	0.247^{*}	0.140^{*}	0.299	0.293^{*}	0.298	0.300	0.301	$0.053^{\dagger}$
	M3Care	0.299	0.226^{*}	0.151^{*}	0.299	0.291^{*}	0.299	0.299	0.299	$0.048^{\dagger}$
	FuseMoE	0.300	0.216^{*}	0.161^{*}	0.300	0.287^{*}	0.300	0.299	0.300	$0.048^{\dagger}$
IHD	Concat	0.355	0.289^{*}	0.174^{*}	0.353^{*}	0.343^{*}	0.355	0.354	0.352^{*}	$0.055^{\dagger}$
	M3Care	0.336	0.268^{*}	0.160^{*}	0.336	0.326^{*}	0.337	0.336	0.337	$0.051^{\dagger}$
	FuseMoE	0.332	0.258^{*}	0.198^{*}	0.333	0.326^{*}	0.332	0.333	0.332	$0.059^{\dagger}$
NHD	Concat	0.434	0.129^{*}	0.383^{*}	0.426^{*}	0.414^{*}	0.432	0.433	0.431^{*}	$0.030^{\dagger}$
	M3Care	0.421	0.126^{*}	0.366^{*}	0.420	0.397^{*}	0.421^{*}	0.421	0.420^{*}	$0.026^{\dagger}$
	FuseMoE	0.418	0.106^{*}	0.375^{*}	0.418	0.402^{*}	0.418	0.419	0.418	$0.029^{\dagger}$
currWindow										
EXT-ST	Concat	0.301	0.224^{*}	0.135^{*}	0.294^{*}	0.288^{*}	–	–	–	0.006
	M3Care	0.300	0.212^{*}	0.127^{*}	0.299	0.287^{*}	–	–	–	0.008
	FuseMoE	0.288	0.213^{*}	0.124^{*}	0.288	0.279^{*}	–	–	–	0.007
IHD	Concat	0.346	0.274^{*}	0.174^{*}	0.344^{*}	0.337^{*}	–	–	–	$0.029^{\dagger}$
	M3Care	0.354	0.295^{*}	0.165^{*}	0.353	0.338^{*}	–	–	–	$0.027^{\dagger}$
	FuseMoE	0.339	0.257^{*}	0.184^{*}	0.338	0.327^{*}	–	–	–	$0.027^{\dagger}$
NHD	Concat	0.424	0.122^{*}	0.373^{*}	0.424	0.411^{*}	–	–	–	0.015
	M3Care	0.416	0.101^{*}	0.377^{*}	0.416	0.401^{*}	–	–	–	0.016
	FuseMoE	0.424	0.095^{*}	0.380^{*}	0.424	0.400^{*}	–	–	–	0.011

Notes: EXT-ST: extended hospital stay > 7 days; IHD: in-hospital death; NHD: non-home discharge. Permuted MCC values are averaged over 100 shuffles. * indicates a significant reduction relative to unpermuted MCC ($p < 0.05$). † indicates MCC significantly above zero after all modality contents were shuffled ($p < 0.05$). A dash indicates that the modality was not included in the corresponding window configuration.

mortality. In contrast, ECG permutation had little effect across most tasks, with significant reductions observed only for M3Care in mechanical ventilation and cardiac arrest prediction. When all modality contents were shuffled, MCC fell to 0.024-0.033 for mechanical ventilation and ICU-related tasks, remaining significantly above chance in only three of the nine model-task combinations. For cardiac arrest and one-day mortality, MCC was approximately zero. These results indicate that availability patterns alone were associated with little residual predictive performance in the deterioration cohort.

For the hospitalisation cohort (Table 5.12), shuffling current-window static and time-series data produced substantial reductions in MCC across both window configurations. The largest reductions occurred following time-series permutation for extended stay and in-hospital death prediction, and static data permutation for non-home discharge prediction. Changes were considerably smaller for other modalities. Current-window CXR permutation resulted in small but consistent reductions in MCC, whereas ECG permutation had little effect. Shuffling previous-window time-series, CXR and clinical notes produced minimal changes, although a few differences were statistically significant.

The residual performance following permutation of all modality contents differed between the two hospitalisation configurations. For dualWindow models, MCC remained at 0.026-0.059, significantly above chance across all nine model-task combinations. In contrast, currWindow models retained MCC values of 0.006-0.029, with significant residual performance observed only for in-hospital death prediction. These findings suggest that previous-window availability patterns contain some predictive information, even when the recorded modality values are shuffled. However, the residual performance was minimal relative to the unpermuted results.

5.7 Do modality availability and fusion methods affect fairness across patient groups?

Table 5.13: Fairness ratios of the deterioration and hospitalisation models on the test set

			Deterioration					Hospitalisation					
								Dual window			Current window		
Attribute	**Metric**	**Model**	**MV**	**CA**	**ICU-24h**	**ICU**	**MORT-1D**	**EXT-ST**	**IHD**	**NHD**	**EXT-ST**	**IHD**	**NHD**
Sex	DPR	Concat	0.61 [0.51–0.74]	0.35 [0.21–0.53]	0.64 [0.55–0.74]	0.63 [0.55–0.72]	0.76 [0.63–0.94]	0.78 [0.73–0.84]	0.78 [0.71–0.86]	0.86 [0.81–0.92]	0.77 [0.73–0.82]	0.78 [0.71–0.86]	0.89 [0.84–0.94]
		M3Care	0.57 [0.47–0.69]	0.32 [0.21–0.50]	0.67 [0.58–0.77]	0.65 [0.57–0.75]	0.68 [0.55–0.86]	0.78 [0.73–0.84]	0.80 [0.73–0.89]	0.91 [0.86–0.96]	0.78 [0.72–0.83]	0.79 [0.72–0.87]	0.83 [0.78–0.88]
		FuseMoE	0.62 [0.49–0.81]	0.42 [0.30–0.56]	0.67 [0.58–0.76]	0.65 [0.57–0.73]	0.64 [0.49–0.83]	0.76 [0.71–0.82]	0.81 [0.74–0.88]	0.90 [0.85–0.95]	0.83 [0.79–0.88]	0.77 [0.71–0.84]	0.84 [0.79–0.90]
	EOR	Concat	0.59 [0.47–0.76]	0.34 [0.19–0.54]	0.57 [0.43–0.75]	0.56 [0.45–0.69]	0.76 [0.61–0.93]	0.77 [0.71–0.85]	0.77 [0.69–0.85]	0.87 [0.82–0.92]	0.75 [0.69–0.81]	0.78 [0.70–0.86]	0.89 [0.84–0.93]
		M3Care	0.54 [0.41–0.70]	0.30 [0.18–0.49]	0.65 [0.52–0.82]	0.58 [0.47–0.74]	0.68 [0.54–0.87]	0.76 [0.70–0.84]	0.80 [0.72–0.90]	0.88 [0.83–0.93]	0.76 [0.69–0.84]	0.79 [0.71–0.87]	0.84 [0.79–0.89]
		FuseMoE	0.62 [0.44–0.84]	0.42 [0.30–0.58]	0.64 [0.52–0.80]	0.56 [0.45–0.69]	0.64 [0.48–0.83]	0.74 [0.67–0.81]	0.81 [0.74–0.89]	0.88 [0.83–0.94]	0.82 [0.76–0.89]	0.76 [0.70–0.84]	0.84 [0.78–0.89]
	EOP	Concat	0.97 [0.91–1.00]	0.96 [0.91–1.00]	0.96 [0.88–1.00]	0.96 [0.88–1.00]	0.95 [0.81–1.00]	0.86 [0.79–0.95]	0.94 [0.84–1.00]	0.87 [0.82–0.92]	0.89 [0.83–0.96]	0.91 [0.82–0.99]	0.89 [0.84–0.93]
		M3Care	0.96 [0.89–1.00]	0.94 [0.77–1.00]	0.96 [0.87–1.00]	0.99 [0.98–1.00]	0.91 [0.73–1.00]	0.89 [0.82–0.98]	0.92 [0.81–1.00]	0.88 [0.83–0.93]	0.88 [0.80–0.98]	0.91 [0.82–0.99]	0.84 [0.79–0.89]
		FuseMoE	0.97 [0.90–1.00]	0.98 [0.94–1.00]	0.95 [0.87–1.00]	0.99 [0.96–1.00]	0.91 [0.73–1.00]	0.89 [0.82–0.97]	0.90 [0.82–0.99]	0.88 [0.83–0.94]	0.92 [0.86–0.99]	0.92 [0.82–0.99]	0.84 [0.77–0.90]
Age	DPR	Concat	0.36 [0.23–0.51]	0.12 [0.00–0.36]	0.18 [0.11–0.26]	0.20 [0.13–0.29]	0.11 [0.04–0.20]	0.17 [0.13–0.22]	0.08 [0.05–0.12]	0.06 [0.04–0.08]	0.25 [0.20–0.30]	0.08 [0.05–0.12]	0.05 [0.04–0.07]
		M3Care	0.34 [0.23–0.51]	0.24 [0.07–0.56]	0.27 [0.19–0.37]	0.22 [0.15–0.31]	0.11 [0.04–0.18]	0.18 [0.14–0.23]	0.07 [0.04–0.11]	0.07 [0.05–0.08]	0.23 [0.18–0.29]	0.07 [0.04–0.11]	0.04 [0.03–0.06]
		FuseMoE	0.22 [0.13–0.36]	0.49 [0.38–0.82]	0.25 [0.17–0.34]	0.23 [0.17–0.31]	0.07 [0.02–0.16]	0.20 [0.16–0.25]	0.05 [0.02–0.07]	0.05 [0.03–0.07]	0.23 [0.19–0.28]	0.07 [0.04–0.11]	0.03 [0.02–0.04]
	EOR	Concat	0.36 [0.21–0.53]	0.07 [0.00–0.42]	0.12 [0.05–0.24]	0.15 [0.07–0.25]	–	0.14 [0.10–0.19]	0.08 [0.00–0.13]	0.06 [0.04–0.08]	0.23 [0.18–0.29]	0.08 [0.00–0.13]	0.05 [0.03–0.07]
		M3Care	0.33 [0.19–0.53]	0.24 [0.00–0.67]	0.27 [0.16–0.42]	0.19 [0.10–0.32]	–	0.17 [0.12–0.22]	0.07 [0.00–0.12]	0.06 [0.04–0.08]	0.24 [0.17–0.31]	0.07 [0.00–0.11]	0.04 [0.02–0.05]
		FuseMoE	0.13 [0.03–0.28]	0.50 [0.41–0.81]	0.25 [0.14–0.38]	0.21 [0.13–0.33]	–	0.18 [0.13–0.23]	0.04 [0.00–0.07]	0.05 [0.03–0.07]	0.22 [0.17–0.27]	0.07 [0.00–0.11]	0.03 [0.01–0.04]
	EOP	Concat	0.83 [0.68–0.97]	0.33 [0.00–0.83]	0.79 [0.72–0.94]	0.84 [0.76–0.96]	–	0.40 [0.27–0.58]	0.47 [0.00–0.83]	0.17 [0.07–0.29]	0.46 [0.33–0.61]	0.47 [0.00–0.85]	0.16 [0.07–0.29]
		M3Care	0.84 [0.75–0.98]	0.90 [0.00–1.00]	0.88 [0.84–0.97]	0.84 [0.78–0.96]	–	0.33 [0.21–0.48]	0.56 [0.00–0.88]	0.21 [0.11–0.35]	0.36 [0.22–0.53]	0.70 [0.30–0.88]	0.14 [0.05–0.27]
		FuseMoE	0.80 [0.70–0.94]	0.80 [0.35–1.00]	0.89 [0.86–0.99]	0.82 [0.75–0.93]	–	0.41 [0.28–0.59]	0.47 [0.00–0.88]	0.16 [0.06–0.28]	0.40 [0.28–0.56]	0.69 [0.19–0.88]	0.10 [0.02–0.21]
Ethnicity	DPR	Concat	0.07 [0.01–0.14]	0.00 [0.00–0.00]	0.17 [0.10–0.28]	0.21 [0.12–0.31]	0.05 [0.01–0.10]	0.68 [0.55–0.82]	0.55 [0.43–0.69]	0.48 [0.42–0.58]	0.72 [0.60–0.85]	0.57 [0.44–0.73]	0.53 [0.46–0.62]
		M3Care	0.09 [0.04–0.16]	0.00 [0.00–0.00]	0.18 [0.10–0.28]	0.18 [0.10–0.28]	0.05 [0.01–0.11]	0.70 [0.59–0.84]	0.53 [0.41–0.69]	0.49 [0.42–0.58]	0.55 [0.43–0.68]	0.51 [0.37–0.65]	0.46 [0.39–0.55]
		FuseMoE	0.03 [0.00–0.09]	0.00 [0.00–0.00]	0.19 [0.11–0.29]	0.20 [0.12–0.29]	0.05 [0.02–0.11]	0.60 [0.49–0.73]	0.47 [0.35–0.61]	0.48 [0.39–0.58]	0.64 [0.53–0.75]	0.67 [0.55–0.83]	0.39 [0.30–0.49]
	EOR	Concat	0.07 [0.00–0.22]	–	0.29 [0.21–0.51]	0.32 [0.19–0.50]	–	0.66 [0.53–0.81]	0.52 [0.42–0.74]	0.48 [0.40–0.63]	0.72 [0.62–0.87]	0.61 [0.54–0.79]	0.47 [0.35–0.58]
		M3Care	0.12 [0.00–0.25]	–	0.28 [0.18–0.46]	0.26 [0.14–0.44]	–	0.72 [0.64–0.87]	0.52 [0.42–0.70]	0.45 [0.33–0.57]	0.53 [0.40–0.69]	0.53 [0.45–0.70]	0.42 [0.33–0.55]
		FuseMoE	0.00 [0.00–0.00]	–	0.29 [0.22–0.44]	0.32 [0.24–0.51]	–	0.60 [0.48–0.75]	0.48 [0.38–0.65]	0.46 [0.35–0.59]	0.60 [0.47–0.73]	0.66 [0.56–0.84]	0.34 [0.22–0.45]
	EOP	Concat	0.67 [0.00–0.88]	–	0.31 [0.09–0.61]	0.50 [0.25–0.74]	–	0.84 [0.77–0.96]	0.58 [0.25–0.88]	0.72 [0.60–0.88]	0.84 [0.76–0.95]	0.70 [0.41–0.92]	0.77 [0.65–0.91]
		M3Care	0.67 [0.00–0.84]	–	0.37 [0.11–0.66]	0.44 [0.20–0.73]	–	0.79 [0.68–0.95]	0.67 [0.34–0.90]	0.76 [0.64–0.91]	0.69 [0.50–0.86]	0.66 [0.33–0.87]	0.72 [0.59–0.88]
		FuseMoE	0.62 [0.50–0.74]	–	0.44 [0.19–0.72]	0.42 [0.18–0.68]	–	0.72 [0.58–0.88]	0.68 [0.38–0.90]	0.72 [0.57–0.86]	0.81 [0.73–0.95]	0.78 [0.54–0.93]	0.69 [0.53–0.88]
CXR availability	DPR	Concat	0.35 [0.29–0.42]	0.26 [0.18–0.38]	0.37 [0.32–0.43]	0.45 [0.40–0.52]	0.33 [0.28–0.40]						
		M3Care	0.33 [0.27–0.40]	0.22 [0.16–0.32]	0.35 [0.30–0.40]	0.38 [0.34–0.44]	0.28 [0.23–0.35]						
		FuseMoE	0.25 [0.20–0.31]	0.45 [0.33–0.59]	0.38 [0.33–0.44]	0.44 [0.39–0.50]	0.31 [0.24–0.40]						
	EOR	Concat	0.42 [0.33–0.53]	0.29 [0.18–0.44]	0.38 [0.29–0.51]	0.52 [0.42–0.65]	0.32 [0.26–0.39]						
		M3Care	0.40 [0.31–0.52]	0.26 [0.17–0.40]	0.33 [0.27–0.41]	0.41 [0.33–0.51]	0.28 [0.22–0.35]						
		FuseMoE	0.26 [0.19–0.37]	0.54 [0.37–0.74]	0.36 [0.30–0.45]	0.49 [0.40–0.61]	0.30 [0.23–0.39]						
	EOP	Concat	0.86 [0.74–0.97]	0.73 [0.40–0.98]	0.69 [0.62–0.77]	0.71 [0.64–0.78]	0.91 [0.73–1.00]						
		M3Care	0.89 [0.76–0.99]	0.86 [0.42–1.00]	0.67 [0.59–0.74]	0.66 [0.59–0.73]	0.76 [0.45–0.99]						
		FuseMoE	0.87 [0.71–0.99]	0.98 [0.94–1.00]	0.75 [0.67–0.81]	0.72 [0.66–0.79]	0.76 [0.45–0.99]						
Insurance	DPR	Concat						0.39 [0.36–0.44]	0.21 [0.12–0.29]	0.17 [0.15–0.19]	0.45 [0.41–0.49]	0.22 [0.13–0.30]	0.18 [0.16–0.20]
		M3Care						0.38 [0.35–0.42]	0.20 [0.10–0.29]	0.21 [0.19–0.23]	0.42 [0.38–0.46]	0.23 [0.13–0.33]	0.14 [0.12–0.16]
		FuseMoE						0.42 [0.39–0.47]	0.15 [0.08–0.24]	0.19 [0.17–0.21]	0.46 [0.43–0.51]	0.23 [0.14–0.33]	0.15 [0.13–0.17]
	EOR	Concat						0.33 [0.29–0.38]	0.19 [0.07–0.27]	0.13 [0.11–0.15]	0.39 [0.35–0.43]	0.19 [0.09–0.29]	0.14 [0.12–0.16]
		M3Care						0.32 [0.28–0.36]	0.16 [0.06–0.27]	0.17 [0.15–0.19]	0.35 [0.31–0.41]	0.21 [0.10–0.31]	0.10 [0.08–0.12]
		FuseMoE						0.36 [0.31–0.41]	0.12 [0.04–0.21]	0.15 [0.13–0.17]	0.40 [0.36–0.45]	0.21 [0.09–0.32]	0.12 [0.09–0.14]
	EOP	Concat						0.71 [0.63–0.81]	0.84 [0.73–0.99]	0.59 [0.54–0.67]	0.76 [0.69–0.85]	0.87 [0.83–0.98]	0.57 [0.44–0.66]
		M3Care						0.69 [0.61–0.77]	0.71 [0.52–0.95]	0.62 [0.55–0.70]	0.73 [0.68–0.85]	0.84 [0.74–0.96]	0.52 [0.46–0.61]
		FuseMoE						0.75 [0.67–0.85]	0.82 [0.73–0.94]	0.60 [0.56–0.67]	0.79 [0.72–0.89]	0.95† [0.97–0.99]	0.52 [0.45–0.61]

Notes: MV: mechanical ventilation; CA: in-hospital cardiac arrest; ICU-24h: ICU admission within 24 hours; ICU: ICU admission during the hospital stay; MORT-1D: one-day mortality. EXT-ST: extended hospital stay > 7 days; IHD: in-hospital death; NHD: non-home discharge. Values are point estimates with 95% BCa confidence intervals (1,000 bootstrap resamples within groups). DPR: demographic parity ratio; EOR: equalised odds ratio; EOP: equal opportunity ratio. **Deterioration outcomes:** MV, mechanical ventilation; CA, in-hospital cardiac arrest; ICU-24h, ICU admission within 24 hours; ICU, ICU admission; MORT-1D, one-day mortality. Hospitalisation outcomes: EXT-ST, extended hospital stay (>7 days); IHD, in-hospital death; NHD, non-home discharge. Dual window: current and previous windows; current window: current window only. Blank: attribute not available in that pathway. –: undefined, because at least one group had no positive patients in the test set (CA: Asian, Hispanic; MORT-1D: under 30, Asian, Hispanic). †: point estimate outside its interval, because 98% of resamples fell below it (the Other insurance group has 5 positive patients).

We assessed fairness across the best-performing runs of concatenation, M3Care and FuseMoE for each outcome in both prediction pathways (Table 5.13). Here, we treat EOP as the primary fairness measure as it compares sensitivity across patient groups. This is particularly relevant to clinical deterioration prediction, where failing to identify patients who subsequently deteriorate could delay clinical intervention. DPR and EOR were also examined to provide additional context on differences in patient selection rates.

Overall, fairness varied more by prediction outcome and patient attribute than by fusion method. The 95% confidence intervals for EOP overlapped across the three models in 43 of 44 outcome-attribute combinations. The only exception was insurance type for in-hospital death prediction using the current window. The attributes with the lowest EOP were also consistent across models. In the hospitalisation pathway, age had the lowest EOP across all six outcome-window combinations, with particularly large disparities for non-home discharge (EOP 0.10–0.21). This may reflect the influence of age-related functional and social care needs on discharge destination, which may not be adequately captured by early ED observations. Insurance-related disparities were also greater for non-home discharge than for the other hospitalisation outcomes. In the deterioration pathway, ethnicity showed the greatest disparity for ICU admission within 24 hours, ICU admission during the hospital stay and mechanical ventilation. Across deterioration outcomes, EOP was highest for sex, followed by age, CXR availability and ethnicity. For hospitalisation outcomes, sex again showed the highest EOP, followed by ethnicity and insurance, while age was substantially lower.

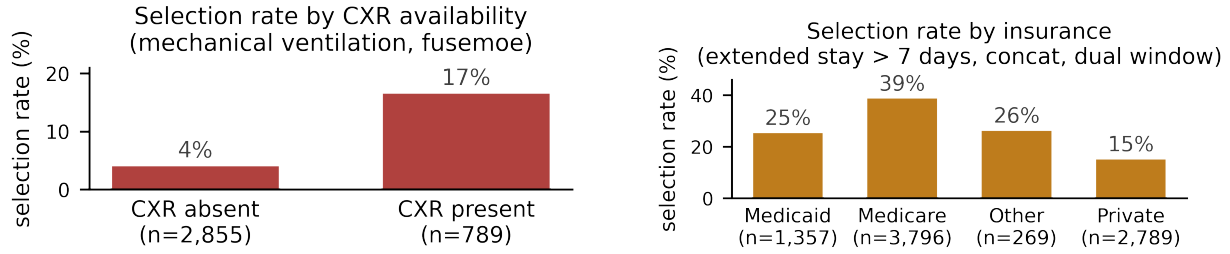


Figure 5.6: Selection rate for mechanical ventilation by CXR availability (left), and extended hospital stay by insurance type (right).

Note: Differences in patient selection rates provide further context for these fairness gaps (Figure 5.6). For mechanical ventilation prediction using FuseMoE, ED stays with a chest X-ray were over four times more likely to be classified as positive than those without one. In fact, the underlying prevalence of mechanical ventilation was also higher among patients with a chest X-ray (10.4% vs 2.5%). This suggests that the difference in selection rates partly reflects the underlying outcome distribution and the clinical process through which imaging was requested, rather than modality availability alone. A similar pattern was observed for extended hospital stay prediction using concatenation with the dual-window configuration. Medicare patients had the highest selection rate, while those with Private insurance had the lowest. The underlying prevalence of extended stay was also higher among Medicare patients (23.4% vs 14.6%), although the difference in selection rates was greater than that in outcome prevalence. In addition, insurance type was also strongly associated with age. 78.7% of Medicare patients aged 65 or over, compared with 10.7% of Private patients. Given that age consistently showed the lowest EOP in the hospitalisation pathway, the observed insurance-related disparities may be partly confounded by differences in age distribution.

Taken together, we could not see evidence that the choice of fusion method substantially altered fairness across patient groups. Instead, disparities were more closely associated with the prediction outcome, patient characteristics and modality availability.

5.8 Model explanations

As described in Section 4.7, we examined model predictions at three levels: modality contributions, individual features and spatial attribution within chest X-rays. MM-SHAP was computed across the full test set for all model runs, while feature-level SHAP provided both global and patient-level explanations. For cases with available chest X-rays, Grad-CAM or SHAP-weighted Grad-CAM heatmaps could be used to visualise spatial contributions to model predictions.

5.8.1 Global modality contribution

Unlike the permutation analysis in Section 5.6, which assessed changes in predictive performance, MM-SHAP quantified the relative contribution of each available modality to the predicted probability. The resulting modality contribution across all selected model runs is presented in Figure 5.7 below.

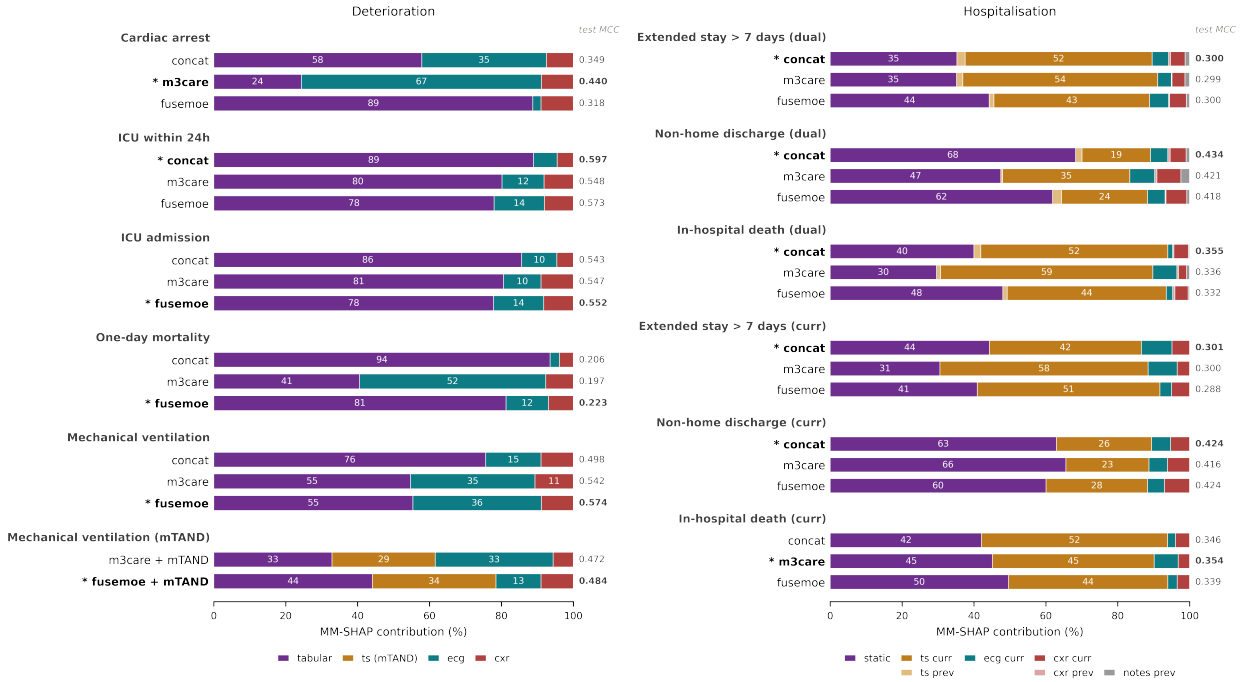


Figure 5.7: **MM-SHAP modality attribution shares across the full test set.** Relative modality contributions (%) for individual model runs, grouped by prediction outcome in the deterioration (left) and hospitalisation (right) cohorts. Each stacked bar represents one run, with modality shares summing to 100%. Missing modalities were assigned zero attribution. Test MCC values are shown on the right of each stacked bar, and the best run of each outcome are highlighted in bold with *.

For the deterioration cohort, tabular data accounted for the largest attribution share in all but two runs. Both exceptions occurred with M3Care, where ECG was the leading modality for cardiac

arrest and one-day mortality prediction. CXR generally received a small attribution share across the full test set, partly because 78% of ED stays lacked CXR data and absent modalities were assigned zero attribution. Models with similar predictive performance generally showed comparable attribution patterns, although cardiac arrest was an exception. The ECG share ranged from a few percent with FuseMoE to approximately two-thirds with M3Care, which achieved the best performance for this outcome. When the analysis was restricted to stays with an available chest X-ray (Figure 5.8), its attribution share increased to approximately 20-50%, often overtaking tabular data.

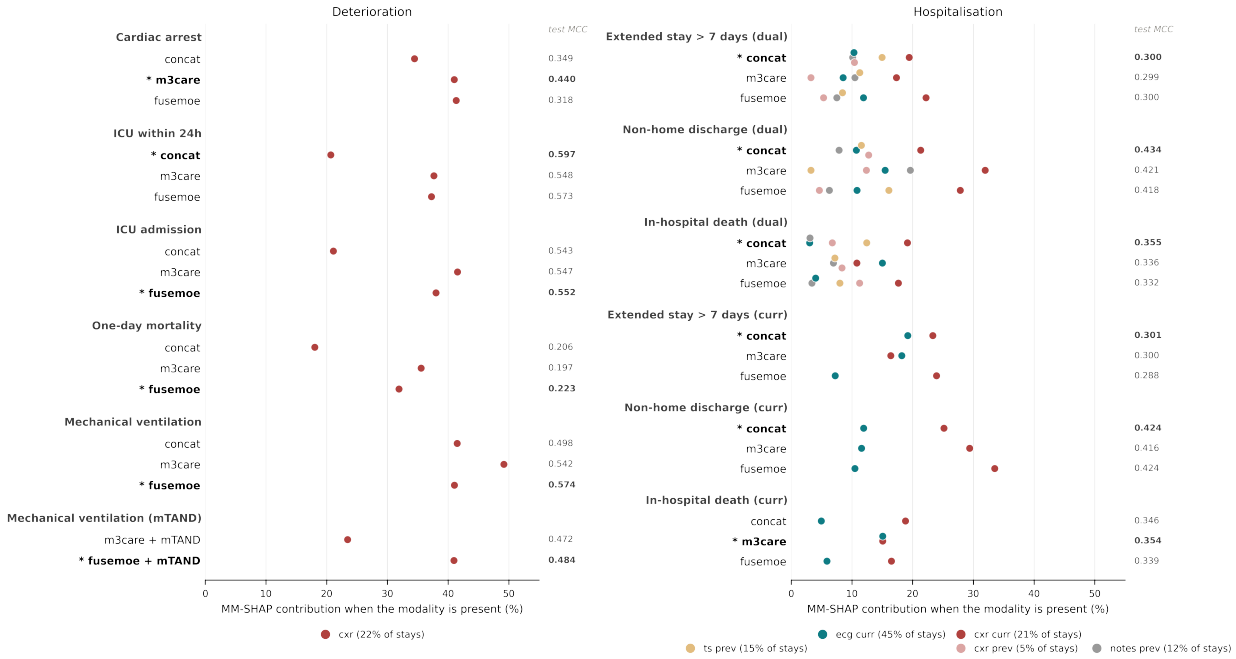


Figure 5.8: **MM-SHAP modality attribution shares conditional on modality availability.** Relative modality contributions calculated using only test cases where the corresponding modality was available, grouped by prediction outcome in the deterioration (left) and hospitalisation (right) cohorts. Each marker represents the attribution share of a modality for an individual model run. Test MCC values are shown on the right of each stacked bar, and the best run of each outcome are highlighted in bold with *.

For the hospitalisation cohort, static data and current-window time series dominated the attribution profiles, together accounting for over 80% of the total share across all model runs. When considering only admissions with each modality available, current-window CXR had the largest contribution among the less frequently available modalities, while current ECG and previous-window modalities generally accounted for around 20% or less. Overall, no individual modality apart from static data and current-window time series exceeded approximately one-third of the total attribution, even when present.

5.8.2 Feature-level explanations

Feature-level SHAP provides a more detailed view of the structured variables underlying the modality contributions reported in Section 5.8.1. For each model run, global and local explanations

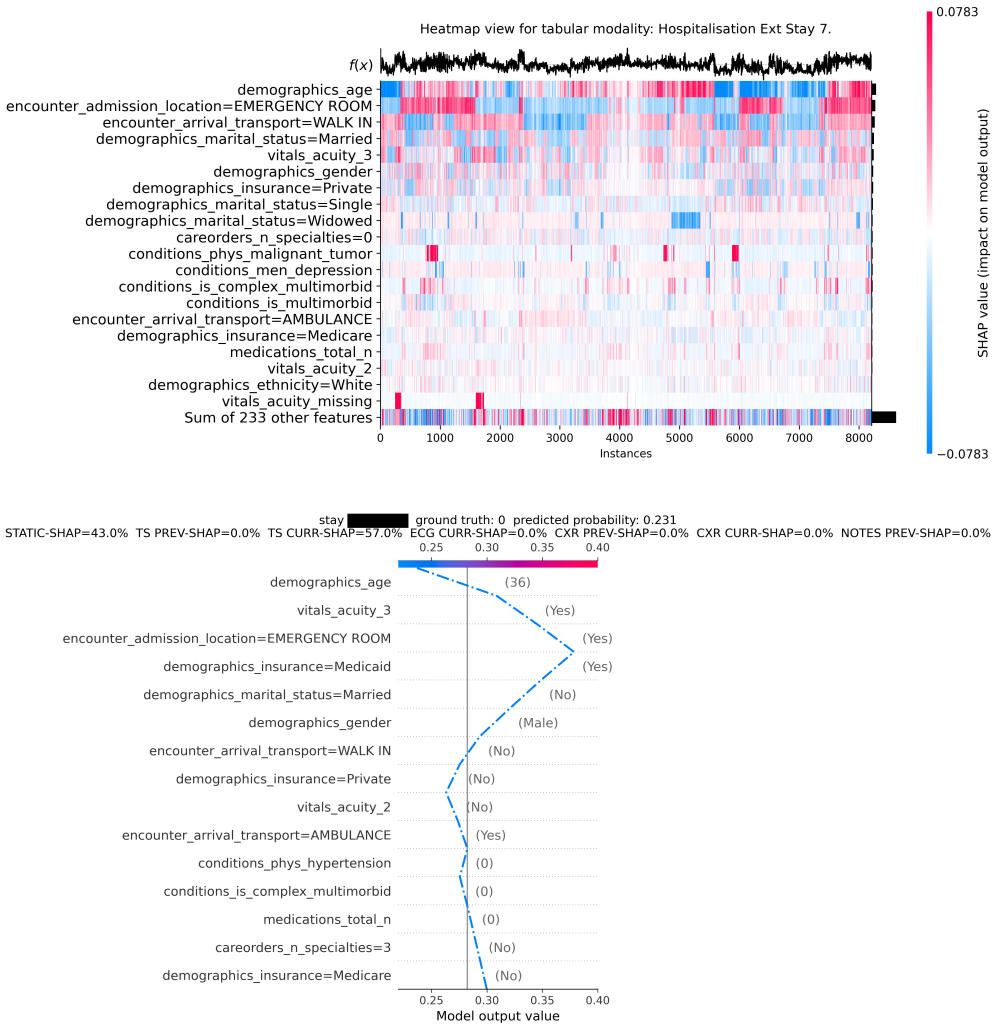
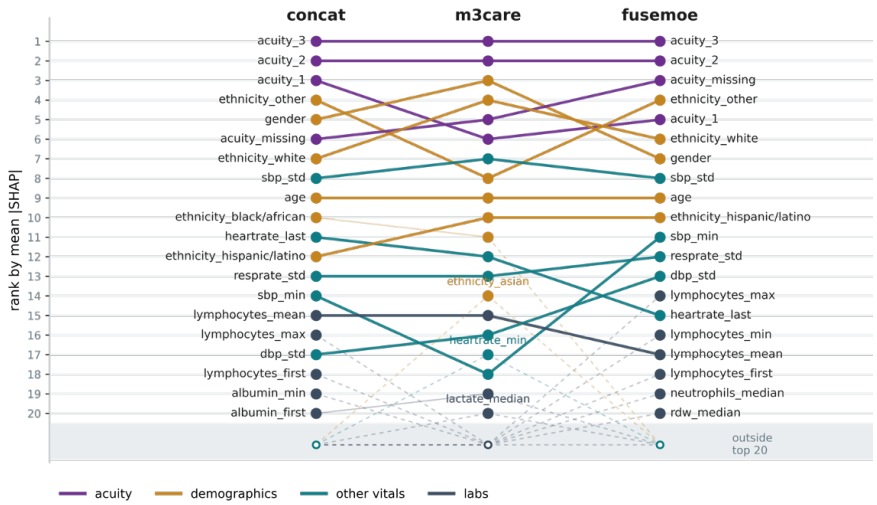


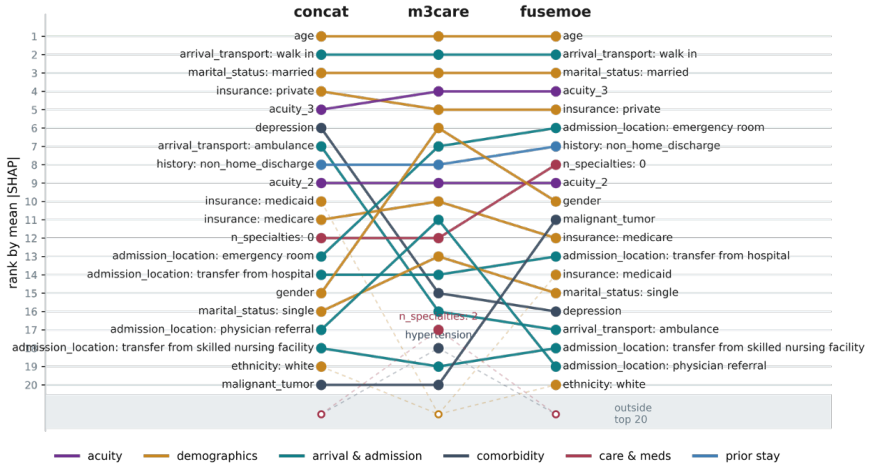
Figure 5.9: **Upper:** Global tabular SHAP values of FuseMoE predicting extended hospital stay in the test set. Only the top 20 features are shown. Columns are stays grouped by hierarchical clustering of their SHAP values. Red raises and blue lowers the predicted probability. f(x) represents the base value plus summed SHAP values. **Lower:** Example decision plot for a patient from FuseMoE predicting extended hospital stay.

can be generated. Figure 5.9 shows one set of examples from a model predicting extended hospital stay. The global heatmap shows that no single feature dominates, with the combined contribution of the over 200 features outside the top 20 exceeding that of any individual feature. In the decision plot of one example patient from the same model, static features accounted for 43% of the total modality attribution, while current-window time-series data contributed the remaining 57%. Younger age (36 years), triage acuity level 3 and emergency-room admission reduced the predicted probability, whereas Medicaid insurance and unmarried status increased it. The associations with insurance and marital status are also relevant to the fairness analysis in Section 5.7.

To examine whether feature importance depended on the fusion architecture, Figure 5.10 compares the top 20 features across concatenation, M3Care and FuseMoE for mechanical ventilation and



(a) Mechanical ventilation



(b) Non-home discharge

Figure 5.10: Top 20 features of models predicting mechanical ventilation (upper) and non-home discharge (lower) across 3 fusion methods. Features are coloured by type.

non-home discharge. For mechanical ventilation, triage acuity levels 3 and 2 consistently occupied the first and second positions across all three models (Figure 5.10a), with level 3 being the most influential feature. As described in Section 3.4.1, triage acuity represents an emergency severity index ranging from 1 (most urgent) to 5 (least urgent). Level 3 indicates patients requiring multiple clinical resources but not immediate life-saving intervention, and was the second most common category in the cohort (see Table 5.1). No laboratory feature appeared among the top 10 in any model, although several were represented lower in the rankings. For non-home discharge, age, walk-in arrival and married status consistently occupied the first three positions across all three fusion methods (Figure 5.10b). Other features, including triage acuity, insurance and comorbidities, appeared throughout the top 20, although their rankings varied between models. Overall, the leading features were consistent across fusion architectures for both outcomes, while greater variation was observed among the lower-ranked features.

5.8.3 Spatial explanations

To understand the spatial contribution of CXR modality to model predictions, we generated SHAP-weighted Grad-CAM maps. This extends the modality-level explanations in Section 5.8.1. These maps retain only positive responses and are individually normalised from a coarse 7×7 grid. They therefore allow only qualitative comparison of spatial contributions to model predictions, instead of attribution strength or precise anatomical localisation.

Qualitative inspection[1] suggested that spatial response patterns were broadly similar between FuseMoE and M3Care for the same prediction outcome, but varied across tasks. For cardiac arrest, both models showed stronger positive responses around the central chest, whereas ICU admission predictions showed more prominent responses in the lower lung fields. These responses represent positive SHAP-weighted contributions relative to the training background, rather than absolute predicted probabilities. Consequently, positive responses may be present even when a model correctly predicts a negative outcome.

[1] Representative CXR spatial explanation maps are not reproduced because the MIMIC-CXR Data Use Agreement restricts the redistribution of individual patient images, including de-identified images (see `https://physionet.org/content/mimic-cxr/view-dua/2.1.0/`).

Chapter 6

Discussions

6.1 What the results support

1. Advanced fusion helped on some tasks but did not consistently improve on concatenation
The comparison across the two clinical pathways did not identify a consistently superior fusion architecture. Simple concatenation remained competitive with FuseMoE and M3Care, despite lacking explicit routing or missing-modality imputation mechanisms (Section 5.2). In deterioration, FuseMoE achieved the highest MCC for three of the five outcomes, but only the improvements from FuseMoE and M3Care on mechanical ventilation remained statistically significant after BH correction. Conversely, concatenation significantly outperformed both for ICU admission within 24 hours (Table 5.4). In hospitalisation, concatenation achieved the highest or joint-highest MCC in five of the six outcome-window comparisons, with both significant differences favouring concatenation over FuseMoE (Table 5.5).

These findings should also be considered in the context of model development. Mechanical ventilation was the initial task used to develop the advanced fusion methods, including their architectural configurations. These configurations were subsequently applied to the remaining deterioration outcomes. The largest improvement over concatenation therefore occurred on the task that guided model development, while the other outcomes provided a broader assessment of their applicability. Taken together, the results suggest that additional architectural complexity does not guarantee better prediction. Its value depends on the clinical outcome, available information and the conditions under which the models were developed.

2. Additional learning strategies and temporal representations brought no consistent gain

Adding CL to M3Care did not improve MCC. Performance reduced for mechanical ventilation and extended hospital stay under both window configurations (Tables 5.9 and 5.10). Its effects on embedding quality were also inconsistent. GCN-to-original embedding similarity decreased for CXR in deterioration, but improved for current CXR and ECG in dual-window hospitalisation (Figures 5.3 and 5.4. Even where similarity improved, MCC remained below the baseline. Furthermore, the learned cross-modal targets showed greater similarity to the original embeddings than the GCN outputs. This suggests that the target information was not fully transferred to the imputed representations. Two-stage curriculum learning included a complete-case warm-up for mechanical ventilation and current-window extended stay, but neither setting showed improvement

over its baseline. However, as single-stage training was only investigated under the dual-window configuration, the effects of training strategy and observation window cannot be separated.

Time-aware modelling similarly provided no improvement for mechanical ventilation. Temporal FuseMoE and temporal M3Care underperformed their static counterparts (Figure 5.5). For FuseMoE, recall and AUROC improved slightly, but specificity declined, indicating a shift towards identifying more positive cases at the expense of additional false positives. One possible explanation is the limited temporal information within the 90-minute ED observation window, where repeated ECG and CXR recordings were uncommon and most stays lacked CXR (Table 5.1). However, the temporal variants also used individual records rather than pooled embeddings and treated clinical time series as a separate modality. The performance reduction therefore cannot be attributed to mTAND alone. Overall, more elaborate training objectives and temporal representations did not improve MCC in the investigated settings, although their effectiveness may differ with denser observations or alternative model configurations.

3. Missing-modality mechanisms behaved as intended, but their predictive benefit was modest

The architectural analyses showed that the missing-modality mechanisms operated as intended, although their predictive benefits were modest. For FuseMoE, the joint router achieved the highest mechanical ventilation MCC among the investigated layouts (Table 5.6). Its expert selections were associated with modality availability without explicit supervision, while the Flex-MoE-inspired [39] combination router followed the observed availability patterns but performed less well. This suggests that learned routing may be preferable to enforcing modality-combination assignments. M3Care also benefited modestly from GCN-based imputation across both representative tasks (Table 5.8). Overall, the routing and imputation mechanisms behaved as intended, but provided only modest predictive gains that did not consistently justify their additional complexity.

4. Predictive performance depended more on modality content than availability

The permutation experiments further revealed that predictive performance depended primarily on recorded clinical information rather than modality availability alone. Permuting CXR content also reduced predictive performance for most deterioration outcomes, whereas ECG permutation had relatively little effect (Table 5.11). This does not suggest that ECG was clinically unimportant, but rather that its representations provided limited additional predictive information under the investigated configurations. When modality contents were shuffled entirely while preserving availability patterns, predictive performance fell to near zero across deterioration outcomes (Table 5.11). A small residual signal remained in dual-window hospitalisation models, suggesting that the previous admission's modality availability also contained some predictive information (Table 5.12). This is consistent with the association between modality availability and adverse outcomes observed in both cohorts (Tables 5.1 and 5.2).

5. Explanations at several levels gave complementary views of model behaviour, but did not establish clinically meaningful reasoning

The explanation analyses examined which information sources influenced model predictions. Across both pathways, structured clinical information accounted for most of the predictive signal. In deterioration, tabular data, including summarised time-series measurements, dominated MM-SHAP attribution in nearly all model runs. In hospitalisation, static EHR data and current-window time series were both important, together accounting for most of the attribution (Figure 5.7). The contribution of other modalities depended partly on their availability. CXR received a relatively

small attribution share across the full deterioration cohort, but its contribution increased substantially among stays with imaging (Figure 5.8). These patterns were broadly consistent across fusion architectures. At the feature level, newly introduced variables, particularly triage acuity and arrival transport, ranked among the most influential features for hospitalisation tasks (Figure 5.10b). This highlights that feature selection may be as important as the fusion strategy.

Spatial explanations further showed that the image regions contributing to predictions varied between outcomes. Together, these analyses provided a broader assessment of information utilisation than any individual explanation method. However, attribution patterns and expert-routing behaviour describe statistical relationships within the trained models, rather than establishing causal explanations or clinically validated reasoning.

6. Fairness disparities followed outcomes and patient attributes more than fusion architecture

Fairness disparities were more strongly associated with prediction outcomes and patient characteristics than with fusion architecture. Across the three models, EOP confidence intervals overlapped in almost all outcome-attribute comparisons (Table 5.13). Age showed the greatest disparities in hospitalisation, particularly for non-home discharge, while ethnicity was associated with larger gaps for several deterioration outcomes. Differences in selection rates also partly reflected underlying outcome prevalence and modality availability (Figure 5.6). These findings suggest that subgroup disparities cannot be attributed to fusion mechanisms alone, but must be considered alongside clinical characteristics and data collection practices. As fairness mitigation was not investigated, the study cannot establish whether modifying the fusion mechanisms would reduce these disparities.

7. Some findings held across both pathways, but the advantage of advanced fusion did not generalise

Although the two pathways differed in their clinical objectives and modality composition, several findings were consistent. Structured clinical information remained the principal contributor to prediction, the clinical information recorded within each modality was more informative than its availability alone, and fairness disparities were more closely related to outcomes and patient attributes than to the fusion architecture. However, the predictive advantage of advanced fusion did not generalise consistently. Its clearest improvement occurred for mechanical ventilation, the initial development outcome, and was not reproduced across most remaining deterioration outcomes or in hospitalisation. The latter also presented greater modality heterogeneity, with substantially more availability patterns (Figure 5.2). These findings suggest that the effectiveness of fusion architectures depends on the clinical task and data configuration. They support evaluating multi-modal methods under different clinical conditions, retaining simple baselines alongside assessments of robustness, fairness and explainability. Comparisons with previous research are discussed in Section 6.2, followed by implications for NHS multi-modal research in Section 6.4.

6.2 Comparison with previous work

The two prediction pathways were built on previous studies: **MDS-ED** [4] for deterioration and **MM-HealthFair** [2, 3] for hospitalisation. Both provide useful reference results for related clinical outcomes. However, differences in cohort construction, input modalities and model selection criteria limit direct comparison. In particular, our study used validation MCC for checkpoint selection, whereas **MM-HealthFair** used validation loss. In this section, we compare test AUROC values as a common measure of discrimination, rather than as evidence of model superiority.

Table 6.1: Comparison of test AUROC with the **MDS-ED** benchmark [4] for the deterioration pathway.

Prediction outcome	MDS-ED	M3Care	FuseMoE
CA	0.986	0.978	0.979
ICU-24h	0.915	0.895	0.904
ICU	0.898	0.883	0.890
MV	0.959	0.945	0.941
MORT-1D	0.960	0.947	0.956

Notes: CA: in-hospital cardiac arrest; ICU-24h: ICU admission within 24 hours; ICU: ICU admission during the hospital stay; MV: mechanical ventilation; MORT-1D: one-day mortality. **MDS-ED** results come from their best models that combine ECG waveform and clinical routine data.

The deterioration cohort followed the **MDS-ED** framework but was reconstructed using **MIMIC-IV** v3.1 instead of v2.2, with further refinements in outcome extraction and an additional CXR modality (described in Section 3.3). As shown in Table 6.1, M3Care and FuseMoE achieved generally comparable, although slightly lower, AUROC across the five prediction outcomes. The differences were relatively small, particularly for cardiac arrest and one-day mortality. Unlike **MDS-ED**, our models used pre-computed representations from frozen ECG-FM and DenseNet foundation models, rather than training one S4 waveform encoder end-to-end for each deterioration task [4]. This allowed embeddings to be reused for different prediction tasks flexibly.

Table 6.2: Comparison of test AUROC with the **MM-HealthFair** benchmark [3] for the hospitalisation pathway under dual-window and current-window configurations.

		Dual-window		Current-window	
Prediction outcome	**MM-HealthFair**	**M3Care**	**FuseMoE**	**M3Care**	**FuseMoE**
In-hospital death	0.73	0.889	0.900	0.897	0.900
Extended stay $>$ 7 days	0.73	0.733	0.732	0.728	0.737
Non-home discharge	0.69	0.824	0.825	0.825	0.825

Notes: **MM-HealthFair** results come from their fully fused IF-EHR+TS+NT model. Their results were reported in 2 decimal places. Dual-window includes current and historical modalities, whereas current-window excludes separate historical modalities.

The hospitalisation cohort extended **MM-HealthFair** with a broader cohort definition and additional modalities. The original study required historical clinical information and a minimum number of recorded measurements [2, 3]. After removing these requirements, the cohort increased from 20,596 to 82,107 patients. ECG and CXR modalities were introduced, alongside information from the current ED stay and recent hospitalisation history where available. As shown in Table 6.2, our models achieved comparable or numerically higher AUROC than **MM-HealthFair**. The largest differences were observed for in-hospital death and non-home discharge, while extended stay prediction remained broadly comparable. Performance was also similar between the current-window and dual-window configurations, suggesting limited additional predictive value from the historical modalities in this setting.

These comparisons should be interpreted cautiously. Differences in patient selection, observation windows, feature preparation and model selection procedures may contribute to the observed variation in AUROC. The higher values for hospitalisation cannot therefore be attributed solely to the proposed fusion architectures. Nevertheless, the results demonstrate that the extended pipeline

achieved discrimination comparable to or above the reported reference values, while accommodating more heterogeneous clinical information and incomplete modalities.

6.3 Limitations of our work

Despite providing a systematic evaluation of multimodal fusion approaches across two clinical prediction pathways, this study has several limitations concerning data generalisability, modality representations, model selection, experimental design and evaluation.

6.3.1 Generalisability and data availability

Our work relied solely on the MIMIC database family, which contains retrospective clinical records from a single medical centre in the United States. Although the study investigated multiple prediction outcomes across the two clinical pathways, all experiments were based on data originating from the same healthcare institution. Therefore, the findings may not generalise to other hospitals, healthcare systems or patient populations. Differences between US and UK healthcare systems, including clinical workflows, investigation practices, admission procedures and discharge policies, may affect both the availability of clinical information and its relationship with patient outcomes. The modality-importance analyses in Section 5.8 indicated that structured EHR features and time-series measurements were particularly influential for many of the investigated outcomes. However, these patterns may depend on the selected prediction tasks, observation windows and available modalities, and may not generalise to other clinical applications.

Data availability and cohort selection introduce additional limitations. As discussed in Chapter 3, the availability of clinical modalities varied considerably across patients and pathways. Missingness in routinely collected clinical data is unlikely to be entirely random, as investigations such as ECGs, CXRs and laboratory tests are ordered according to patients' clinical presentations and healthcare needs. Modality availability may be associated with patient characteristics, clinical decisions and subsequent outcomes. Although the study investigated the predictive contribution of modality availability and content, further research is needed to fully disentangle the underlying clinical and institutional factors driving missingness. Moreover, patients with available imaging, physiological signals or historical clinical records may represent selected subpopulations, potentially limiting the generalisability of findings obtained from these modalities.

6.3.2 Modality representations and fusion architectures

The modelling framework relied largely on pre-trained modality encoders to generate representations for subsequent multimodal fusion. While this approach enabled the integration of heterogeneous clinical data with manageable computational requirements, the resulting embeddings were not necessarily optimised for the specific prediction outcomes investigated. The pre-trained encoders were not systematically fine-tuned for the downstream tasks, potentially limiting their ability to capture clinically relevant information. Furthermore, mean pooling was commonly applied to obtain fixed-dimensional representations from sequential or spatially structured embeddings, including ECG, CXR and clinical text representations. This aggregation may discard informative temporal, spatial or contextual relationships, reducing the information available to the fusion architectures.

The architectural comparison was also restricted to two advanced fusion frameworks, FuseMoE and M3Care, alongside concatenation. Although these models represent different approaches to

multimodal integration and missing-modality handling, other fusion and imputation strategies were not investigated. The adopted models also have specific limitations. For M3Care, inspection of the selected models revealed that the learnable similarity mixing coefficients and graph thresholds remained close to their initial values after training, as discussed in Section 5.4.1. This suggests limited adaptation of these graph-construction parameters, though the underlying patient similarity representations could still change during training. GCN-based imputation constructed patient similarity graphs using the available mini-batch, without a fixed reference cohort during inference. The resulting imputed embeddings may be sensitive to batch composition, and the stability of predictions under different inference batches was not comprehensively evaluated. For FuseMoE, expert routing and load balancing introduced additional architectural complexity, while the extent to which individual experts learned distinct and clinically meaningful representations remained difficult to establish.

6.3.3 Experimental design and model selection

Several aspects of the experimental design may limit the robustness of the reported comparisons. Hyperparameter tuning and architectural ablations were conducted using mechanical ventilation and extended stay as representative outcomes for the deterioration and hospitalisation pathways, as described in Section 4.4. The selected configurations were subsequently applied to the remaining outcomes without independent optimisation. Although this approach reduced computational requirements and supported consistent comparisons, configurations optimised for the representative outcomes may not be optimal for other prediction tasks. In addition, architectural configurations were selected according to test-set MCC during the ablation studies. Where the same test partition was subsequently used for final performance reporting, this introduces a risk of optimistic performance estimates arising from repeated test-set-informed model selection. The reported bootstrap confidence intervals quantify uncertainty in the estimated metrics but do not account for this potential selection bias.

Our experimental scope was also limited by computational constraints. While the training hyperparameter search incorporated a three-seed stability assessment, the final model comparisons and most architectural ablations were conducted using a single random seed. Variability observed during hyperparameter tuning indicates that model performance can depend on random initialisation and training stochasticity, so the reported results may not fully reflect the variability across independent training runs. Additionally, artificial modality dropout was not systematically investigated during training for concatenation and M3Care, while only a limited range of dropout configurations was explored for FuseMoE. This restricted the assessment of whether deliberate exposure to incomplete modality combinations could improve robustness beyond learning from naturally missing clinical data.

The uncertainty estimates were also subject to limitations arising from the bootstrap resampling procedure described in Section 4.5.2. The 95% confidence intervals were calculated by resampling individual prediction records with replacement, without accounting for potential clustering of encounters within patients. This is particularly relevant to the deterioration pathway, where multiple ED stays from the same patient could be included in the cohort. As repeated encounters from the same patient may be statistically dependent, treating them as independent observations during bootstrap resampling may underestimate the uncertainty in predictive performance estimates.

The main predictive comparisons also employed a fixed classification threshold of 0.5, without systematic threshold optimisation for individual outcomes. Consequently, the reported threshold-dependent metrics may not represent the best achievable operating points for each model. Although MCC, AUROC, balanced accuracy, recall and specificity provided complementary measures of predictive performance, probability calibration and clinical utility were not comprehensively evaluated. Therefore, the results do not establish whether the predicted risks would support clinical decision-making at practically relevant intervention thresholds.

6.3.4 Contrastive learning and temporal modelling

The investigation of CL in M3Care was exploratory and restricted to selected training strategies, auxiliary objectives and loss-weight configurations. Although several CL-related ablations were conducted, the hyperparameter search was not exhaustive, and the reasons for the inconsistent effects on predictive performance and imputation quality were not fully established. In particular, the current-window hospitalisation experiments employed two-stage curriculum learning, whereas the dual-window experiments used single-stage joint training. Since these configurations also differed in their included modalities, the effects of training strategy and modality composition could not be isolated. The learned cross-modal targets generally exhibited higher similarity to the original embeddings than the GCN-generated representations, but due to time constraints, the mechanisms limiting the transfer of this information to the imputation process were not comprehensively investigated.

The evaluation of mTAND-based temporal encoding was similarly limited to mechanical ventilation prediction in the deterioration pathway. Although mTAND incorporated observation timestamps, the experiments did not systematically examine the effects of modality-specific sampling frequencies, observation density or temporal coverage within the initial 90-minute ED observation window. Additionally, the original fusion models used the `averageRecord` representation, whereas the mTAND extensions used `allRecord` to preserve individual observations and their timestamps. The observed performance differences could not be attributed exclusively to the temporal encoder, as the underlying time-series representation also differed.

6.3.5 Explainability and fairness evaluation

Although modality-level explanations and feature attribution for structured EHR data were investigated, fine-grained explanations were not developed for time-series measurements and clinical notes. This limited the ability to identify influential observation times, individual measurements and clinically relevant text segments within the multimodal predictions. For CXR images, Grad-CAM visualisations were constrained by the spatial resolution and learned representations of the frozen DenseNet encoder. The resulting heatmaps provided approximate localisation of influential image regions rather than precise anatomical or pathological explanations.

Fairness evaluation was also affected by differences in outcome prevalence and subgroup sample sizes. For relatively rare outcomes, small numbers of positive cases within demographic subgroups could produce unstable fairness estimates and wide confidence intervals, limiting the reliability of subgroup comparisons. Also, the analyses focused on selected demographic attributes and group-level fairness metrics, without thoroughly examining intersectional subgroups or the underlying causes of observed disparities. Potential biases may also arise across patient subgroups or from

patterns embedded in non-tabular modalities, such as CXR, ECG and clinical notes, which were not specifically investigated in this study.

6.4 Implications for NHS multi-modal research

The findings of this study have several implications for the development of multi-modal AI within NHS research. An important consideration is that clinical records reflect not only a patient's underlying condition, but also the care they receive. Investigations such as ECGs, CXRs and laboratory tests are performed according to clinical needs, resulting in incomplete records and irregularly sampled observations. These data collection patterns may therefore reflect clinical decisions alongside changes in patient health [70]. This raises questions about whether models learn transferable clinical relationships or patterns specific to the environment in which they were trained. Fusion methods should accommodate heterogeneous clinical data while allowing these effects to be examined. For NHS research, model evaluation should therefore consider dependence on local clinical practices alongside predictive performance, particularly when transferring between hospitals or care pathways.

The recent Single Patient Record initiative aims to bring together fragmented patient information across NHS services, improving access to clinical records across care settings [93, 94]. This creates opportunities for developing multi-modal patient representations, as explored in previous research on learning from incomplete clinical data [36]. However, a unified clinical record does not necessarily translate into a single embedding that is equally useful across different applications. Representations optimised for mortality or deterioration prediction may emphasise different aspects of patient information from those needed for patient similarity retrieval or phenotype discovery. Information that contributes little to one task may still be relevant to another. The modular framework developed in this study provides a starting point for investigating this question. While pre-trained modality embeddings and software components can be reused across prediction tasks, the transferability of the learned fused representations remains to be evaluated.

Fairness should also be considered throughout the development of multi-modal patient representations, rather than solely at the prediction stage. Differences in the availability and frequency of clinical measurements may arise from patient characteristics, clinical needs and healthcare practices [69, 70]. Such differences may introduce disparities before data reach the fusion model. Fairness evaluation could therefore extend beyond subgroup prediction metrics to examine differences in modality coverage and sampling patterns. However, differences in data availability do not necessarily indicate unfairness, as they may also reflect differences in clinical needs. Understanding these patterns in their clinical context is therefore important when assessing whether differences in data collection contribute to unequal model performance.

A final consideration is the transferability of these findings to NHS settings. Our experiments used MIMIC-IV data collected in the United States, where patient populations, clinical workflows and recording practices may differ from those in UK hospitals. Validation using NHS datasets is crucial to determine whether the observed performance, robustness and fairness characteristics generalise across healthcare systems [66]. Such evaluation should consider not only differences in patient populations, but also whether models remain reliable when clinical data collection practices change.

6.5 Future work

Building on the findings and limitations of this study, several directions could be explored to further improve multimodal representation learning, robustness to incomplete clinical information and the applicability of the developed framework to NHS and broader healthcare settings.

6.5.1 Improving modality representations and temporal modelling

As discussed in Section 6.3.2, the current framework largely relied on pre-trained feature extractors, with limited adaptation to the investigated clinical outcomes. Future research could explore foundation models pre-trained on clinical tasks more closely related to the target outcomes, alongside end-to-end or parameter-efficient fine-tuning of modality encoders, such as LoRA [95]. Alternative aggregation strategies, such as attention-based pooling or hierarchical representation learning, could also be investigated to preserve temporal, spatial and contextual information that may be lost through mean pooling.

Further improvements could focus on the modelling of longitudinal clinical information. Although mTAND enabled the incorporation of irregular observation timestamps, its evaluation was restricted to mechanical ventilation prediction within the initial 90-minute ED observation window. The relatively short observation period and sparse repeated measurements, particularly for ECG and CXR, may have limited the benefits of explicit temporal modelling. Future studies could investigate longer observation windows and more longitudinal prediction tasks, incorporating repeated measurements collected at different sampling frequencies. Alternative temporal architectures, including Transformer-based encoders, could also be evaluated to investigate whether more expressive temporal representations improve predictive performance.

6.5.2 Extending multimodal fusion and representation learning

Future research could extend the architectural comparison to alternative fusion and missing-modality handling approaches. For example, DrFuse [34] learns disentangled shared and modality-specific representations to address missing modalities and inconsistencies between clinical data sources. MUSE [36] employs patient-modality bipartite graphs and mutual-consistent contrastive learning to support representation learning under missing modalities and labels. These approaches could provide complementary perspectives to the expert-routing and graph-based imputation mechanisms investigated in FuseMoE and M3Care.

Recent developments in LLMs also provide opportunities for alternative multimodal integration strategies. For example, UniPACT [96] combines structured EHR information with representations derived from raw ECG waveforms within an LLM-based prognostic question-answering framework, demonstrating its applicability to clinical prediction tasks in the MDS-ED benchmark. Investigating such approaches could help establish whether language-model-based integration provides additional benefits over conventional embedding-level fusion, particularly for combining structured clinical measurements with textual and physiological information.

Another interesting direction is to extend the current single-task modelling framework to multitask learning. Shared modality encoders and fusion representations could be jointly optimised across related prediction tasks, such as mechanical ventilation, ICU admission and mortality. This may improve representation efficiency and facilitate the learning of clinically relevant information shared

across outcomes. However, potential negative transfer between tasks with different characteristics would require careful evaluation.

Further investigation of M3Care's graph-based imputation and contrastive-learning mechanisms is also warranted. The current experiments showed that learned cross-modal targets could achieve relatively high similarity to original modality embeddings, but this information was not consistently reflected in the GCN-generated representations or downstream predictive performance. Future work could explore alternative contrastive objectives, target construction strategies, loss-weight schedules and training procedures to improve the alignment between learned targets and imputed embeddings. Improvements to graph construction could also be investigated, including alternative patient-similarity measures, more effective optimisation of learnable graph parameters and the use of fixed reference cohorts to improve the stability of imputation during inference.

6.5.3 Improving robustness and experimental validation

Future experiments could more systematically investigate robustness to incomplete and changing modality availability. Artificial modality dropout during training could be evaluated across all fusion architectures, alongside simulated changes in modality coverage and observation patterns. Such experiments would help determine whether models trained under the missingness patterns of one healthcare environment remain reliable when deployed under different clinical investigation practices.

More rigorous experimental validation would also strengthen the reliability of model comparisons. Future studies could adopt validation-based architectural selection followed by evaluation on an independent, previously unused test set, alongside repeated training across multiple random seeds. Patient-level clustered bootstrap resampling could be employed where repeated encounters are included, while probability calibration, outcome-specific decision thresholds and clinically relevant operating points could be investigated to assess the practical usefulness of model predictions beyond discrimination metrics.

6.5.4 External validation and adaptation to NHS settings

A major priority is to evaluate the generalisability of the framework beyond the single-centre MIMIC database family. External validation using independent datasets would help establish whether the observed relationships between modality availability, fusion mechanisms and predictive performance are transferable across clinical populations and institutions. For example, the INSPECT dataset [97], which incorporates longitudinal EHR data and medical imaging for pulmonary embolism prediction, could provide an alternative multimodal setting for evaluating the flexibility of the framework.

Most importantly, future research could investigate the adaptation of the developed pipeline to UK clinical datasets and patient pathways. Differences in healthcare delivery, clinical coding, investigation practices, admission procedures and patient populations may influence both the available modalities and their predictive value. Potential next steps include the construction of comparable NHS prediction cohorts and assessing model performance, calibration and robustness under local data conditions. Such validation would be necessary before considering the application of these models to NHS clinical decision support.

6.5.5 Fairness and explainability

Further work could investigate methods for identifying and mitigating disparities in multimodal predictions across patient subgroups. Potential approaches include subgroup reweighting, fairness-aware regularisation, adversarial debiasing and post-processing adjustments [98]. These methods could be evaluated alongside predictive performance to understand potential trade-offs between fairness criteria and model discrimination.

Fairness analyses could also be extended to intersectional patient groups and modality-specific sources of bias. Future studies could examine whether differences in the availability, quality or representations of CXR images, ECG recordings and clinical notes contribute to unequal model behaviour beyond the demographic and structured EHR characteristics investigated in this study. More fine-grained explanations for temporal measurements and clinical text could additionally support the investigation of influential clinical information and potential sources of bias within individual modalities.

6.5.6 Extending multimodal representations to patient retrieval

Beyond clinical outcome prediction, the developed framework could be extended to similarity-based patient retrieval. Rather than using multimodal representations exclusively for binary classification, future systems could retrieve historical or reference cases with similar clinical characteristics, presenting their previous care trajectories and observed outcomes as contextual information for clinical decision support.

Such retrieval could also provide a complementary form of model explanation, allowing predictions to be considered alongside examples of comparable cases. However, similarity in the learned embedding space does not necessarily correspond to clinically meaningful similarity, particularly when representations have been optimised for individual prediction outcomes. Future research would therefore need to evaluate retrieval relevance, robustness to missing modalities and the clinical interpretability of the retrieved cases.

Ultimately, these extensions could support the development of more transferable multimodal patient representations that accommodate heterogeneous and incomplete clinical information, while providing a foundation for applications beyond individual prediction tasks.

Chapter 7

Conclusion

This study developed and evaluated a multimodal clinical prediction framework for integrating heterogeneous and incomplete electronic health records (EHRs) using the MIMIC-IV open dataset. Building on the previous **MM-HealthFair** framework, the project extended the range of clinical modalities to include ECG recordings and chest X-ray images alongside structured EHR data, time-series measurements and clinical notes. Two prediction pathways were investigated: short-term deterioration following emergency department attendance and outcomes associated with inpatient hospitalisation.

The study systematically compared simple concatenation with two advanced multimodal fusion architectures, FuseMoE and M3Care, across multiple clinical prediction outcomes. The results demonstrated that more sophisticated fusion mechanisms and missingness mitigation strategies did not consistently outperform concatenation, though improvements were observed for selected tasks. Additional investigations of M3Care's graph-based imputation and contrastive learning, as well as mTAND-based temporal encoding highlighted the challenges of learning effective representations from incomplete and irregularly sampled clinical data. In particular, improved embedding alignment and more expressive temporal modelling did not necessarily translate into better downstream predictive performance. Beyond predictive performance, the developed framework supported the investigation of modality contributions, robustness to missing information, explainability and fairness. These analyses highlighted the importance of routinely collected structured clinical information and demonstrated that both modality content and availability can influence model predictions.

Our findings emphasise that the effectiveness of multimodal AI depends not only on fusion architecture, but also on the quality, availability and clinical relevance of the underlying data. While further validation and methodological development are required before clinical application, this project provides a reusable and extensible foundation for investigating multimodal representation learning and clinical risk prediction. Future research could build on this framework to develop more robust and transferable patient representations, supporting applications across NHS clinical pathways and beyond.

Bibliography

[1] Luis R. Soenksen et al. “Integrated Multimodal Artificial Intelligence Framework for Healthcare Applications”. In: *npj Digital Medicine* 5.1 (2022), p. 149. DOI: 10.1038/s41746-022-00689-4.

[2] Sophie Martin and Jonathan Pearson. *Understanding fairness and explainability in multimodal approaches within healthcare*. NHS England PhD Internship report. NHS England, 2024. URL: `https://github.com/nhsengland/mm-healthfair/blob/main/report/MMHealthFair_v1_SM_Final_Report.pdf`.

[3] Konstantin Georgiev. *MMHealthFair: Evaluating Fairness in Multimodal Learning Approaches for Risk Prediction*. NHS England PhD Internship report. NHS England, 2025. URL: `https://github.com/nhsengland/mm-healthfair/blob/main/report/MMHealthFair_v2_KG_Final_Report.pdf`.

[4] Juan Miguel Lopez Alcaraz, Hjalmar Bouma, and Nils Strodthoff. “Enhancing Clinical Decision Support with Physiological Waveforms — A Multimodal Benchmark in Emergency Care”. In: *Computers in Biology and Medicine* 192 (2025), p. 110196. DOI: 10.1016/j.compbiomed.2025.110196.

[5] Felix Krones et al. “Review of multimodal machine learning approaches in healthcare”. In: *Information Fusion* 114 (2025), p. 102690. DOI: 10.1016/j.inffus.2024.102690.

[6] Benjamin Shickel et al. “Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record (EHR) Analysis”. In: *IEEE journal of biomedical and health informatics* 22 (5 2018), pp. 1589–1604. DOI: 10.1109/jbhi.2017.2767063.

[7] Cao Xiao, Edward Choi, and Jimeng Sun. “Opportunities and challenges in developing deep learning models using electronic health records data: a systematic review”. In: *Journal of the American Medical Informatics Association* 25 (10 2018), pp. 1419–1428. DOI: 10.1093/jamia/ocy068.

[8] Siyue Yang et al. “Machine learning approaches for electronic health records phenotyping: a methodical review”. In: *Journal of the American Medical Informatics Association : JAMIA* 30 (2 2023), pp. 367–381. DOI: 10.1093/jamia/ocac216.

[9] Sercan Arık and Tomas Pfister. “TabNet: Attentive Interpretable Tabular Learning”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 35 (8 2021), pp. 6679–6687. DOI: 10.1609/aaai.v35i8.16826.

[10] Yuqi Si et al. “Deep representation learning of patient data from Electronic Health Records (EHR): A systematic review”. In: *Journal of Biomedical Informatics* 115 (2021), p. 103671. DOI: 10.1016/j.jbi.2020.103671.

[11] Sarah Pungitore and Vignesh Subbian. “Assessment of Prediction Tasks and Time Window Selection in Temporal Modeling of Electronic Health Record Data: a Systematic Review”. In: *Journal of Healthcare Informatics Research* 7 (3 2023), pp. 313–331. DOI: 10.1007/s41666-023-00143-4.

[12] Abhidnya Patharkar et al. “Predictive modeling of biomedical temporal data in healthcare applications: review and future directions”. In: *Frontiers in Physiology* 15 (2024), p. 1386760. DOI: 10.3389/fphys.2024.1386760.

[13] Feng Xie et al. “Deep learning for temporal data representation in electronic health records: A systematic review of challenges and methodologies”. In: *Journal of Biomedical Informatics* 126 (2022), p. 103980. DOI: 10.1016/j.jbi.2021.103980.

[14] Beanbonyka Rim et al. “Deep Learning in Physiological Signal Data: A Survey”. In: *Sensors (Basel, Switzerland)* 20 (4 2020), p. 969. DOI: 10.3390/s20040969.

[15] U. Sumalatha et al. “Deep Learning Applications in ECG Analysis and Disease Detection: An Investigation Study of Recent Advances”. In: *IEEE Access* 12 (2024), pp. 126258–126284. DOI: 10.1109/access.2024.3447096.

[16] Ali Amirahmadi, Mattias Ohlsson, and Kobra Etminani. “Deep learning prediction models based on EHR trajectories: A systematic review”. In: *Journal of Biomedical Informatics* 144 (2023), p. 104430. DOI: 10.1016/j.jbi.2023.104430.

[17] Yunendah Nur Fuadah and Ki Moo Lim. “Advances in cardiovascular signal analysis with future directions: a review of machine learning and deep learning models for cardiovascular disease classification based on ECG, PCG, and PPG signals”. In: *Biomedical Engineering Letters* 15.4 (2025), pp. 619–660. DOI: 10.1007/s13534-025-00473-9.

[18] Barsha Abhisheka et al. “Recent trend in medical imaging modalities and their applications in disease diagnosis: a review”. In: *Multimedia Tools and Applications* 83.14 (2023), pp. 43035–43070. DOI: 10.1007/s11042-023-17326-1.

[19] Jiaji Wang, Shuihua Wang, and Yudong Zhang. “Deep learning on medical image analysis”. In: *CAAI Transactions on Intelligence Technology* 10 (1 2025), pp. 1–35. DOI: 10.1049/cit2.12356.

[20] Laura Quinn et al. “Interobserver variability studies in diagnostic imaging: a methodological systematic review”. In: *The British Journal of Radiology* 96 (1148 2023), p. 20220972. DOI: 10.1259/bjr.20220972.

[21] Most Nilufa Yeasmin et al. “Advances of AI in image-based computer-aided diagnosis: A review”. In: *Array* 23 (2024), p. 100357. DOI: 10.1016/j.array.2024.100357.

[22] Sanad Aburass et al. “Vision Transformers in Medical Imaging: a Comprehensive Review of Advancements and Applications Across Multiple Diseases”. In: *Journal of Imaging Informatics in Medicine* 38.6 (2025), pp. 3928–3971. DOI: 10.1007/s10278-025-01481-y.

[23] Amirhossein Kazerouni et al. “Diffusion models in medical imaging: A comprehensive survey”. In: *Medical Image Analysis* 88 (2023), p. 102846. DOI: 10.1016/j.media.2023.102846.

[24] Shaoting Zhang and Dimitris Metaxas. “On the challenges and perspectives of foundation models for medical image analysis”. In: *Medical Image Analysis* 91 (2024), p. 102996. DOI: 10.1016/j.media.2023.102996.

[25] Licheng Jiao et al. “Foundation Models Meet Medical Image Interpretation”. In: *Research* 9 (2026). DOI: 10.34133/research.1024.

[26] Elias Hossain et al. “Natural Language Processing in Electronic Health Records in relation to healthcare decision-making: A systematic review”. In: *Computers in Biology and Medicine* 155 (2023), p. 106649. DOI: 10.1016/j.compbiomed.2023.106649.

[27] Katrin Klug et al. “From admission to discharge: a systematic review of clinical natural language processing along the patient journey”. In: *BMC Medical Informatics and Decision Making* 24 (1 2024), p. 238. DOI: 10.1186/s12911-024-02641-w.

[28] Jinhyuk Lee et al. “BioBERT: a pre-trained biomedical language representation model for biomedical text mining”. In: *Bioinformatics* 36 (4 2020), pp. 1234–1240. DOI: 10.1093/bioinformatics/btz682.

[29] Felix Busch et al. “Current applications and challenges in large language models for patient care: a systematic review”. In: *Communications Medicine* 5.1, 26 (2025). DOI: 10.1038/s43856-024-00717-2.

[30] Paul Pu Liang, Amir Zadeh, and Louis Philippe Morency. “Foundations & Trends in Multimodal Machine Learning: Principles, Challenges, and Open Questions”. In: *ACM Computing Surveys* 56 (10 2024). DOI: 10.1145/3656580.

[31] Siwar Chaabene et al. “An overview of methods and techniques in multimodal data fusion with application to healthcare”. In: *International Journal of Data Science and Analytics* 20 (4 2025), pp. 3093–3117. DOI: 10.1007/s41060-025-00715-0.

[32] Shih Cheng Huang et al. “Fusion of medical imaging and electronic health records using deep learning: a systematic review and implementation guidelines”. In: *npj Digital Medicine* 3.1, 136 (2020). DOI: 10.1038/s41746-020-00341-z.

[33] Kimberly Amador et al. “A cross-attention-based deep learning approach for predicting functional stroke outcomes using 4D CTP imaging and clinical metadata”. In: *Medical Image Analysis* 99 (2025), p. 103381. DOI: 10.1016/j.media.2024.103381.

[34] Wenfang Yao et al. “DrFuse: Learning Disentangled Representation for Clinical Multi-Modal Fusion with Missing Modality and Modal Inconsistency”. In: *Proceedings of the AAAI Conference on Artificial Intelligence* 38 (15 2024), pp. 16416–16424. DOI: 10.1609/aaai.v38i15.29578.

[35] Chaohe Zhang et al. “M3Care: Learning with Missing Modalities in Multimodal Healthcare Data”. In: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 1 (2022), pp. 2418–2428. DOI: 10.1145/3534678.3539388.

[36] Zhenbang Wu et al. “Multimodal Patient Representation Learning with Missing Modalities and Labels”. In: *International Conference on Learning Representations*. 2024. URL: `https://proceedings.iclr.cc/paper_files/paper/2024/hash/f49d76cf84df83a611883c621c96d2d9-Abstract-Conference.html`.

[37] Noam Shazeer et al. “Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer”. In: *5th International Conference on Learning Representations, ICLR 2017 - Conference Track Proceedings* (2017). URL: `https://arxiv.org/pdf/1701.06538`.

[38] Xing Han et al. *FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion*. 2025. DOI: 10.48550/arxiv.2402.03226. arXiv: `2402.03226 [cs]`.

[39] Sukwon Yun et al. *Flex-MoE: Modeling Arbitrary Modality Combination via the Flexible Mixture-of-Experts*. 2024. DOI: 10.48550/arxiv.2410.08245. arXiv: `2410.08245 [cs]`.

[40] Noah Cohen Kalafut, Xiang Huang, and Daifeng Wang. “Joint variational autoencoders for multimodal imputation and embedding”. In: *Nature Machine Intelligence* 5.6 (2023), pp. 631–642. DOI: 10.1038/s42256-023-00663-z.

[41] Junxin Wang et al. “VAE-Driven Multimodal Fusion for Early Cardiac Disease Detection”. In: *IEEE Access* 12 (2024), pp. 90535–90551. DOI: 10.1109/access.2024.3420444.

[42] Wasifur Rahman et al. “Integrating Multimodal Information in Large Pretrained Transformers”. In: *Proceedings of the Annual Meeting of the Association for Computational Linguistics* (2020), pp. 2359–2369. DOI: 10.18653/v1/2020.acl-main.214.

[43] Rachael A. Hughes et al. “Accounting for missing data in statistical analyses: multiple imputation is not always the answer”. In: *International Journal of Epidemiology* 48 (4 2019), pp. 1294–1304. DOI: 10.1093/ije/dyz032.

[44] Camillo Maria Caruso, Valerio Guarrasi, and Paolo Soda. “Resilient Vision-Tabular Multimodal Learning under Modality Missingness”. In: *arXiv preprint arXiv:2605.12031* (2026). URL: `https://arxiv.org/abs/2605.12031`.

[45] Lien P. Le et al. “Multimodal missing data in healthcare: A comprehensive review and future directions”. In: *Computer Science Review* 56 (2025), p. 100720. DOI: 10.1016/j.cosrev.2024.100720.

[46] Aashish Bhandari and Sonika Tyagi. “A comparative evaluation of handling missing data points and modalities in electronic health records”. In: *International Journal of Medical Informatics* 210 (2026), p. 106302. DOI: 10.1016/j.ijmedinf.2026.106302.

[47] Gargi Joshi, Rahee Walambe, and Ketan Kotecha. “A Review on Explainability in Multimodal Deep Neural Nets”. In: *IEEE Access* 9 (2021), pp. 59800–59821. DOI: 10.1109/access.2021.3070212.

[48] Nikolaos Rodis et al. “Multimodal Explainable Artificial Intelligence: A Comprehensive Review of Methodological Advances and Future Research Directions”. In: *IEEE Access* 12 (2024), pp. 159794–159820. DOI: 10.1109/access.2024.3467062.

[49] Scott M Lundberg and Su-In Lee. “A Unified Approach to Interpreting Model Predictions”. In: *Advances in Neural Information Processing Systems*. Vol. 30. Curran Associates, Inc., 2017. URL: `https://papers.nips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html`.

[50] Letitia Parcalabescu and Anette Frank. “MM-SHAP: A Performance-agnostic Metric for Measuring Multimodal Contributions in Vision and Language Models & Tasks”. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, 2023, pp. 4032–4059. DOI: 10.18653/v1/2023.acl-long.223.

[51] Sarthak Jain and Byron C. Wallace. “Attention is not Explanation”. In: *NAACL HLT 2019 - 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies - Proceedings of the Conference* 1 (2019), pp. 3543–3556. DOI: 10.18653/v1/n19-1357.

[52] Sarah Wiegreffe and Yuval Pinter. “Attention is not not Explanation”. In: *EMNLP-IJCNLP 2019 - 2019 Conference on Empirical Methods in Natural Language Processing and 9th International Joint Conference on Natural Language Processing, Proceedings of the Conference* (2019), pp. 11–20. DOI: 10.18653/v1/d19-1002.

[53] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. ""Why Should I Trust You?": Explaining the Predictions of Any Classifier". In: *Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2016* (2016), pp. 97–101. DOI: 10.18653/v1/n16-3020.

[54] Mukund Sundararajan, Ankur Taly, and Qiqi Yan. "Axiomatic Attribution for Deep Networks". In: *Proceedings of the 34th International Conference on Machine Learning*. Vol. 70. Proceedings of Machine Learning Research. PMLR, 2017, pp. 3319–3328. URL: https://proceedings.mlr.press/v70/sundararajan17a.html.

[55] Avanti Shrikumar, Peyton Greenside, and Anshul Kundaje. "Learning Important Features Through Propagating Activation Differences". In: *Proceedings of the 34th International Conference on Machine Learning*. Vol. 70. Proceedings of Machine Learning Research. PMLR, 2017, pp. 3145–3153. URL: https://proceedings.mlr.press/v70/shrikumar17a.html.

[56] Alexander Binder et al. "Layer-wise relevance propagation for deep neural network architectures". In: *Information Science and Applications, ICISA 2016* 376 (2016), pp. 913–922. DOI: 10.1007/978-981-10-0557-2_87.

[57] Edoardo Mosca et al. "SHAP-Based Explanation Methods: A Review for NLP Interpretability". In: *Proceedings of the 29th International Conference on Computational Linguistics*. International Committee on Computational Linguistics, 2022, pp. 4593–4603. URL: https://aclanthology.org/2022.coling-1.406/.

[58] Karen Simonyan, Andrea Vedaldi, and Andrew Zisserman. "Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps". In: *2nd International Conference on Learning Representations, ICLR 2014 - Workshop Track Proceedings* (2013). URL: https://arxiv.org/pdf/1312.6034.

[59] Ramprasaath R. Selvaraju et al. "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization". In: *2017 IEEE International Conference on Computer Vision (ICCV)*. 2017, pp. 618–626. DOI: 10.1109/iccv.2017.74.

[60] Aditya Chattopadhay et al. "Grad-CAM++: Improved Visual Explanations for Deep Convolutional Networks". In: *Proceedings - 2018 IEEE Winter Conference on Applications of Computer Vision, WACV 2018* 2018-January (2017), pp. 839–847. DOI: 10.1109/wacv.2018.00097.

[61] Matthew D. Zeiler and Rob Fergus. "Visualizing and Understanding Convolutional Networks". In: *Computer Vision – ECCV 2014*. Vol. 8689. Lecture Notes in Computer Science. Springer Verlag, 2014, pp. 818–833. DOI: 10.1007/978-3-319-10590-1_53.

[62] Vitali Petsiuk, Abir Das, and Kate Saenko. "RISE: Randomized Input Sampling for Explanation of Black-box Models". In: *British Machine Vision Conference 2018, BMVC 2018* (2018). URL: https://arxiv.org/pdf/1806.07421.

[63] Daiju Ueda et al. "Fairness of Artificial Intelligence in Healthcare: Review and Recommendations". In: *Japanese Journal of Radiology* 42.1 (2024), pp. 3–15. DOI: 10.1007/s11604-023-01474-3.

[64] Deepshikha Charan Ashana et al. "Equitably Allocating Resources during Crises: Racial Differences in Mortality Prediction Models". In: *American Journal of Respiratory and Critical Care Medicine* 204.2 (2021), pp. 178–186. DOI: 10.1164/rccm.202012-4383oc.

[65] Laleh Seyyed-Kalantari et al. “Underdiagnosis Bias of Artificial Intelligence Algorithms Applied to Chest Radiographs in Under-Served Patient Populations”. In: *Nature Medicine* 27.12 (2021), pp. 2176–2182. DOI: 10.1038/s41591-021-01595-0.

[66] Eliane Röösli, Selen Bozkurt, and Tina Hernandez-Boussard. “Peeking into a Black Box, the Fairness and Generalizability of a MIMIC-III Benchmarking Model”. In: *Scientific Data* 9.1 (2022), p. 24. DOI: 10.1038/s41597-021-01110-7.

[67] Yongshuo Zong, Yongxin Yang, and Timothy Hospedales. *MEDFAIR: Benchmarking Fairness for Medical Imaging*. 2023. DOI: 10.48550/arxiv.2210.01725. arXiv: 2210.01725 [cs.LG].

[68] Agostina J. Larrazabal et al. “Gender Imbalance in Medical Imaging Datasets Produces Biased Classifiers for Computer-Aided Diagnosis”. In: *Proceedings of the National Academy of Sciences* 117.23 (2020), pp. 12592–12594. DOI: 10.1073/pnas.1919012117.

[69] Ziad Obermeyer et al. “Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations”. In: *Science* 366.6464 (2019), pp. 447–453. DOI: 10.1126/science.aax2342.

[70] Denis Agniel, Isaac S. Kohane, and Griffin M. Weber. “Biases in Electronic Health Record Data Due to Processes within the Healthcare System: Retrospective Observational Study”. In: *BMJ* 361 (2018), k1479. DOI: 10.1136/bmj.k1479.

[71] Cynthia Dwork et al. *Fairness Through Awareness*. 2011. DOI: 10.48550/arxiv.1104.3913. arXiv: 1104.3913 [cs.CC].

[72] Matt J. Kusner et al. *Counterfactual Fairness*. 2018. DOI: 10.48550/arxiv.1703.06856. arXiv: 1703.06856 [stat.ML].

[73] Moritz Hardt, Eric Price, and Nathan Srebro. “Equality of Opportunity in Supervised Learning”. In: *Advances in Neural Information Processing Systems*. Vol. 29. 2016. URL: https://proceedings.neurips.cc/paper_files/paper/2016/hash/6a9659feb1216f14f7384ba499518b38-Abstract.html.

[74] Rachel Bellamy et al. “AI Fairness 360: An Extensible Toolkit for Detecting and Mitigating Algorithmic Bias”. In: *IBM Journal of Research and Development* (2019). DOI: 10.1147/jrd.2019.2942287.

[75] Hilde Weerts et al. “Fairlearn: Assessing and Improving Fairness of AI Systems”. In: *Journal of Machine Learning Research* 24.257 (2023), pp. 1–8. URL: https://www.jmlr.org/papers/v24/23-0389.html.

[76] Pedro Saleiro et al. *Aequitas: A Bias and Fairness Audit Toolkit*. 2019. DOI: 10.48550/arxiv.1811.05577. arXiv: 1811.05577 [cs.LG].

[77] Eoin Delaney et al. *OxonFair: A Flexible Toolkit for Algorithmic Fairness*. 2024. DOI: 10.48550/arxiv.2407.13710. arXiv: 2407.13710 [cs.CY].

[78] Tom Pollard et al. “PhysioNet as a global platform for biomedical research”. In: *Nature Health* 1.8 (2026), pp. 792–795. DOI: 10.1038/s44360-026-00096-z.

[79] Alistair Johnson et al. *MIMIC-IV*. Tech. rep. PhysioNet, 2024. DOI: 10.13026/kpb9-mt58.

[80] Alistair Johnson et al. *MIMIC-IV-ED*. Tech. rep. PhysioNet, 2023. DOI: 10.13026/5ntk-km72.

[81] Asad Aali et al. “MIMIC-IV-Ext-BHC: Labeled Clinical Notes Dataset for Hospital Course Summarization”. In: *PhysioNet* (2025). Version 1.2.0. DOI: 10.13026/5gte-bv70.

[82] Alistair Johnson et al. “MIMIC-IV-Note: Deidentified free-text clinical notes”. In: *PhysioNet* (2023). Version 2.2. DOI: 10.13026/1n74-ne17.

[83] Alistair Johnson et al. “MIMIC-CXR-JPG - chest radiographs with structured labels”. In: *PhysioNet* (2024). Version 2.1.0. DOI: 10.13026/jsn5-t979.

[84] Brian Gow et al. “MIMIC-IV-ECG: Diagnostic Electrocardiogram Matched Subset”. In: *PhysioNet* (2023). Version 1.0. DOI: 10.13026/4nqg-sb35.

[85] Nils Strodthoff, Juan Miguel Lopez Alcaraz, and Wilhelm Haverkamp. *MIMIC-IV-ECG-Ext-ICD: Diagnostic Labels for MIMIC-IV-ECG*. Tech. rep. PhysioNet, 2024. DOI: 10.13026/ypt5-9d58.

[86] Joseph Paul Cohen et al. “TorchXRayVision: A Library of Chest X-ray Datasets and Models”. In: *Proceedings of The 5th International Conference on Medical Imaging with Deep Learning*. Vol. 172. Proceedings of Machine Learning Research. PMLR, 2022, pp. 231–249. URL: https://proceedings.mlr.press/v172/cohen22a.html.

[87] Kaden McKeen et al. “ECG-FM: An Open Electrocardiogram Foundation Model”. In: *JAMIA Open* 8.5 (2025), ooaf122. DOI: 10.1093/jamiaopen/ooaf122.

[88] Emily Alsentzer et al. *Publicly Available Clinical BERT Embeddings*. 2019. DOI: 10.48550/arxiv.1904.03323. arXiv: 1904.03323 [cs.CL].

[89] Mark Neumann et al. “ScispaCy: Fast and Robust Models for Biomedical Natural Language Processing”. In: *Proceedings of the 18th BioNLP Workshop and Shared Task*. 2019, pp. 319–327. DOI: 10.18653/v1/w19-5034. arXiv: 1902.07669 [cs.CL].

[90] Satya Narayan Shukla and Benjamin M. Marlin. *Multi-Time Attention Networks for Irregularly Sampled Time Series*. arXiv:2101.10318 [cs.LG]. 2021. DOI: 10.48550/arxiv.2101.10318.

[91] Damai Dai et al. *DeepSeekMoE: Towards Ultimate Expert Specialization in Mixture-of-Experts Language Models*. arXiv:2401.06066 [cs.CL]. 2024. DOI: 10.48550/arxiv.2401.06066.

[92] Gabriel Erion et al. “Improving performance of deep learning models with axiomatic attribution priors and expected gradients”. In: *Nature Machine Intelligence* 3.7 (2021), pp. 620–631. DOI: 10.1038/s42256-021-00343-w.

[93] NHS England. *NHS England » Single Patient Record – your health at your fingertips*. URL: https://www.england.nhs.uk/digitaltechnology/the-single-patient-record/.

[94] Department of Health and Social Care. *Estimated impact of the introduction of the single patient record: methodology*. 2026. URL: https://www.gov.uk/government/publications/estimated-impact-of-the-introduction-of-the-single-patient-record/estimated-impact-of-the-introduction-of-the-single-patient-record-methodology.

[95] Edward J. Hu et al. *LoRA: Low-Rank Adaptation of Large Language Models*. 2021. arXiv: 2106.09685 [cs.CL]. URL: https://arxiv.org/abs/2106.09685.

[96] Jialu Tang et al. “UniPACT: A Multimodal Framework for Prognostic Question Answering on Raw ECG and Structured EHR”. In: *ICASSP 2026 - 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2026, pp. 22537–22541. DOI: 10.1109/ICASSP55912.2026.11461333.

[97] Shih-Cheng Huang et al. “INSPECT: A Multimodal Dataset for Pulmonary Embolism Diagnosis and Prognosis”. In: *arXiv preprint arXiv:2311.10798* (2023).

[98] Tiago P. Pagano et al. “Bias and Unfairness in Machine Learning Models: A Systematic Review on Datasets, Tools, Fairness Metrics, and Identification and Mitigation Methods”. In: *Big Data and Cognitive Computing* 7 (1 2023), p. 15. DOI: 10.3390/BDCC7010015/S1.